



AREA III

DIAGNOSTIC EXAM



AIM TO TOP
REGION TWO TRAINING CENTER

ILIGAN



Bayaw, remind yourself that





AIM TO TOP
REGIONAL TRAINING CENTER

ILIGAN

QUESTION NO. 1

In psychrometric chart, these are represented by lines at diagonals.

- a. Dry Bulb Temperature
- b. Wet Bulb Temperature
- c. Specific Enthalpy
- d. Specific Volume

PSYCHROMETRY

- the science of studying the properties of air-water mixtures.
- Psychrometrics is a word which derived from the GREEK words, **Psychro**, and **Metrics**. **Psychro** means '*to breathe*' or '*to blow*' or '*to make cold*'. **Metrics** means to 'measure'.
- Hence, psychrometry is used to measure the properties of moist air, their control, the effect on materials, and human comfort.



PSYCHROMETRIC CHART

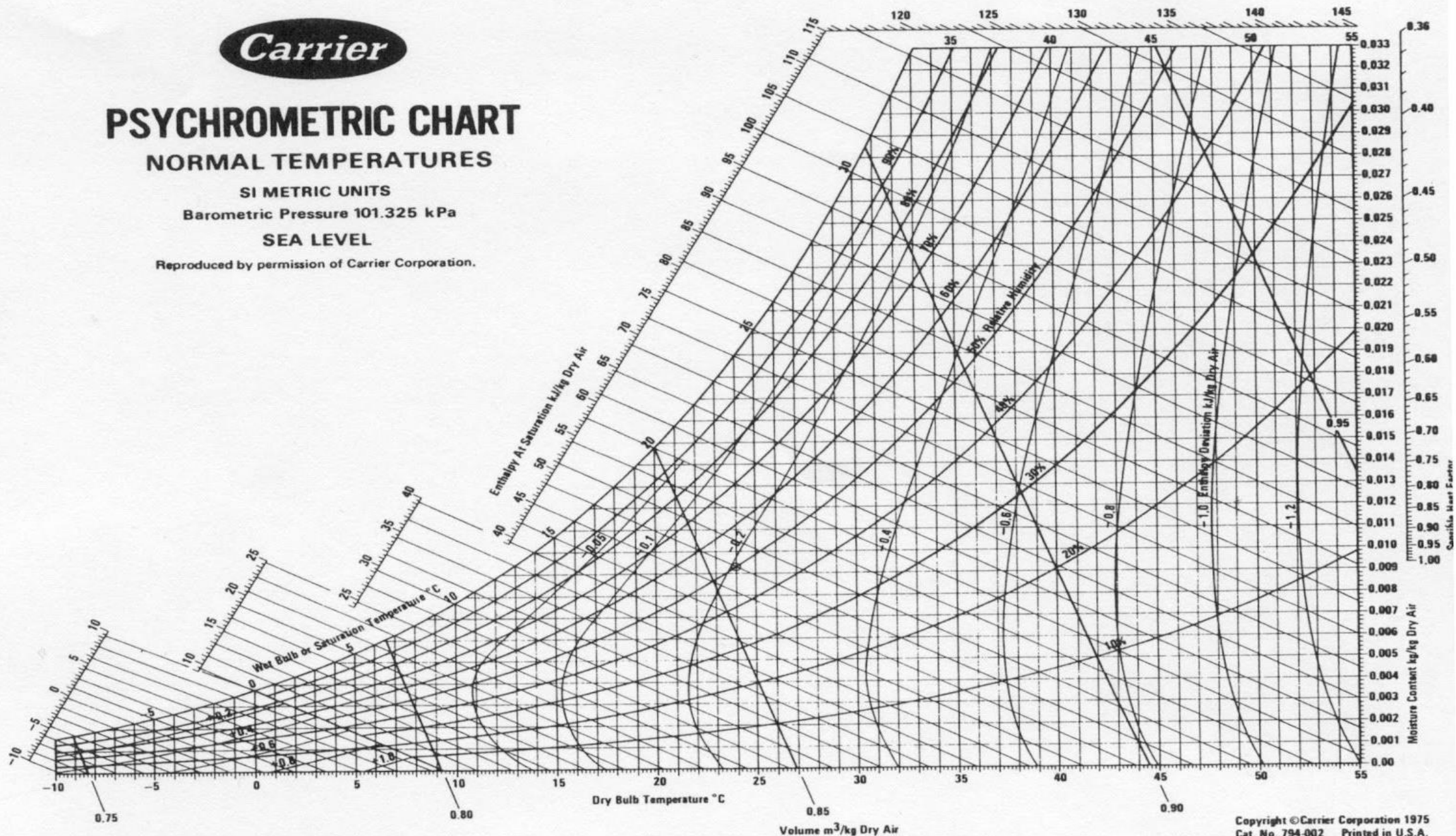
NORMAL TEMPERATURES

SI METRIC UNITS

Barometric Pressure 101.325 kPa

SEA LEVEL

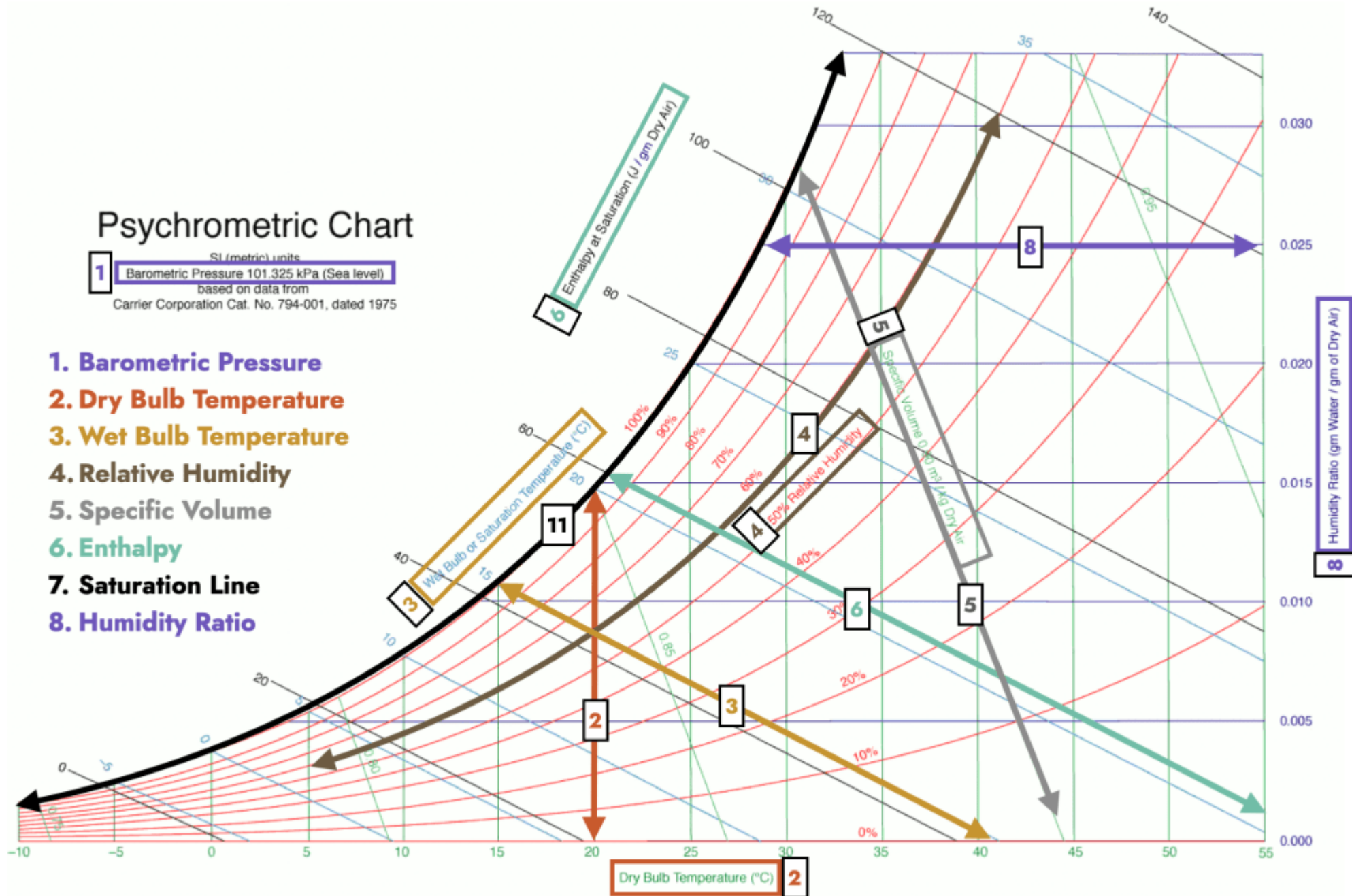
Reproduced by permission of Carrier Corporation.



Psychrometric Chart

1 SI (metric) units
Barometric Pressure 101.325 kPa (Sea level)
based on data from
Carrier Corporation Cat. No. 794-001, dated 1975

1. Barometric Pressure
2. Dry Bulb Temperature
3. Wet Bulb Temperature
4. Relative Humidity
5. Specific Volume
6. Enthalpy
7. Saturation Line
8. Humidity Ratio



QUESTION NO. 1

In psychrometric chart, these are represented by lines at diagonals.

- a. Dry Bulb Temperature
- b. **Wet Bulb Temperature**
- c. Specific Enthalpy
- d. Specific Volume

QUESTION NO. 2

This law states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

- a. 1st Law of Thermodynamics
- b. 2nd Law of Thermodynamics
- c. 3rd Law of Thermodynamics
- d. 0th Law of Thermodynamics

Laws of Thermodynamics

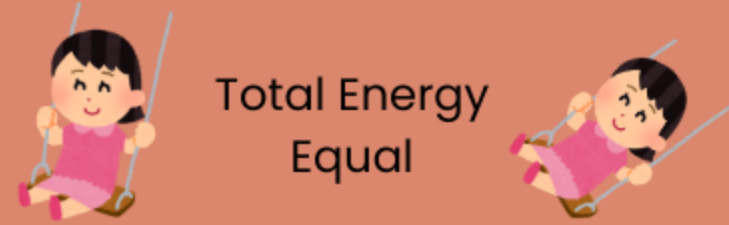
Zeroth Law

Defines temperature equality: if two bodies are each with a third, they're in equilibrium.



First Law

Energy remains constant; it transforms but cannot be created or destroyed.



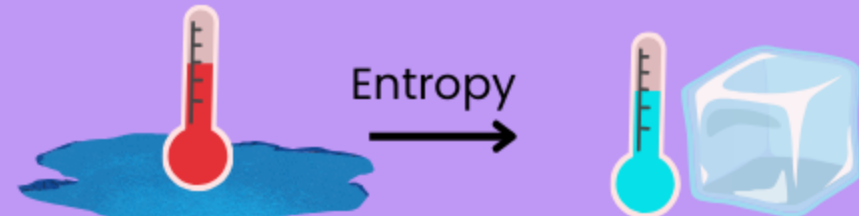
Second Law

Entropy increases in isolated systems; heat flows from hot to cold.



Third Law

Entropy approaches a constant value as temperature approaches absolute zero.



QUESTION NO. 2

This law states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

- a. 1st Law of Thermodynamics
- b. 2nd Law of Thermodynamics
- c. **3rd Law of Thermodynamics**
- d. 0th Law of Thermodynamics

QUESTION NO. 3

In PAES 421:2009, What is the minimum carriage way of one-lane and two-lane FMR, excluding shoulders?

- a. 2-4 m
- b. 4-6 m
- c. 2.5-5 m
- d. 3-6 m

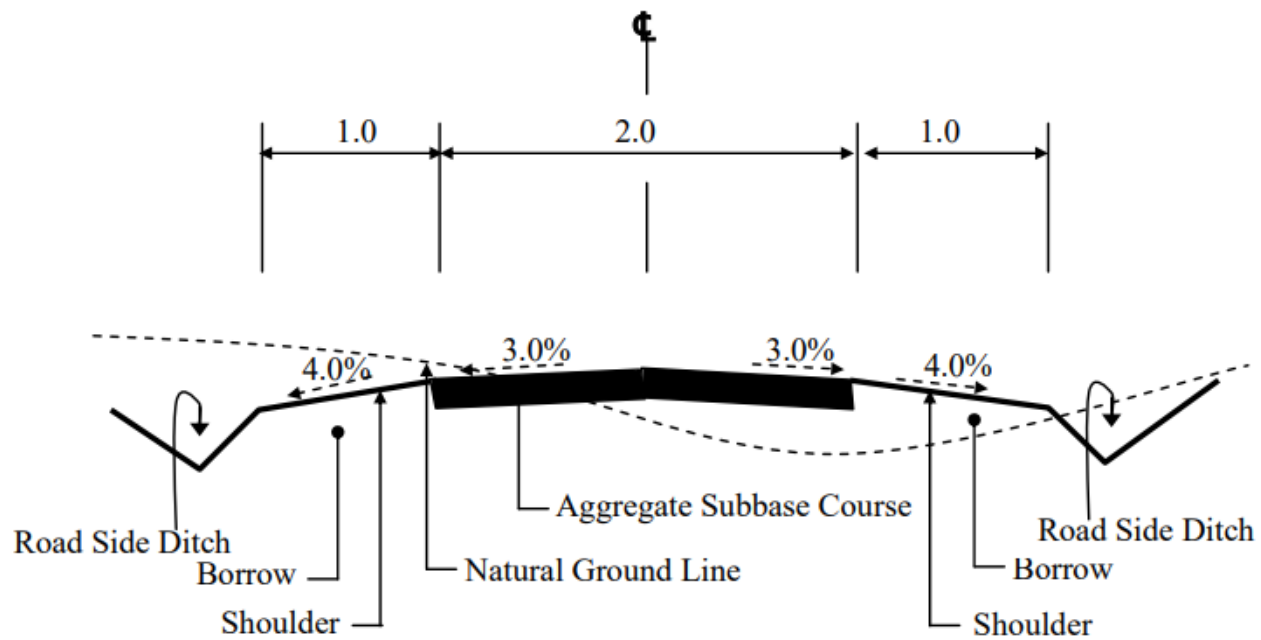


Figure 3 – Typical Cross Section of One-Lane Earth Road

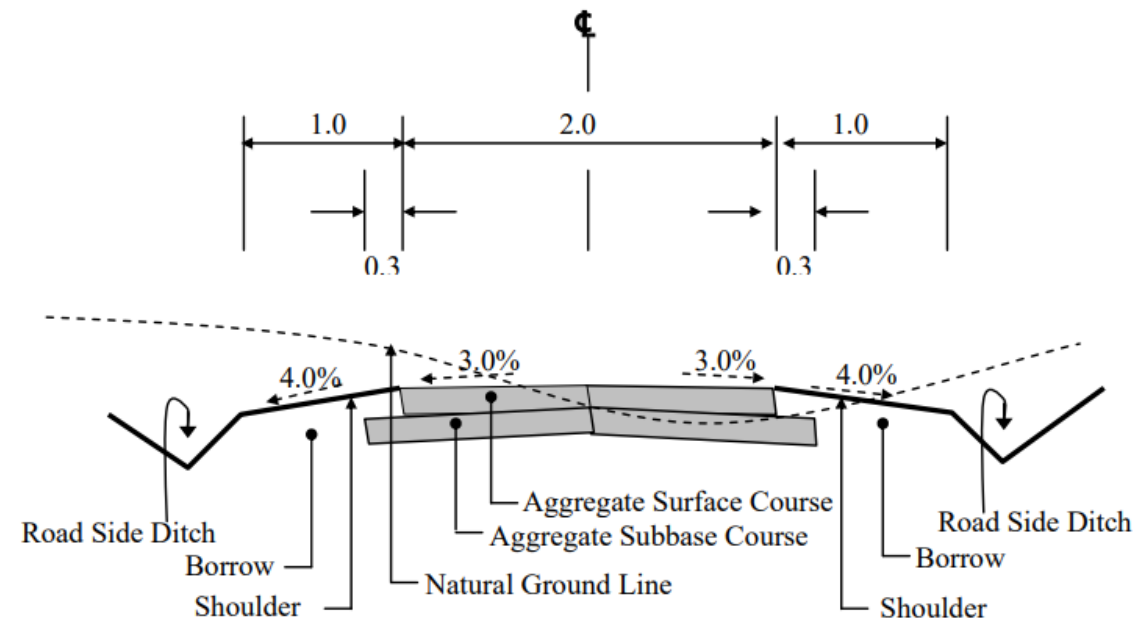


Figure 4 - Typical Cross Section of One-Lane Gravel Road



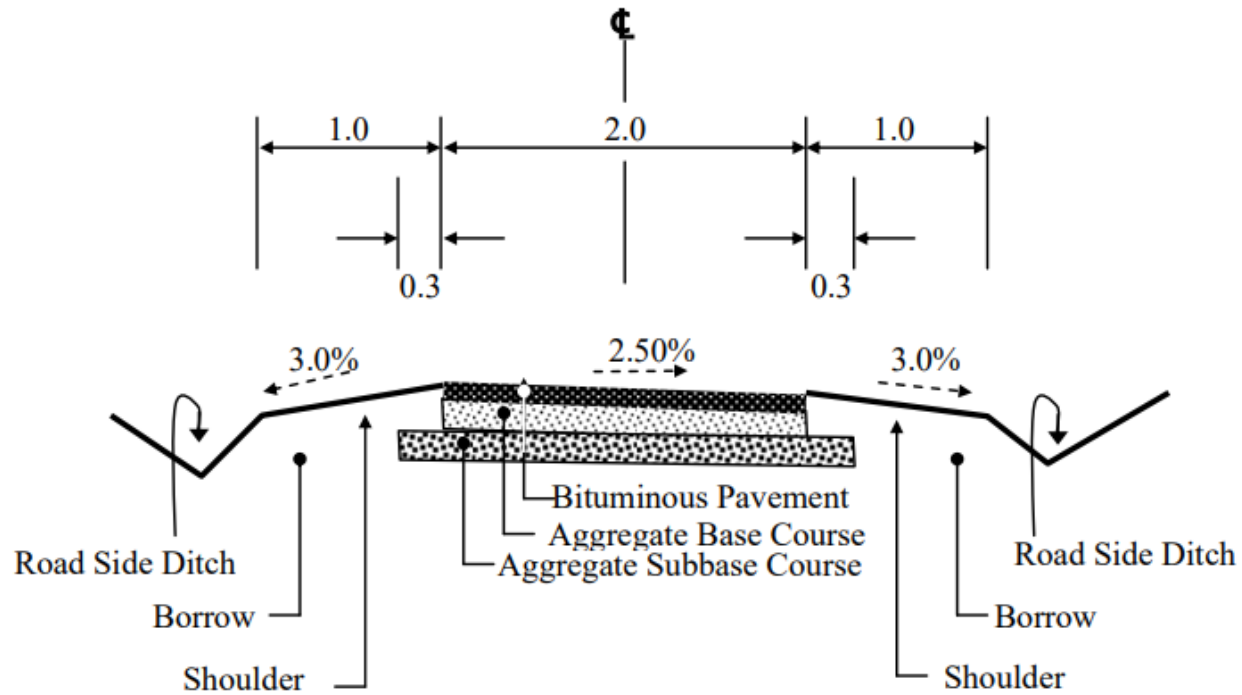


Figure 5 - Typical Cross Section of One-Lane Bituminous Pavement

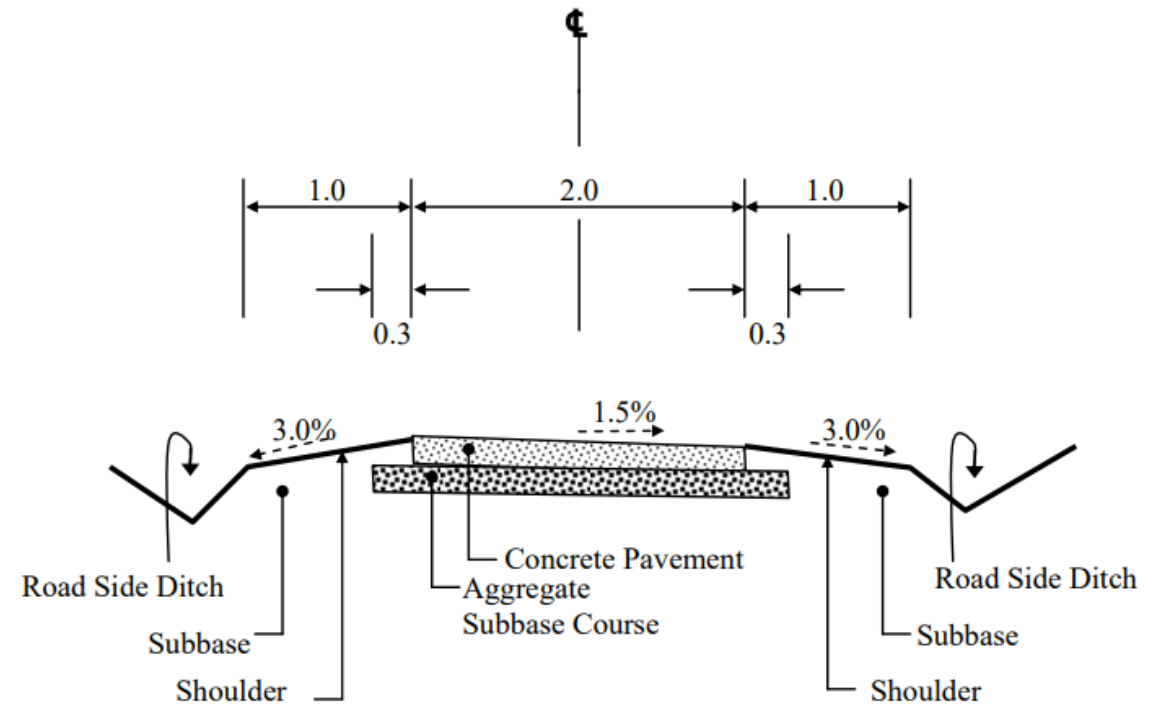
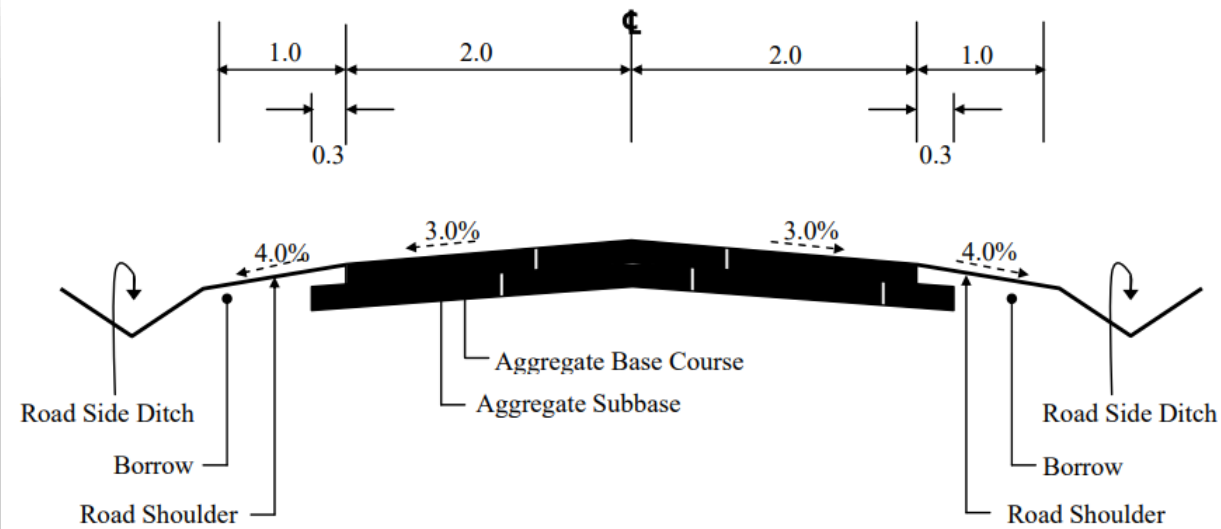
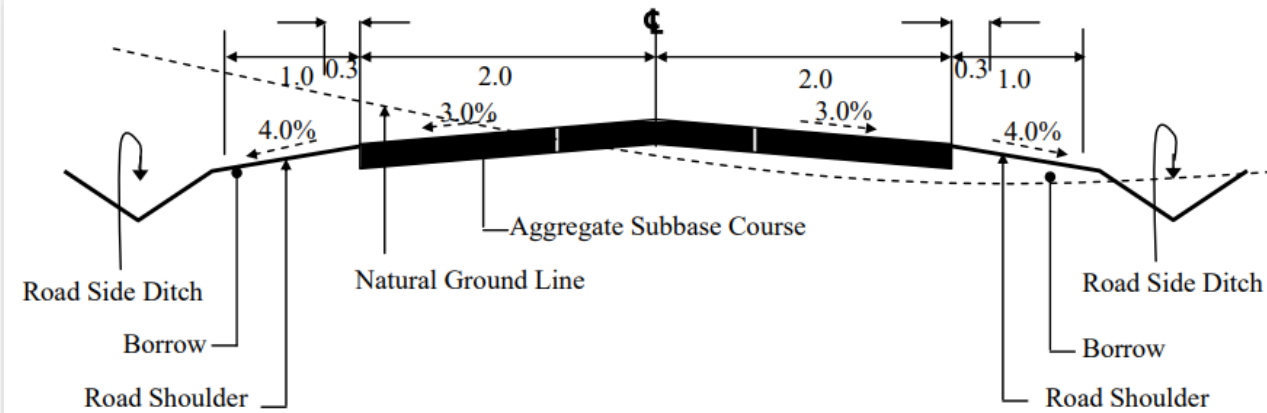


Figure 6 - Typical Cross-Section of One Lane Concrete Road





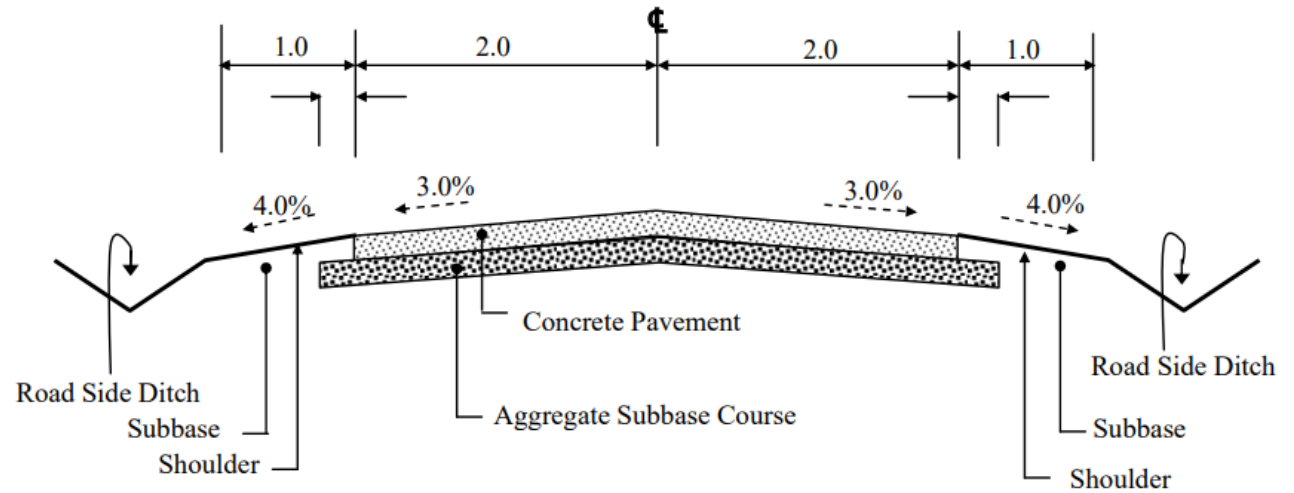


Figure 14 -Typical Cross-Section of a Two-Lane Concrete Road

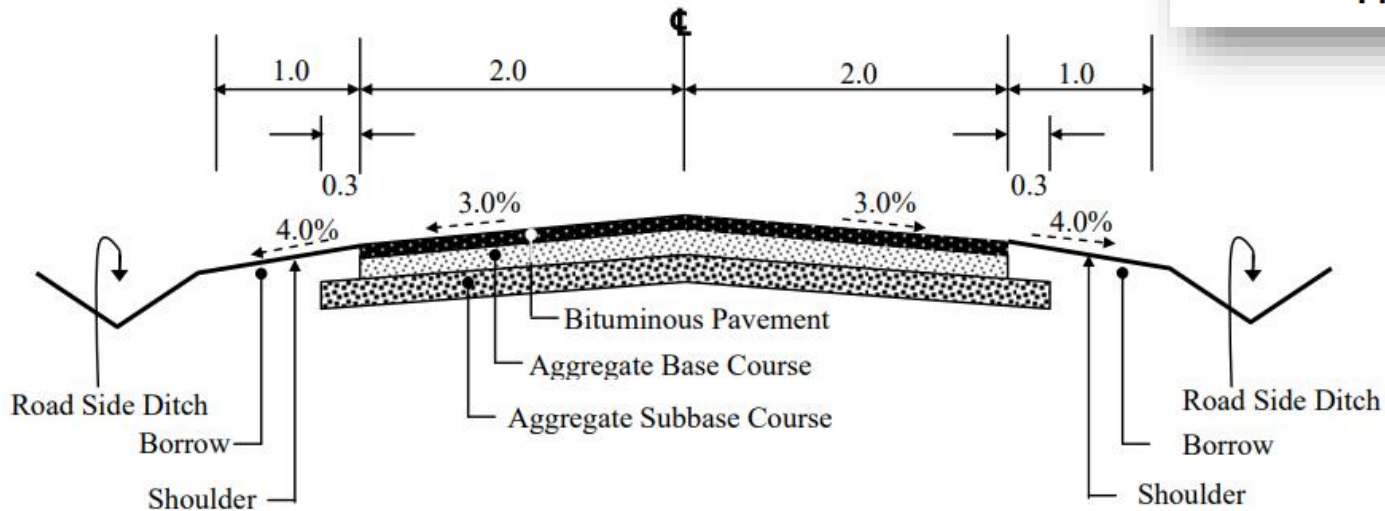


Figure 13 – Typical-Cross-Section of Two-Lane Bituminous Road



AIM TO TOP
ILIGAN



6.2.10 Road carriageway

The road carriageway shall provide space for the safety and convenient movement of the vehicles. The minimum road carriageway width requirement for one-lane and two-lane roadway is summarized in Table 6.

Table 6 – Minimum road carriageway and road shoulder width requirement

Road lane	ADT	Minimum pavement width (m)	Minimum road shoulder width (m)
One lane	<50	2.5	0.75
Two lane	50-400	5	1.5

Road carriageway width shall be measured from the inner edge of the roadway shoulder to the opposite side of the inner edge of the roadway shoulder.

The minimum crossfall for the carriageway shall be 1.5 %. The typical cross-section for a two-lane concrete road is illustrated in Figure 8.

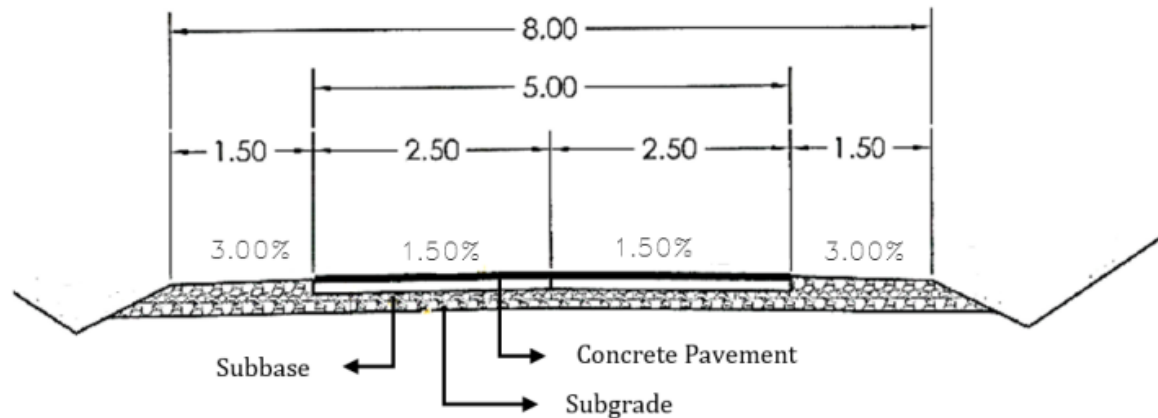


Figure 8 – Typical cross-section for two-lane concrete road

QUESTION NO. 3

In PAES 421:2009, What is the minimum carriage way of one-lane and two-lane FMR, excluding shoulders?

- a. **2-4 m**
- b. 4-6 m
- c. 2.5-5 m
- d. 3-6 m

QUESTION NO. 4

In PAES 402:2000, the recommended brooding temperature for 8-14 days chicks.

- a. provide heat when necessary
- b. 29-32 °C
- c. 32-35 °C
- d. 27-29 °C

Table 2 – Recommended brooding temperature

Age of chicks days	Temperature ° C
1 – 7	32 - 35
8 – 14	29 – 32
14 – 21	27 – 29
Beyond 21 days	Provide heat only when necessary



QUESTION NO. 4

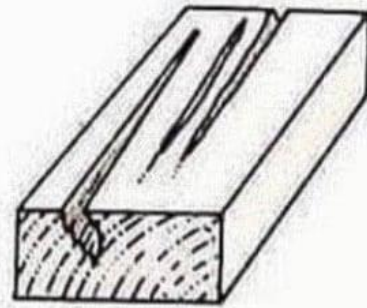
In PAES 402:2000, the recommended brooding temperature for 8-14 days chicks.

- a. provide heat when necessary
- b. **29-32 °C**
- c. 32-35 °C
- d. 27-29 °C

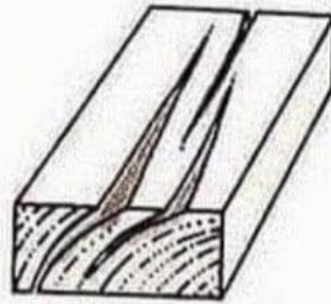
QUESTION NO. 5

What is the defect of wood that causes abnormal growth which occurs at the starting point of a limb or branch of a tree?

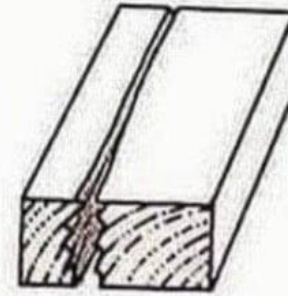
- a. shake
- b. knot
- c. check
- d. wane



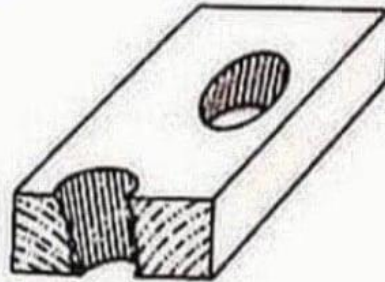
Checks



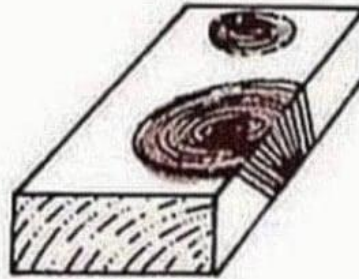
Shakes



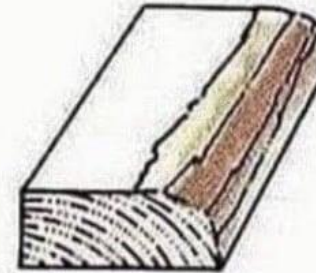
Splits



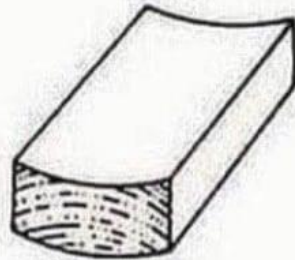
Knot holes



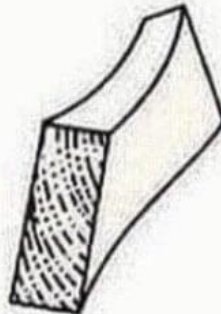
Knots



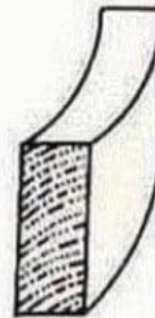
Wane



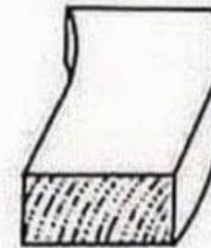
Cupped



Twisted



Bowed



Crooked

QUESTION NO. 5

What is the defect of wood that causes abnormal growth which occurs at the starting point of a limb or branch of a tree?

- a. shake
- b. **knot**
- c. check
- d. wane

QUESTION NO. 6

In PAES 410:2000, the lairage shall be provided with lighting intensity of ____ lux.

- a. 200
- b. 100
- c. 500
- d. 110

6.3.8 Lighting

6.3.8.1 Illumination of 110 lux shall be provided within the lairage.

6.3.8.2 For all suspect pens, lighting intensity of not less than 540 lux shall be provided.



QUESTION NO. 6

In PAES 410:2000, the lairage shall be provided with lighting intensity of ____ lux.

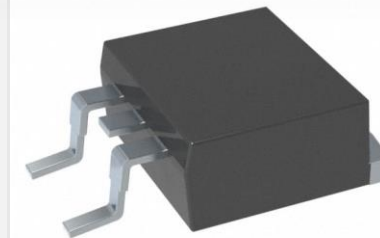
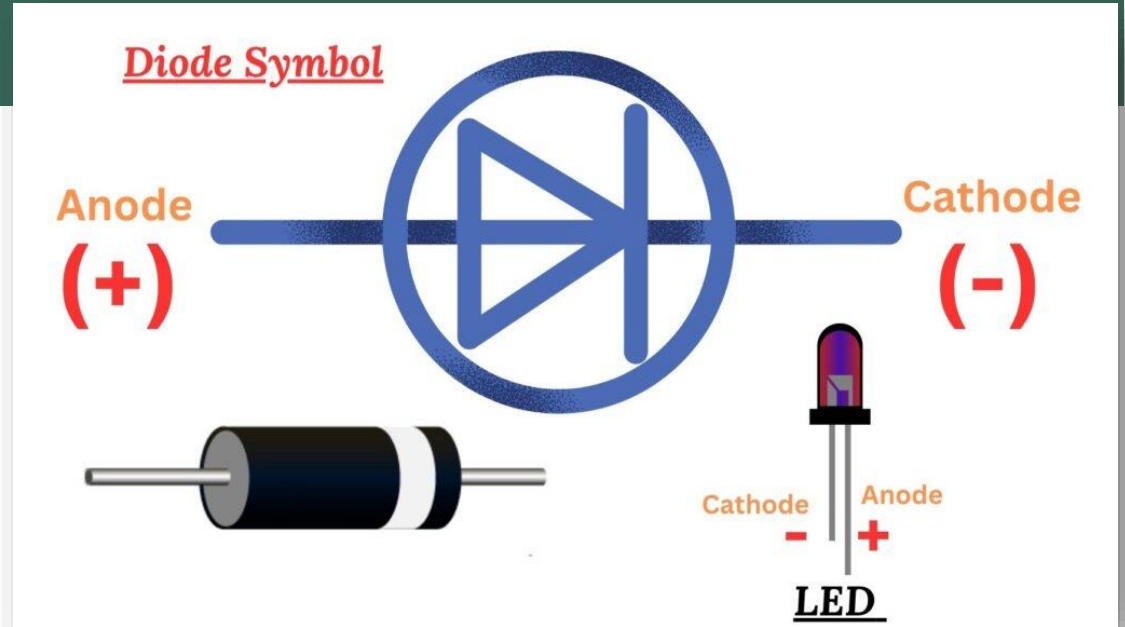
- a. 200
- b. 100
- c. 500
- d. **110**

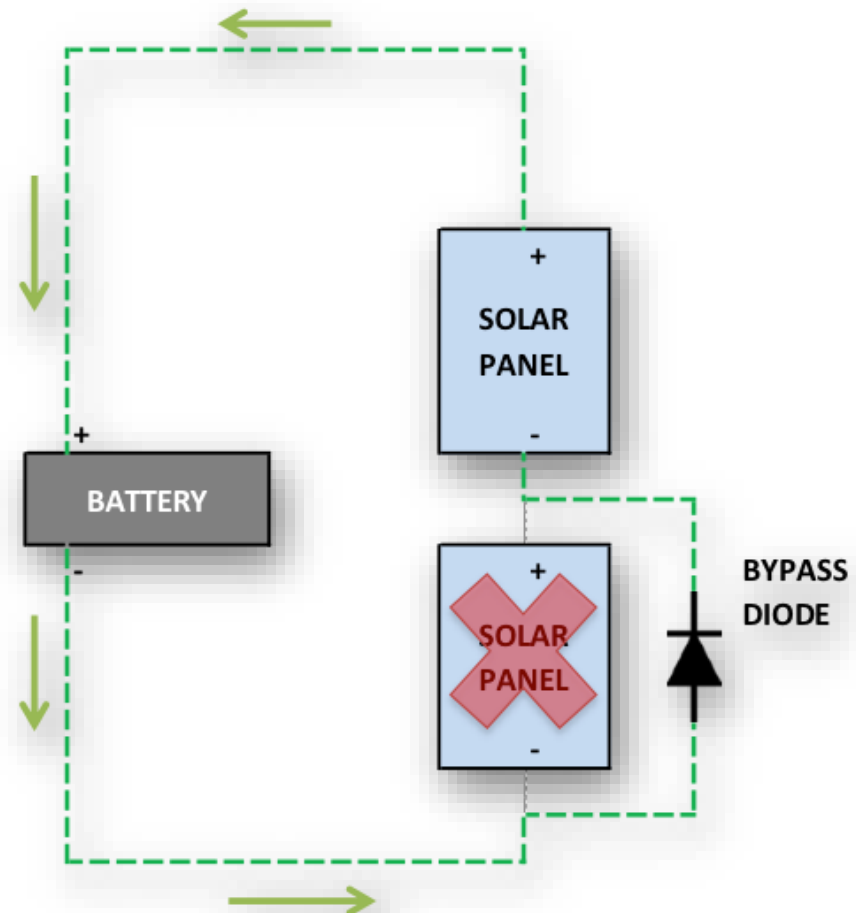
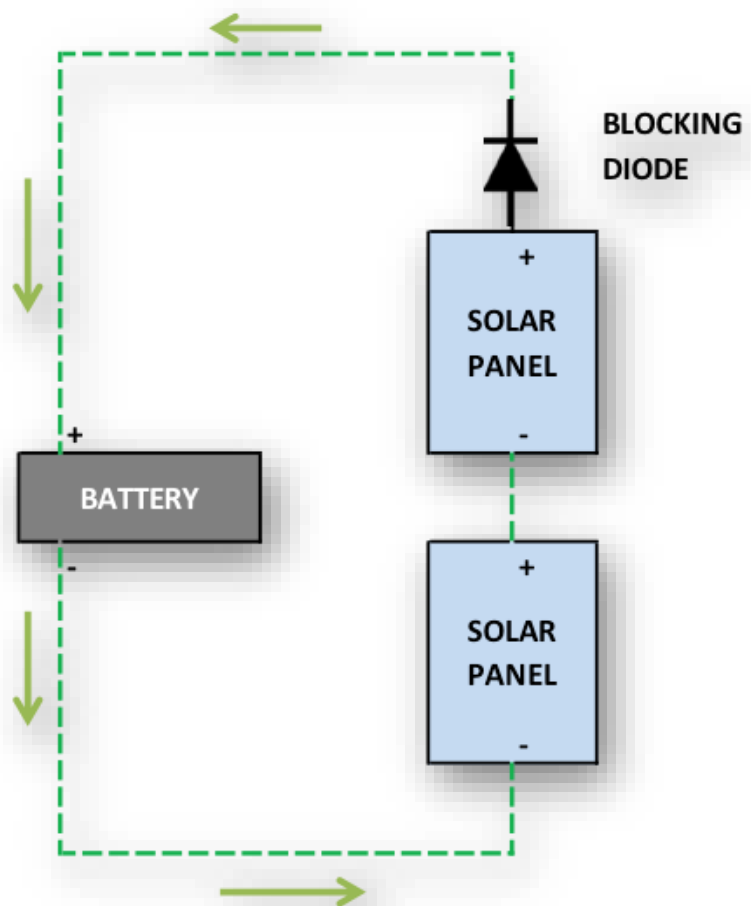
QUESTION NO. 7

What diode is needed as protection against lightning in solar installations?

- a. Bypass diode
- b. Blocking diode
- c. Shading diode
- d. Lightning diode

Diodes are electrical components that allow current to **flow in one direction** while blocking it in the opposite direction.





QUESTION NO. 7

What diode is needed as protection against lightning in solar installations?

- a. Bypass diode
- b. Blocking diode
- c. Shading diode
- d. Lightning diode

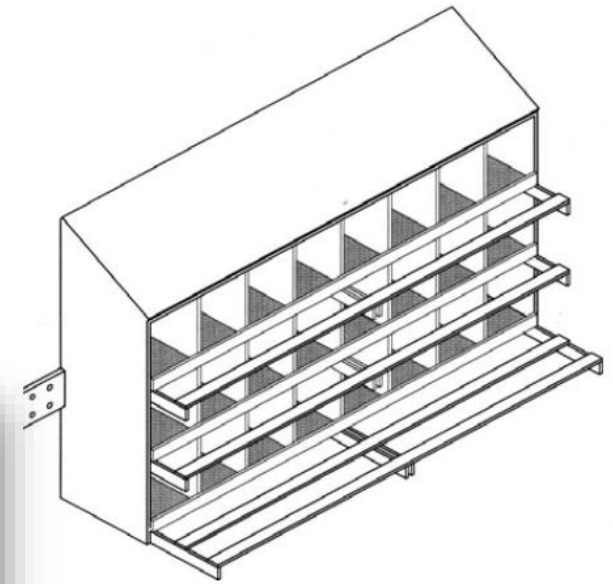
QUESTION NO. 8

In PAES 403:2001, under the individual nest subsection, there shall be one nest for each ___ to ___ layer.

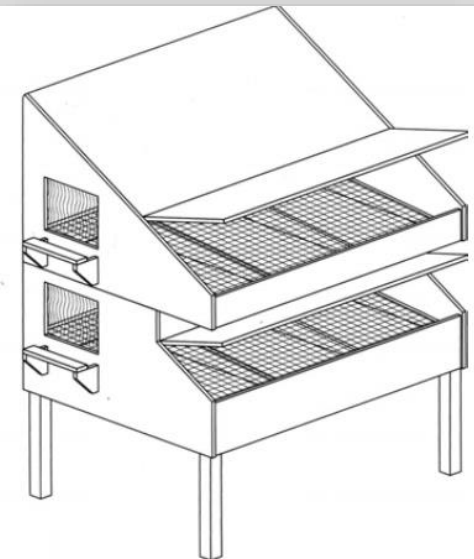
- a. 5 – 6
- b. 8 – 10
- c. 7 – 8
- d. 4 – 6

8.3.2.1 Individual nests (Figure 2a): Individual nest shall be 310 mm wide, 310 mm high and 310 mm - 360 mm long. The nest should be constructed with wire mesh at the back or with the back open for adequate air movement. The bottom of the nests shall be removable for easy cleaning. There shall be **one nest** for each **5 to 6 layer**.

8.3.2.2 Community nest (Figure 2b): These are unpartitioned boxes about 600 mm wide and 2.4 m long, with 200 mm opening at each end through which the layers enter and leave. Each shall have a sloping cover that is hinged so that it may be opened. The nest should be constructed with wire mesh at the back or with the back open for adequate air movement. The laying floor should be removable and should be made of 13 mm galvanized wire mesh. The laying floor shall be sloping 7% towards the other side, which is provided with foam rubber bumper pad. One nest box should be provided for 20 – 25 layers.



a. Individual nest



b. Communal nest



QUESTION NO. 8

In PAES 403:2001, under the individual nest subsection, there shall be one nest for each ___ to ___ layer.

- a. 5 – 6
- b. 8 – 10
- c. 7 – 8
- d. 4 – 6

QUESTION NO. 9

In CPES rating, the maximum weight rating for Environmental, Safety and Health criteria during construction.

- a. 0.40
- b. 0.15
- c. 0.07
- d. 0.30



Criteria for Evaluation

1. **During Construction** (60% for Horizontal Projects and 70% for Vertical Projects)

Criteria for Evaluation

Maximum Rating

Workmanship	0.40
Materials	0.30
Time	0.15
Facilities	0.03
Environmental, Safety and Health	0.07
Resources Deployment	0.05
	===
T O T A L	1.00

2. **Upon Completion** (40% for Horizontal Projects and 30% for Vertical Projects)

Criteria for Evaluation

Maximum Rating

Workmanship	0.50
Materials	0.20
Time	0.30
	===
T O T A L	1.00



AIM TO TOP
REGIONAL TRAINING CENTER
ILIGAN

QUESTION NO. 9

In CPES rating, the maximum weight rating for Environmental, Safety and Health criteria during construction.

- a. 0.40
- b. 0.15
- c. 0.07
- d. 0.30



AIM TO TOP

ILIGAN

QUESTION NO. 10

A complete lighting unit consisting of a lamp or lamps together with parts designed to distribute the light, to position and protect the lamps, and to connect the lamps to the power supply.

- a. lumen
- b. luminaire
- c. luminous
- d. lighting fixture

QUESTION NO. 10

A complete lighting unit consisting of a lamp or lamps together with parts designed to distribute the light, to position and protect the lamps, and to connect the lamps to the power supply.

- a. lumen
- b. luminaire
- c. luminous
- d. lighting fixture

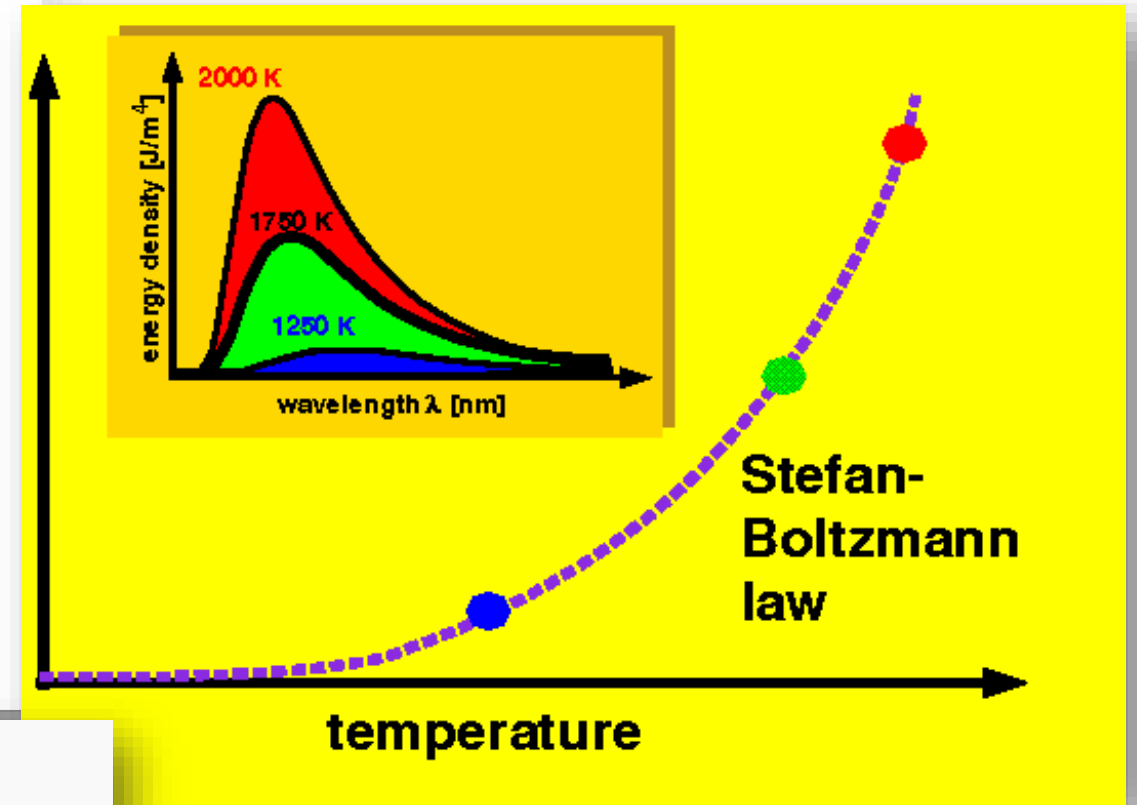
QUESTION NO. 11

The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

- a. Stefan-Boltzmann Law
- b. Newton's Law of Cooling
- c. Fourier's Law
- d. Planck's Law

Definition:

The Stefan-Boltzmann Law is a cornerstone of thermodynamics that relates the thermal radiation emitted by a black body to its temperature.



$$P = e\sigma AT^4$$

Diagram illustrating the variables in the Stefan-Boltzmann Law equation:

- Power radiated (Watts) points to P .
- emissivity (no units) points to e .
- Surface area (m^2) points to A .
- Stefan-Boltzmann constant $5.67 \times 10^{-8} W m^{-2} K^{-4}$ points to σ .
- Temperature (Kelvins) points to T .

Stefan-Boltzmann Law

- The Stefan-Boltzmann law states that a blackbody radiates electromagnetic waves with a total energy flux E directly proportional to the fourth power of the Kelvin temperature T of the object:

$$E = \sigma T^4$$



AIM TO TOP

ILIGAN

Newton's Law of Cooling

Newton's Law of Cooling states that the rate of cooling of an object is proportional to the temperature difference between the object and its surroundings, provided that this difference is not too large. (This law also applies to warming.)

If we let $T(t)$ be the temperature of the object at time t and T_s be the temperature of the surroundings, then we can formulate Newton's Law of Cooling as a differential equation:

$$\frac{dT}{dt} = k(T - T_s)$$

where k is a constant.

28

Newton's Law of Cooling

Before
Temperature at 60°C



Surrounding
Temperature 25°C

After some time
Temperature at 25°C



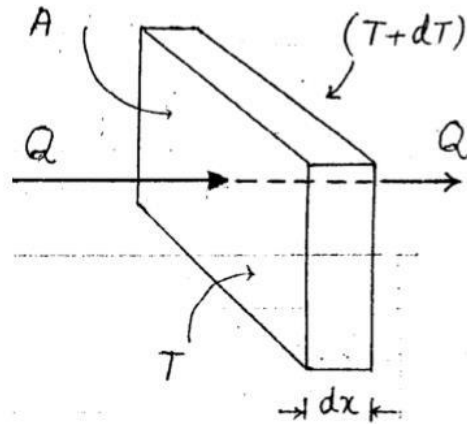
Surrounding
Temperature 25°C



AIM TO TOP
ILIGAN

Fourier's law of heat conduction

The rate of flow of heat through a single homogeneous solid is directly proportional to the area of the section at right angles to the direction of heat flow, and to the change of temperature with respect to the length of the path of the heat flow (temperature gradient).

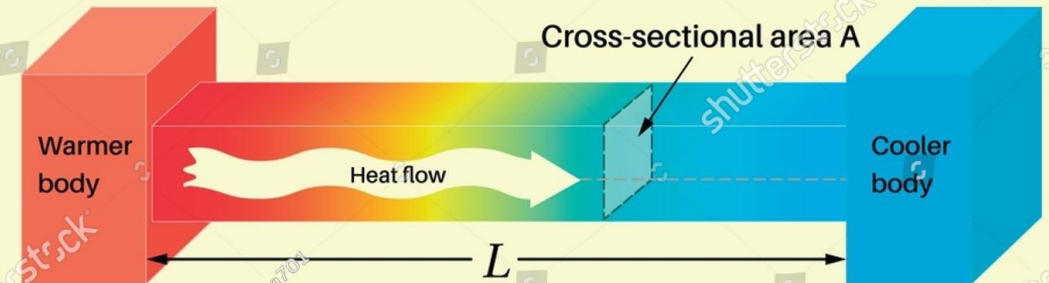


Joseph Fourier



20

Fourier's Law



$$Q = kA \left(\frac{\Delta T}{L} \right) t$$

- Q = heat transferred
- k = thermal conductivity
- A = cross-sectional area
- ΔT = temperature difference
- L = length
- t = duration of heat transfer



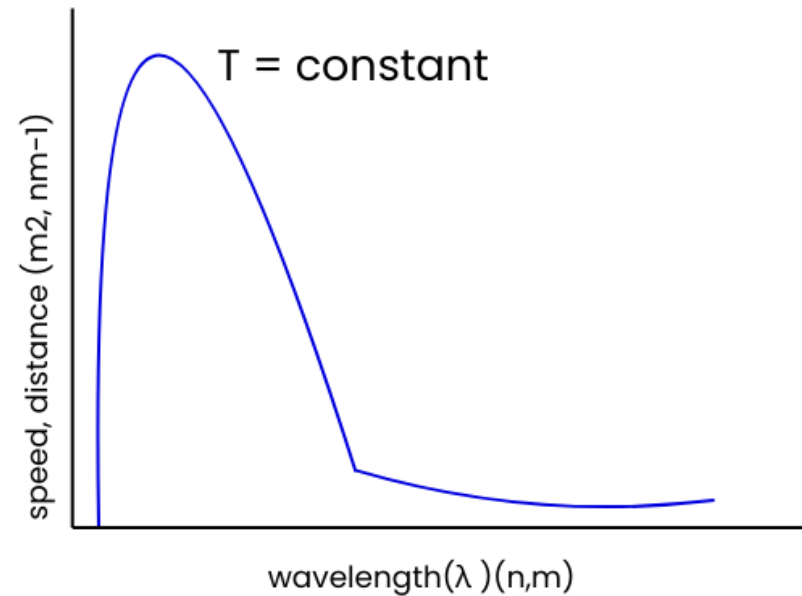
AIM TO TOP
ILIGAN

PLANCKS LAW

Definition

Planck's Law describes how the intensity and distribution of electromagnetic radiation emitted by a black body depend on its temperature.

$$B(\nu, T) = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{(e^{h\nu/kT} - 1)}$$



PLANCK EQUATION

Frequency of radiation, sometimes written as f giving expression $E = hf$.

$$E = h\nu$$

Quantum energy of a photon

$$h = \text{Planck's constant} = 6.626 \times 10^{-34} \text{ Joule}\cdot\text{sec} = 4.136 \times 10^{-15} \text{ eV}\cdot\text{s}$$

QUESTION NO. 11

The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

- a. Stefan-Boltzmann Law
- b. Newton's Law of Cooling
- c. Fourier's Law
- d. Planck's Law

QUESTION NO. 12

In CPES rating, what is the percentage for time upon completion?

- a. 0.10
- b. 0.15
- c. 0.40
- d. 0.30



AIM TO TOP

ILIGAN

Criteria for Evaluation

1. **During Construction** (60% for Horizontal Projects and 70% for Vertical Projects)

Criteria for Evaluation

Maximum Rating

Workmanship	0.40
Materials	0.30
Time	0.15
Facilities	0.03
Environmental, Safety and Health	0.07
Resources Deployment	0.05
===	
T O T A L	1.00

2. **Upon Completion** (40% for Horizontal Projects and 30% for Vertical Projects)

Criteria for Evaluation

Maximum Rating

Workmanship	0.50
Materials	0.20
Time	0.30
===	
T O T A L	1.00



AIM TO TOP
REGIONAL TRAINING CENTER
ILIGAN

QUESTION NO. 12

In CPES rating, what is the percentage for time upon completion?

- a. 0.10
- b. 0.15
- c. 0.40
- d. 0.30



AIM TO TOP

ILIGAN

QUESTION NO. 13

What is the size of the wire AWG 10?

- a. 5.0 mm^2
- b. 5.5 mm^2
- c. 6.0 mm^2
- d. 8.8 mm^2

QUESTION NO. 13

What is the size of the wire AWG 10?

- a. 5.0 mm²
- b. 5.5 mm²
- c. 6.0 mm²
- d. 8.8 mm²

QUESTION NO. 14

Amount of heat added or removed measured in change of temperature without changing state.

- a. Latent Heat
- b. Sensible Heat
- c. Specific Heat
- d. Innate Heat



AIM TO TOP

ILIGAN

LATENT HEAT VERSUS SENSIBLE HEAT

Latent heat is the amount of energy that is either absorbed or released during a phase transition at a constant temperature

Defined for a system with phase transition

Describes the changes in the internal energy of matter

Related to a system at a constant temperature

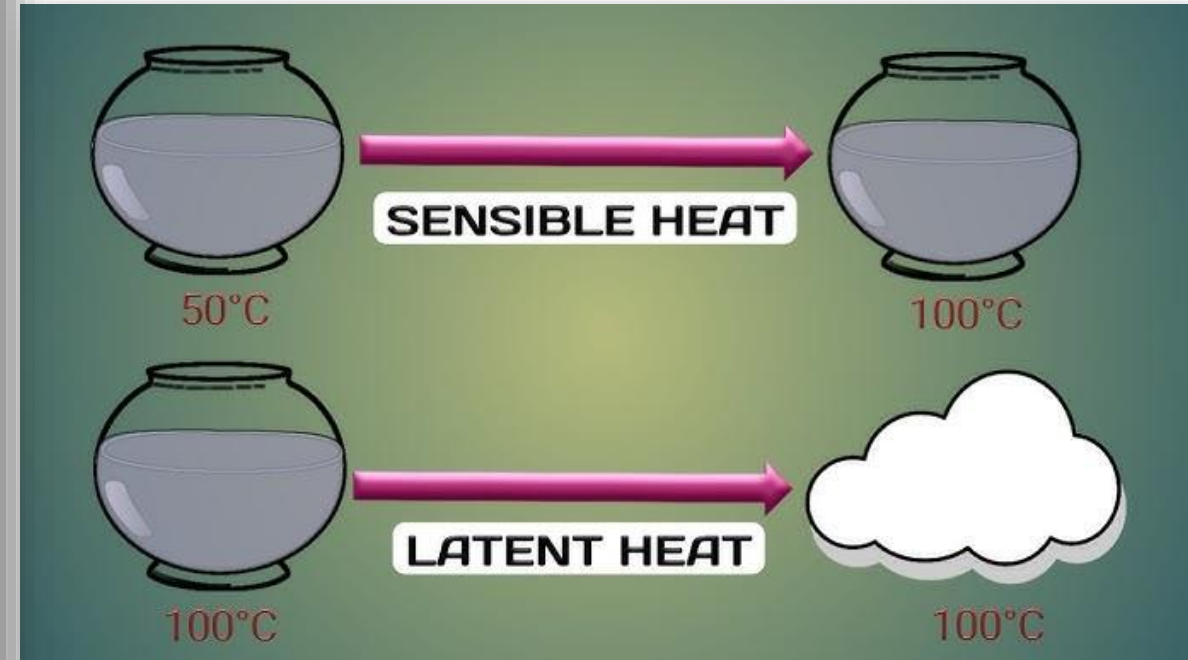
Sensible heat is the energy required to change the temperature of a substance with no phase change

Defined for a system without phase transition

Describes the energy exchange between matter and surroundings

Related to a system with changes in temperature

Visit www.pediaa.com



QUESTION NO. 14

Amount of heat added or removed measured in change of temperature without changing state.

- a. Latent Heat
- b. Sensible Heat
- c. Specific Heat
- d. Innate Heat



QUESTION NO. 15

Measures the ability of the machine to remove hulls with minimum breakage.

- a. Coefficient of wholeness
- b. Hulling Coefficient
- c. Hulling Efficiency
- d. Fineness Modulus

3.4

coefficient of hulling

measure of the ability of the machine to remove the hulls

3.5

coefficient of wholeness

measure of the ability of the machine to remove the hulls without breaking the grain

3.8

hulling efficiency

product of the coefficient of hulling and the coefficient of wholeness of grains, expressed in percent



QUESTION NO. 15

Measures the ability of the machine to remove hulls with minimum breakage.

- a. Coefficient of wholeness
- b. Hulling Coefficient
- c. Hulling Efficiency
- d. Fineness Modulus

QUESTION NO. 16

The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

- a. Carbon credit value
- b. Emission value
- c. Carbon trading value
- d. Carbon emission value

QUESTION NO. 16

The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

- a. Carbon credit value
- b. Emission value
- c. Carbon trading value
- d. Carbon emission value

QUESTION NO. 17

What is the peak amperage of a solar battery charging station equipped with 16 units of 75 watt-peak 4.16-ampere solar module? The system is divided into 4 charging channels, each channel has 4 modules connected in parallel.

- a. 32.2 A
- b. 25.4 A
- c. 16.6 A
- d. 8.3 A



QUESTION NO. 17

What is the peak amperage of a solar battery charging station equipped with 16 units of 75 watt-peak 4.16-ampere solar module? The system is divided into 4 charging channels, each channel has 4 modules connected in parallel.

- a. 32.2 A
- b. 25.4 A
- c. 16.6 A
- d. 8.3 A

QUESTION NO. 18

What is the minimum methane content of biogas in order to operate biogas-fueled generators?

- a. 50%
- b. 65%
- c. 85%
- d. 90%



AIM TO TOP

ILIGAN

QUESTION NO. 18

What is the minimum methane content of biogas in order to operate biogas-fueled generators?

- a. 50%
- b. 65%
- c. 85%
- d. 90%

QUESTION NO. 19

Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

- a. Aerobic Biogas
- b. Anaerobic Biogas
- c. Facultative Biogas
- d. Activated Biogas



AIM TO TOP

ILIGAN

Facultative Lagoon

Surface
Aeration O_2



Sunlight
Energy

Aerobic Zone
(sunlight, DO)

O_2 ← Algae

Aerobic Bacteria → CO_2

Facultative Zone
(some DO, some sunlight)

Facultative Bacteria

Anaerobic Zone
(no DO, no sunlight)

Nutrients
Gases

Anaerobic Bacteria ↑



AIM TO TOP
ILIGAN

QUESTION NO. 19

Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

- a. Aerobic Biogas
- b. Anaerobic Biogas
- c. **Facultative Biogas**
- d. Activated Biogas



QUESTION NO. 20

The wastewater, either treated or untreated, discharged from a reservoir, basin or treatment plant.

- a. Effluent
- b. Influent
- c. Turbid Water
- d. Hard Water

3.7

effluent

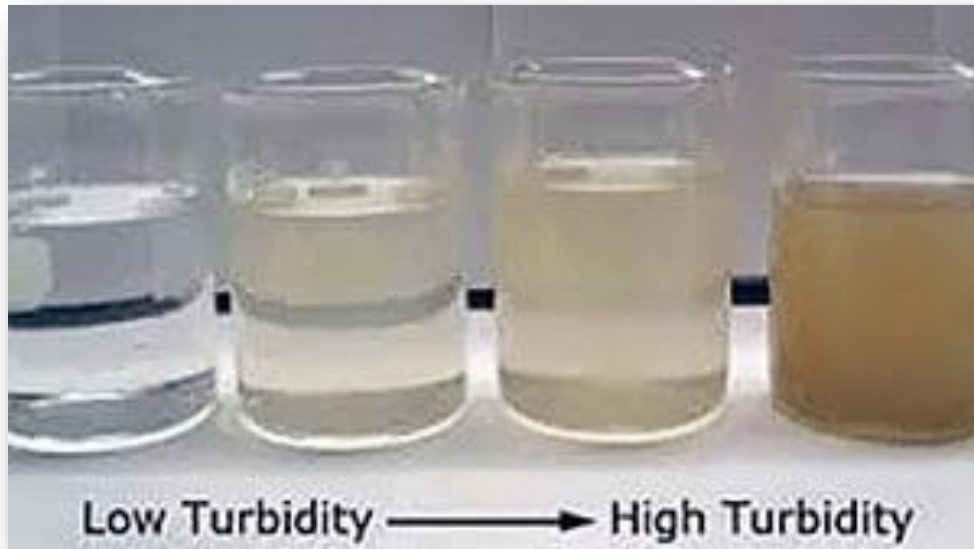
liquid waste, partially or completely treated, flowing out of a reservoir, basin, or wastewater treatment plant

3.12

influent

liquid that flows into a containing space

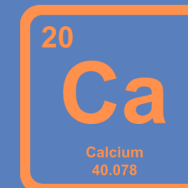




WHAT IS HARD WATER?

Hard water is water that contains an **excess of Calcium & Magnesium**.

The level of hard ions in your water depends on the area which you live in.



THE EFFECTS:

- Leaves Limescale Deposits
- Stains & Damaged Appliances
- Dried Skin & Causes Eczema
- Blocks Pipes & Boilers

QUESTION NO. 20

The wastewater, either treated or untreated, discharged from a reservoir, basin or treatment plant.

- a. **Effluent**
- b. Influent
- c. Turbid Water
- d. Hard Water

QUESTION NO. 21

In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ___ °C.

- a. 80
- b. 120
- c. 180
- d. 200



17.3.2 Equipment

The liquid bituminous material shall be sprayed by means of a pressure distributor of not less than 1000 liters capacity, mounted on pneumatic tires of such width and number that the load produced on the road surface will not exceed 1 kN (1000kgf) per cm width of tire.

The tank shall have heating device able to heat a complete charge of bituminous liquid to **180 °C**. The heating device shall be such that overheating will not occur. Consequently, the flames must not directly touch the casing of the tank containing the bituminous liquid. The liquid shall be insulated in such a way that the drop in



AIM TO TOP
REGIONAL TRAINING CENTER
ILIGAN

QUESTION NO. 21

In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ___ °C.

- a. 80
- b. 120
- c. 180
- d. 200

QUESTION NO. 22

In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ____ - ____ of floor area.

- a. 15% - 30%
- b. 30% - 50%
- c. 25% - 50%
- d. 15% - 25%

8 Ventilating and cooling greenhouses

8.1 Natural ventilating system

8.1.1 Ventilation should either be built-in in the design of the roof and wall configuration.

8.1.2 There should be vent openings on both sides and ridge of a greenhouse.

8.1.3 Large vent openings provide the most ventilation, the total vent area should be 15%-25% of the floor area.



AIM TO TOP
RISING THE EDUCATION STANDARD
ILIGAN

QUESTION NO. 22

In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ____ - ____ of floor area.

- a. 15% - 30%
- b. 30% - 50%
- c. 25% - 50%
- d. 15% - 25%

QUESTION NO. 23

In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

- a. ditches
- b. culvert
- c. channel
- d. conduit



23.2 Roadside Ditch

The data shall be designed on channel dimensions such as the width, depth, slope (Front slope and back slope), velocity, particle size distribution, sediment load, and discharge necessary to maintain stable channel.

This shall be constructed to drain water from the subgrade.

This shall be constructed to collect surface water either from the roadway surface or adjacent.



QUESTION NO. 23

In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

- a. ditches
- b. culvert
- c. channel
- d. conduit

QUESTION NO. 24

In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be ____ m/animal.

- a. 0.5
- b. 1.0
- c. 0.75
- d. 1.25

5 Space requirement

The minimum space requirement shall be $4 \text{ m}^2/\text{animal}$.

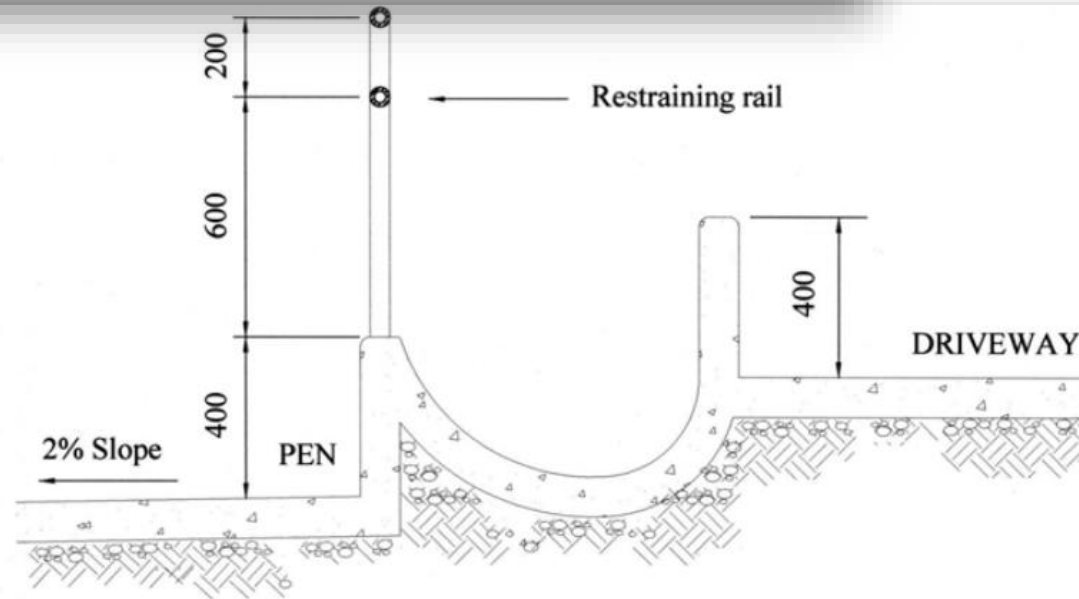


Figure 2 – Sample of a feeding trough

7.2.1.5 The minimum space requirement for feeding shall be 0.750 mm/animal.



QUESTION NO. 24

In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be ____ m/animal.

- a. 0.5
- b. 1.0
- c. 0.75
- d. 1.25

QUESTION NO. 25

The maximum number of outlets for a 15-A circuit.

- a. 8-10 outlets
- b. 10-12 outlets
- c. 14-15 outlets
- d. 6-8 outlets

QUESTION NO. 25

The maximum number of outlets for a 15-A circuit.

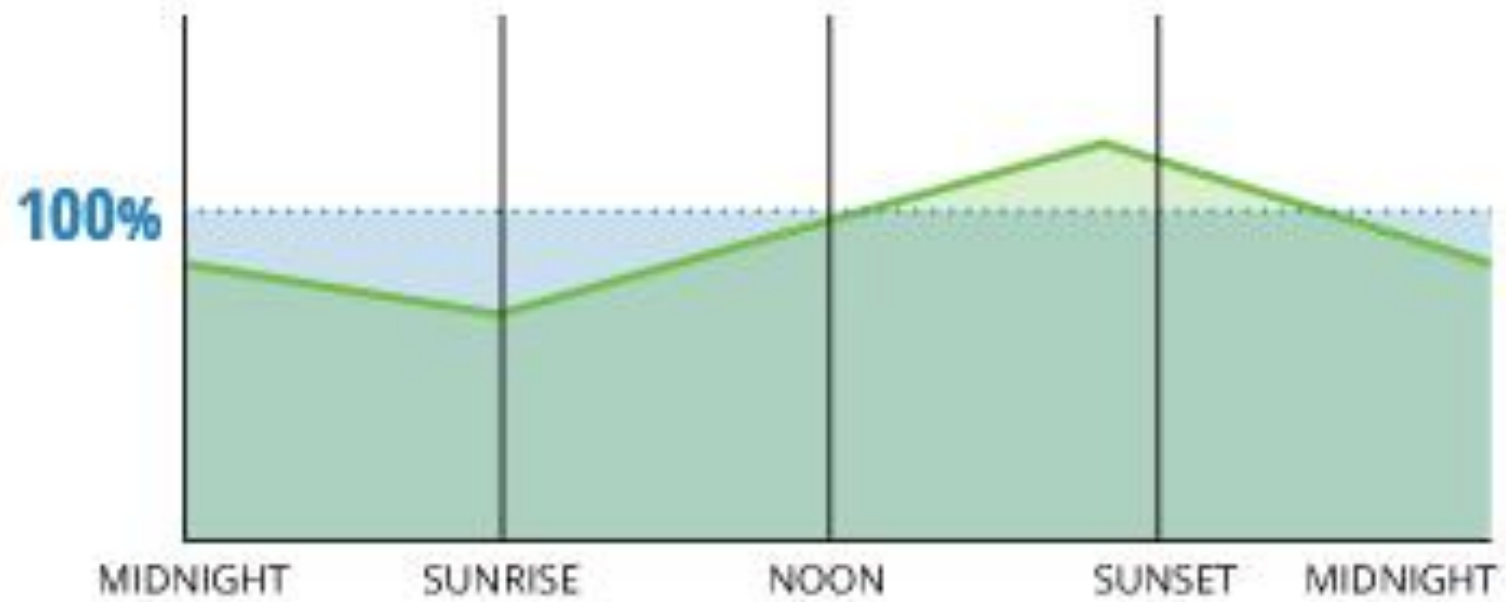
- a. 8-10 outlets
- b. 10-12 outlets
- c. 14-15 outlets
- d. 6-8 outlets

QUESTION NO. 26

Dissolved oxygen concentrations will be the lowest within a waterbody on ____.

- a. Afternoon
- b. Sunrise
- c. Sunset
- d. Noon





QUESTION NO. 26

Dissolved oxygen concentrations will be the lowest within a waterbody on ____.

- a. Afternoon
- b. Sunrise
- c. Sunset
- d. Noon



QUESTION NO. 27

In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of __°C - __°C until they are 3 days old.

- a. 32-35
- b. 29-35
- c. 30-35
- d. 27-35



AIM TO TOP

ILIGAN

8.5.1 Heaters

8.5.1.1 Heaters shall be installed in creep areas to provide newborn pigs its required temperature of **27°C - 35°C** until they are 3 days old.

8.5.1.2 Heat lamps shall be placed 762 mm above the floor or 152 mm above the sow.



QUESTION NO. 27

In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of __°C - __°C until they are 3 days old.

- a. 32-35
- b. 29-35
- c. 30-35
- d. 27-35

QUESTION NO. 28

What is the mechanical power of a windmill with a diameter of 10 meters and velocity of 5 mps?

- a. 55.92 watts
- b. 981.75 watts
- c. 3,926.99 watts
- d. 9,327.56 watts

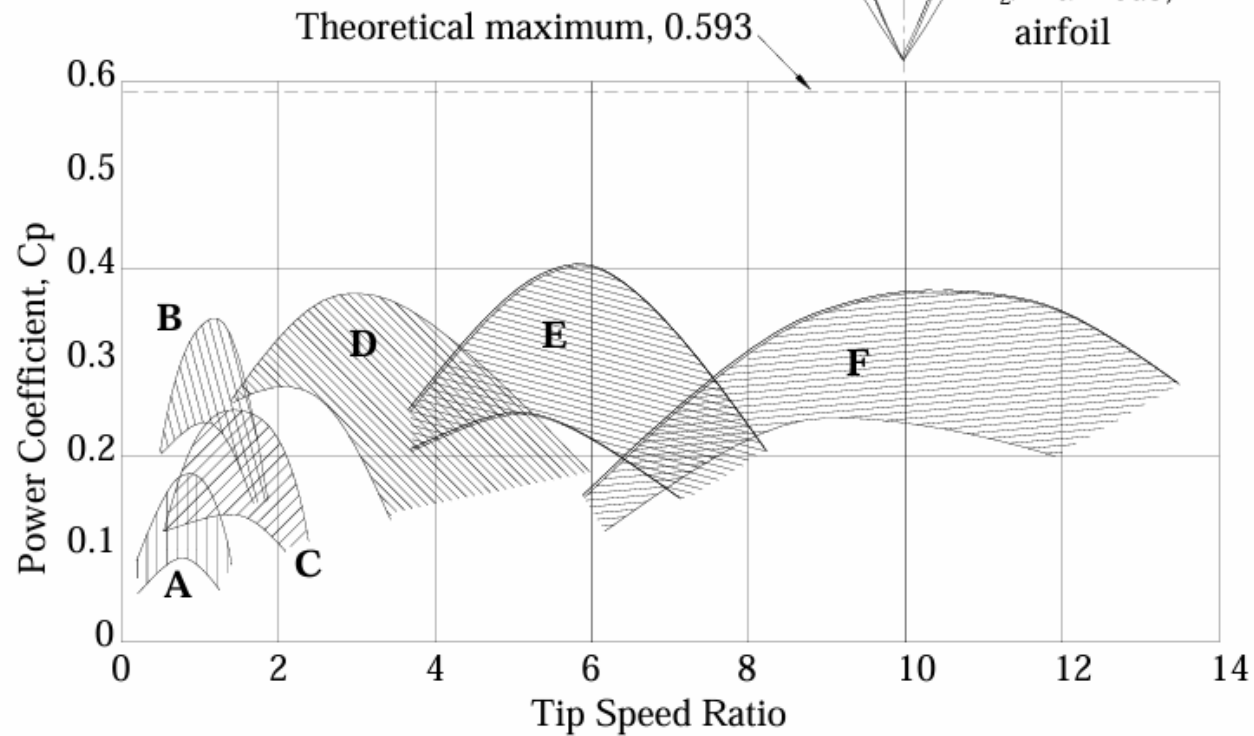
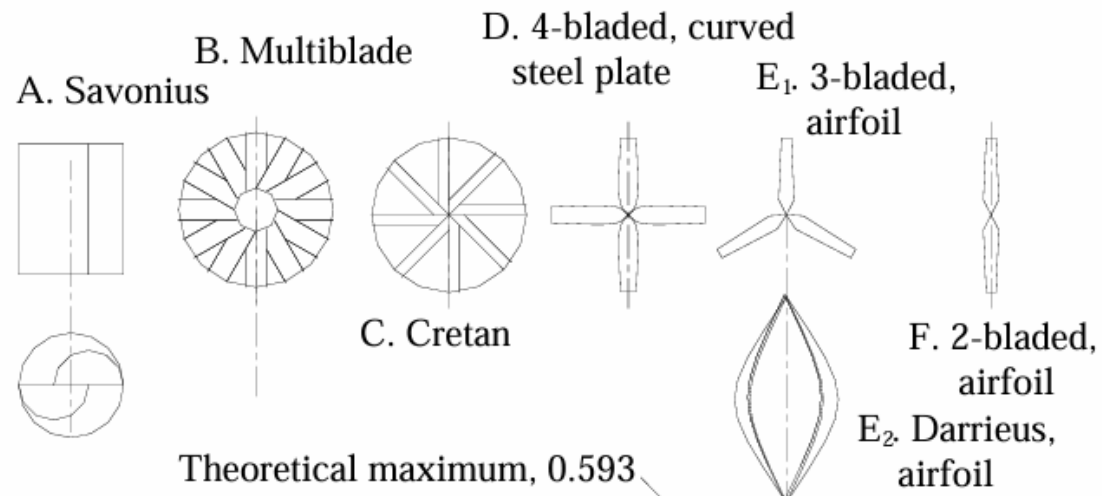
The theoretical power from wind can be given by the formula:

$$\text{Theoretical power} = \left(\rho A V^3 \right) / 2 \quad \text{where: } \rho = \text{air density, } 1.25 \text{ kg/m}^3$$
$$A = \text{rotor area, m}^2$$
$$V = \text{wind speed, m/s}$$

However, using the Froude momentum theory, the maximum energy or power recoverable from wind is only $16/27$ or 0.593^5 times the kinetic energy. Of this available energy, the amount that is actually utilized depends upon the aerodynamic efficiency of the blades, the friction losses in gearing, power transmission, wind velocity gradient due to ground effects, vibration dampening and other factors. For estimation purposes, the following formulas can be used:

$$\text{Available power} = 0.245 A V^3 \quad (\text{for mechanical energy conversion, e. g. pumping})$$

$$\text{Available power} = 0.10 A V^3 \quad (\text{for electric energy conversion})$$



The power coefficient, C_p , of different types of windrotors as function of tip speed ratio.

QUESTION NO. 28

What is the mechanical power of a windmill with a diameter of 10 meters and velocity of 5 mps?

- a. 55.92 watts
- b. 981.75 watts
- c. 3,926.99 watts
- d. 9,327.56 watts

QUESTION NO. 29

A Carnot engine is working between two reservoirs with temperatures of 1000°C and 500°C , what is the thermal efficiency of the system?

- a. 10%
- b. 30%
- c. 50%
- d. 100%



Carnot cycle

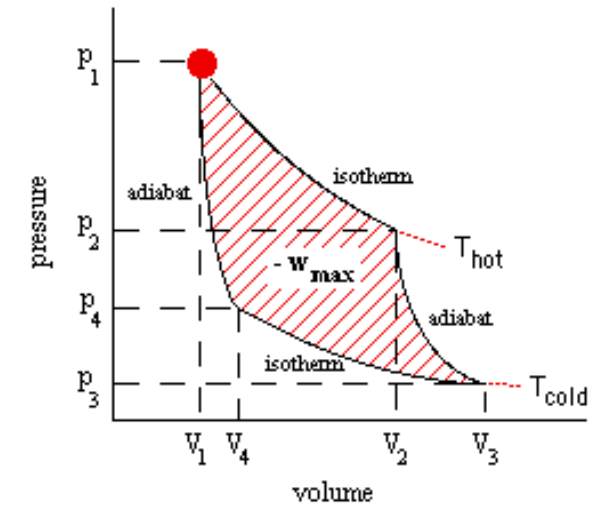
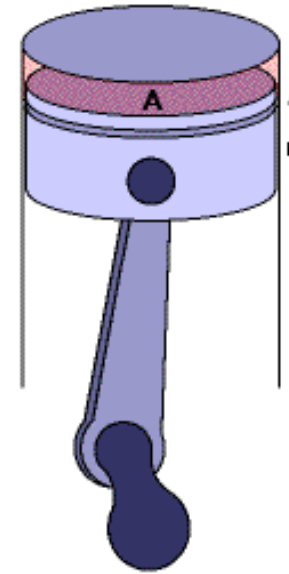
Efficiency
maximum

Temperature
of the cold
reservoir

$$e_{max} = 1 - \frac{T_c}{T_H}$$

Temperature
of the hot
reservoir

In words: The theoretical **maximum efficiency** is equal to 1 minus the **temperature of the cold reservoir** divided by the **temperature of the hot reservoir**.



QUESTION NO. 29

A Carnot engine is working between two reservoirs with temperatures of 1000°C and 500°C , what is the thermal efficiency of the system?

- a. 10%
- b. 30%
- c. 50%
- d. 100%



AIM TO TOP

ILIGAN

QUESTION NO. 30

Environmental-friendly refrigerant

- a. CFC
- b. HFC
- c. HCFC
- d. FC



AIM TO TOP

ILIGAN

BY COMPARISON

Newly calculated data could help people make climate-friendly technology choices

COMPOUND	FORMULA	RADIATIVE EFFICIENCY, W/m ² -ppb	ATMOSPHERIC LIFETIME, YEARS	GLOBAL-WARMING POTENTIAL ^a
CFC-11	CCl ₃ F	0.26	45	4,660
CFC-12	CCl ₂ F ₂	0.32	100	10,200
CFC-113	CCl ₂ FCClF ₂	0.30	85	5,820
HCFC-22	CHClF ₂	0.21	11.9	1,760
HFC-134a	CH ₂ FCF ₃	0.16	13.4	1,300
HFO-1234yf	CF ₃ CF=CH ₂	0.02	0.03	< 1

^a 100-year values, relative to CO₂, which has a value of 1.



QUESTION NO. 30

Environmental-friendly refrigerant

- a. CFC
- b. HFC
- c. HCFC
- d. FC

QUESTION NO. 31

It is the programming language of Arduino microcontrollers.

- a. R
- b. Python
- c. C/C++
- d. Java

QUESTION NO. 31

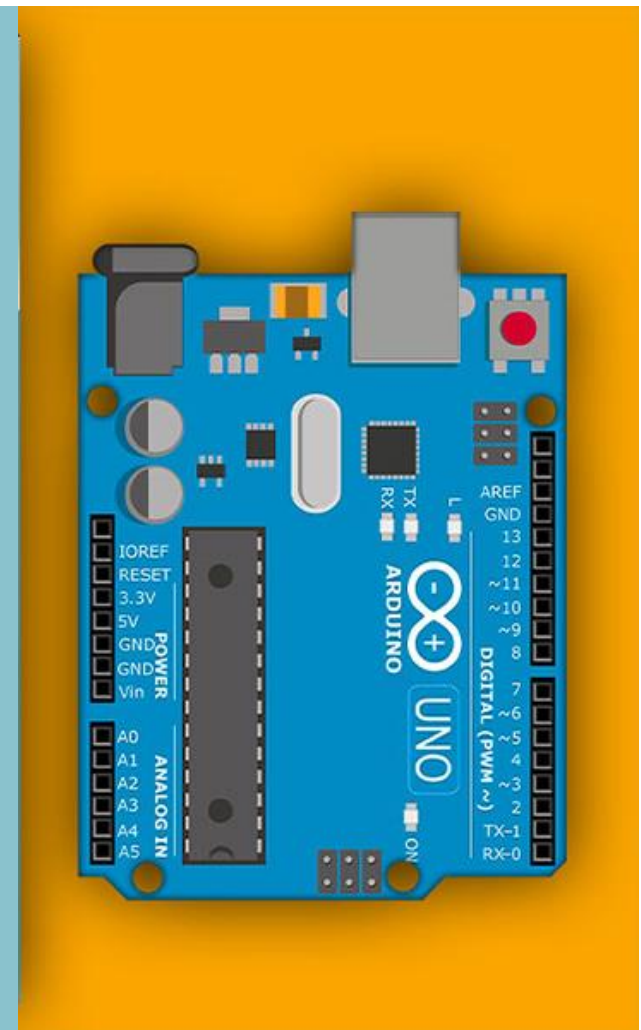
It is the programming language of Arduino microcontrollers.

- a. R
- b. Python
- c. C/C++
- d. Java

QUESTION NO. 32

In Arduino programming, the meaning of IDE is:

- a. Integrated Development Environment
- b. Internal Development Environment
- c. International Development Environment
- d. Interconnected Development Environment



QUESTION NO. 32

In Arduino programming, the meaning of IDE is:

- a. Integrated Development Environment
- b. Internal Development Environment
- c. International Development Environment
- d. Interconnected Development Environment

QUESTION NO. 33

The Feed-in-Tariff (FIT) of solar energy in the BiGSHOW is:

- a. 6.63 PHP/kW-hr
- b. 9.68 PHP/kW-hr
- c. 5.90 PHP/kW-hr
- d. 8.53 PHP/kW-hr

RE Technology	FIT Rate (Php/kWh)	Degression Rate	Installation Target (MW)
Wind	8.53	0.5% after 2 years from effectivity of FIT	200
Biomass	6.63	0.5% after 2 years from effectivity of FIT	250
Solar	9.68	6% after 1 year from effectivity of the FIT	50
Run-of-River Hydropower	5.90	0.5% after 2 years from effectivity of FIT	250



QUESTION NO. 33

The Feed-in-Tariff (FIT) of solar energy in the BiGSHOW is:

- a. 6.63 PHP/kW-hr
- b. 9.68 PHP/kW-hr
- c. 5.90 PHP/kW-hr
- d. 8.53 PHP/kW-hr



AIM TO TOP

ILIGAN

QUESTION NO. 34

It is the gas produced from agricultural waste which is odorless, tasteless, colorless and flammable

- a. CH_4
- b. CO
- c. O_2
- d. CO_2

QUESTION NO. 34

It is the gas produced from agricultural waste which is odorless, tasteless, colorless and flammable

- a. CH_4
- b. CO
- c. O_2
- d. CO_2

QUESTION NO. 35

All are products of pyrolysis except:

- a. Vinegar
- b. Tar
- c. Oil
- d. Syngas

QUESTION NO. 35

All are products of pyrolysis except:

- a. **Vinegar**
- b. Tar
- c. Oil
- d. Syngas

QUESTION NO. 36

What is the electrical energy consumption of a 15-kW vermicast production shredder plugged in a 220-V source if it requires 2 hrs to shred 1.5 tonnes of farm waste?

- a. 0.20 kW-h/kg
- b. 0.02 kW-h/kg
- c. 0.002 kW-h/kg
- d. 0.022 kW-h/kg

QUESTION NO. 36

What is the electrical energy consumption of a 15-kW vermicast production shredder plugged in a 220-V source if it requires 2 hrs to shred 1.5 tonnes of farm waste?

- a. 0.20 kW-h/kg
- b. **0.02 kW-h/kg**
- c. 0.002 kW-h/kg
- d. 0.022 kW-h/kg

QUESTION NO. 37

It is a type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing

- a. friction type
- b. abrasive type
- c. rubber roll type
- d. cone “cono” type

3.21.1

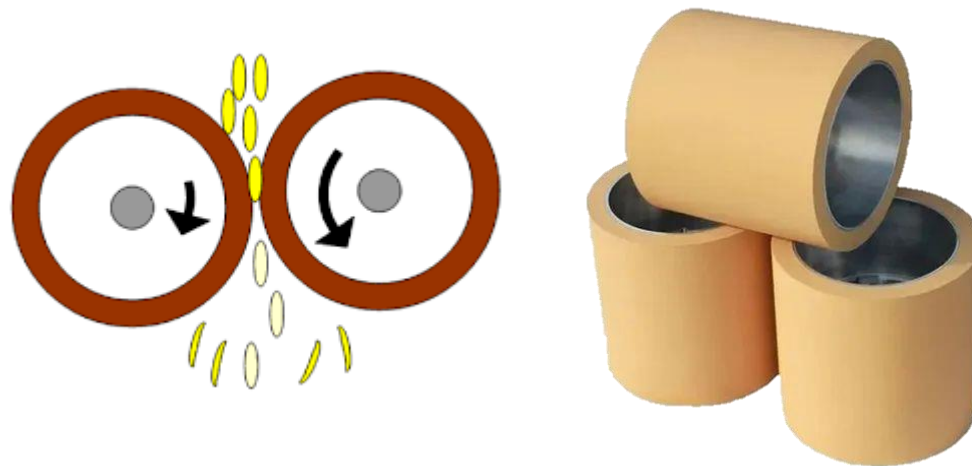
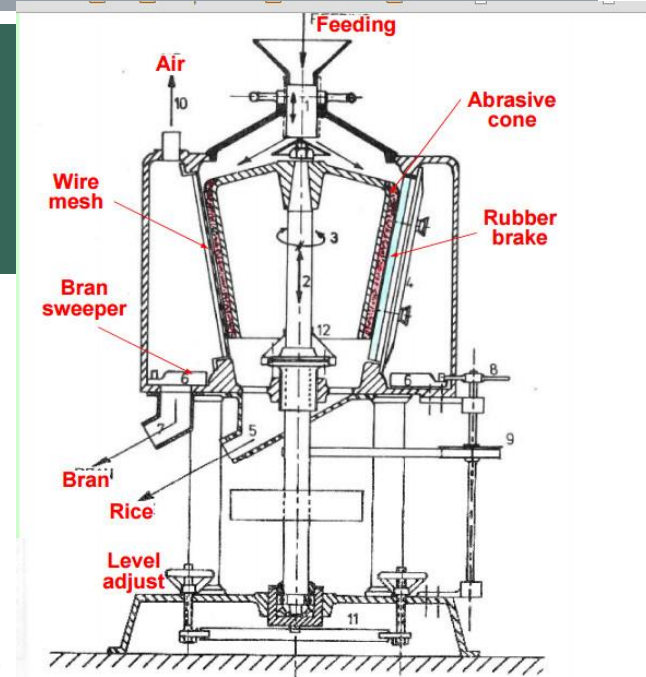
cone “cono” type

type of rice mill having an under-runner stone disc huller and vertical cone whitener

3.21.2

rubber roll type

type of rice mill using rubber roll huller and utilizes friction and/or combination of other types of whitener



3.24.1

abrasive type

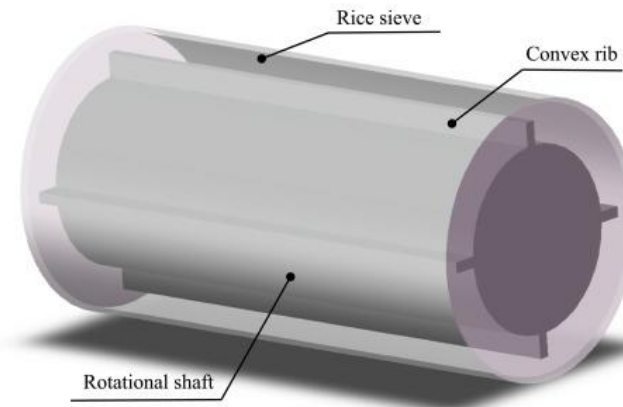
type of whitening machine consisting of a cylinder or cone coated with abrasive material such as emery stone or any similar materials enclosed in a perforated steel housing



3.24.2

friction type

type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing



QUESTION NO. 37

It is a type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing

- a. **friction type**
- b. abrasive type
- c. rubber roll type
- d. cone “cono” type

QUESTION NO. 38

What is the input current to a 5 hp single phase electric motor use to drive a hammer mill at 2200 rpm. The input voltage is 220 volt. Assume a power factor of 0.98

- a. 14.5 amp
- b. 17.3 amp
- c. 20.4 amp
- d. None of the above

QUESTION NO. 38

What is the input current to a 5 hp single phase electric motor use to drive a hammer mill at 2200 rpm. The input voltage is 220 volt. Assume a power factor of 0.98

- a. 14.5 amp
- b. 17.3 amp
- c. 20.4 amp
- d. None of the above

QUESTION NO. 39

When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing:

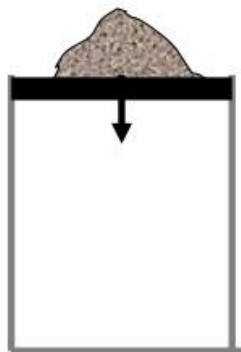
- a. isobaric process
- b. quasi-steady process
- c. isometric process
- d. isentropic process



AIM TO TOP

ILIGAN

Quasi-Static Processes



← See figure for an Example

- A gas is confined to a container of volume V . At the top is a moveable piston. Sand is placed on the top of that to weigh it down.
- Several different types of Quasi-Static Processes can be carried out on this system. These processes & the terminology used to describe them are:

Isochoric Process: ? $V = \text{constant}$

Isobaric Process: ? $P = \text{constant}$

Isothermal Process: ? $T = \text{constant}$

Adiabatic Processes: ? $Q = 0$



QUESTION NO. 39

When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing:

- a. isobaric process
- b. **quasi-steady process**
- c. isometric process
- d. isentropic process

QUESTION NO. 40

A rice thresher that is fed with panicles at one end and discharges threshed product at the other end is called ____.

- a. radial type thresher
- b. hold-on type thresher
- c. axial-flow thresher
- d. cross-flow thresher

hold-on thresher

type of thresher wherein the panicles of the cut plants are fed into the threshing chamber while the stalks are mechanically or manually held during the threshing operation

axial flow thresher

throw-in type of thresher which allows the cut plants to move in a helical manner around the threshing cylinder with a net effect of moving the material axially between the feeding and discharge outlets



AIM TO TOP
ILIGAN

QUESTION NO. 40

A rice thresher that is fed with panicles at one end and discharges threshed product at the other end is called ____.

- a. radial type thresher
- b. hold-on type thresher
- c. axial-flow thresher
- d. cross-flow thresher



QUESTION NO. 41

A wooden pole ($SG=0.65$), 9 cm by 9 cm by 5 m long, hangs vertically from a string in such a way that 3 m is submerged in water and 2 m is above the surface. What is the tension in the string?

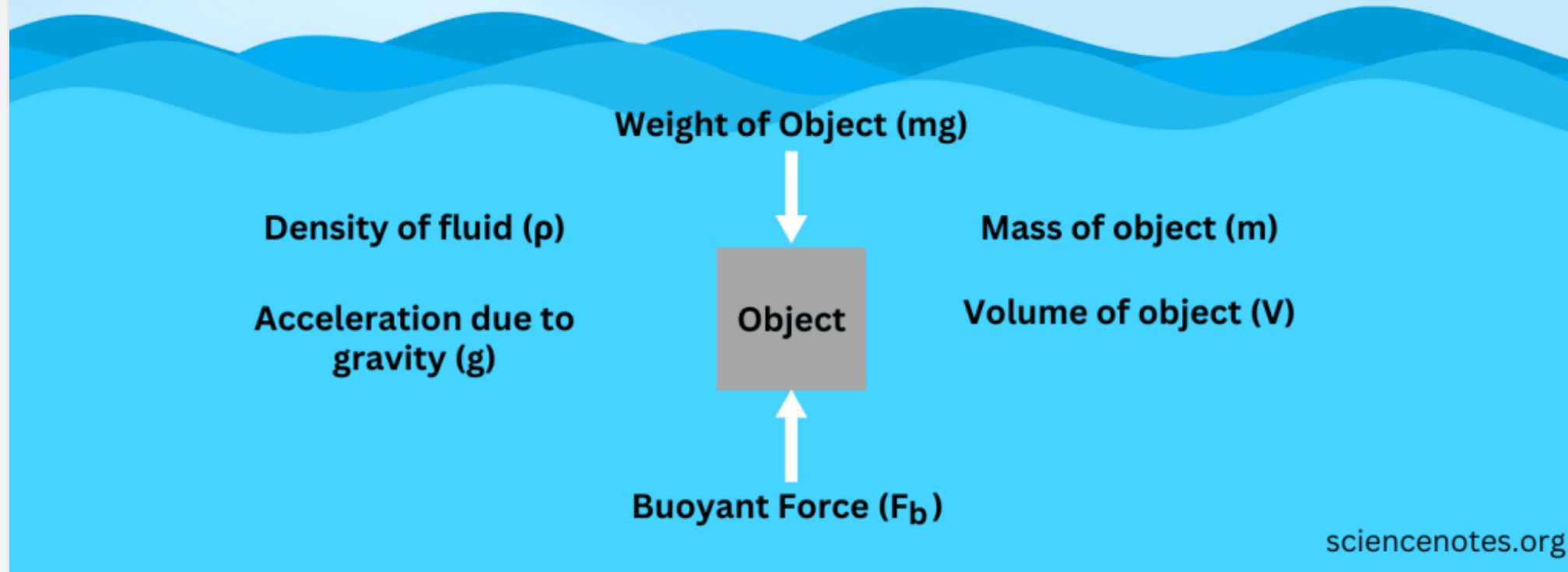
- a. 19.87 N
- b. 17.89 N
- c. 18.96 N
- d. 16.97 N

Buoyancy

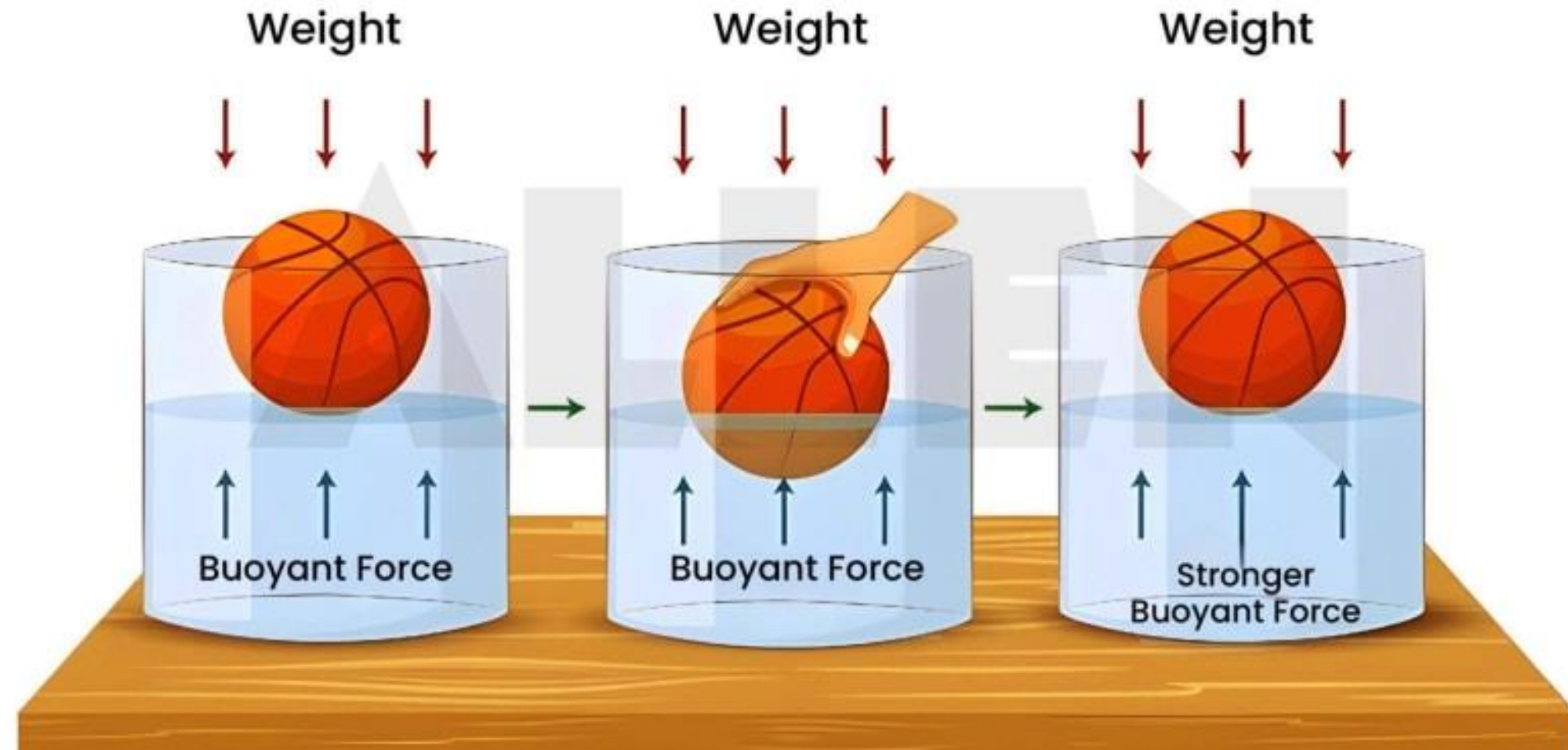
Buoyancy is the upward force exerted by a fluid that opposes the weight of an object in the fluid.

$$F_b = \rho Vg$$

- ρ is the density of the fluid,
- V is the volume of the fluid displaced by the object
- g is the acceleration due to gravity



sciencenotes.org



QUESTION NO. 41

A wooden pole ($SG=0.65$), 9 cm by 9 cm by 5 m long, hangs vertically from a string in such a way that 3 m is submerged in water and 2 m is above the surface. What is the tension in the string?

- a. 19.87 N
- b. 17.89 N
- c. 18.96 N
- d. 16.97 N

QUESTION NO. 42

How much energy is needed for a 5-volt source to move 4 coulombs of charge?

- a. 20 joules
- b. 1.25 joules
- c. 0.8 joules
- d. 80 joules

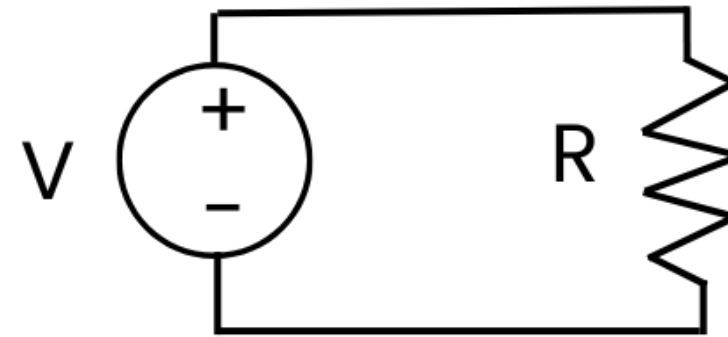


UNITS OF VOLTAGE

Definition:

Voltage, in physics, refers to the electrical potential difference between two points in a circuit or electric field. It represents the amount of energy transferred per unit charge as a charge moves between those points.

$$V = W/Q$$



Ex Examples.com

QUESTION NO. 42

How much energy is needed for a 5-volt source to move 4 coulombs of charge?

- a. 20 joules
- b. 1.25 joules
- c. 0.8 joules
- d. 80 joules

QUESTION NO. 43

It is a hypothetical activity which takes no resource or time to complete. It is represented by broken arrowed line & is used for either distinguishing activities having common starting & finishing events or to identify & maintain proper precedence relationship between activities that are not connected by events

- a. Free Float
- b. Slack Activity
- c. Dummy activity
- d. Zero Slack

QUESTION NO. 43

It is a hypothetical activity which takes no resource or time to complete. It is represented by broken arrowed line & is used for either distinguishing activities having common starting & finishing events or to identify & maintain proper precedence relationship between activities that are not connected by events

- a. Free Float
- b. Slack Activity
- c. **Dummy activity**
- d. Zero Slack

QUESTION NO. 44

In lateral and sub-lateral canal where the available roadway width is 3 meters, the carriageway width shall be 2 meters and the shoulder width shall be 0.5 meters. If the available roadway width is 4 meters, the shoulder shall expand to 1 meter.

- a. road bed lateral canal
- b. roadway on main irrigation
- c. roadway on lateral & sub lateral irrigation canals
- d. Turn-out section canal

QUESTION NO. 44

In lateral and sub-lateral canal where the available roadway width is 3 meters, the carriageway width shall be 2 meters and the shoulder width shall be 0.5 meters. If the available roadway width is 4 meters, the shoulder shall expand to 1 meter.

- a. road bed lateral canal
- b. roadway on main irrigation
- c. roadway on lateral & sub lateral irrigation canals
- d. Turn-out section canal

QUESTION NO. 45

When you want to use objects from a drawing file in another application, you can cut these objects to the Clipboard and then paste them into another application. CUTCLIP copies selected objects to the Clipboard, removing them from the drawing using the shortcut:

- a. Ctrl Y
- b. Ctrl Z
- c. Ctrl X
- d. Ctrl V

QUESTION NO. 45

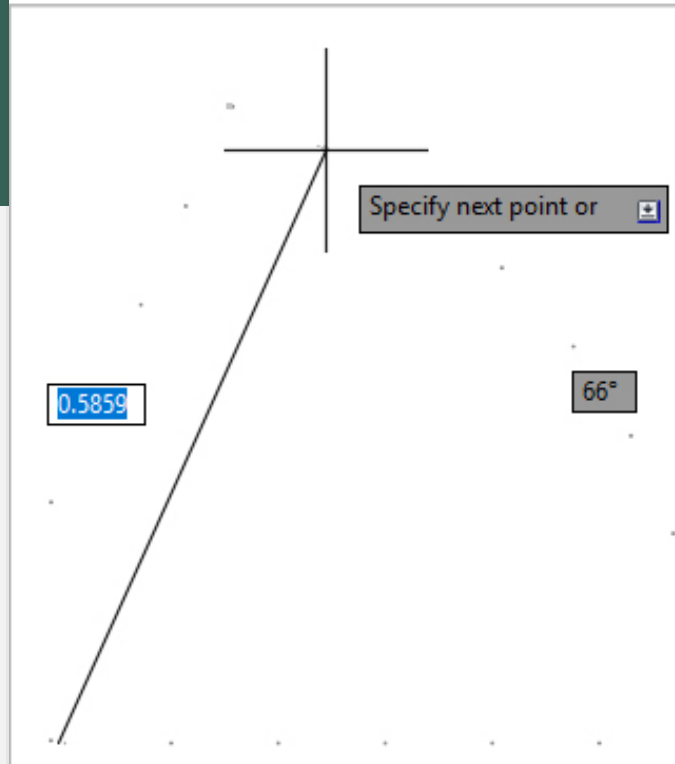
When you want to use objects from a drawing file in another application, you can cut these objects to the Clipboard and then paste them into another application. CUTCLIP copies selected objects to the Clipboard, removing them from the drawing using the shortcut:

- a. Ctrl Y
- b. Ctrl Z
- c. Ctrl X
- d. Ctrl V

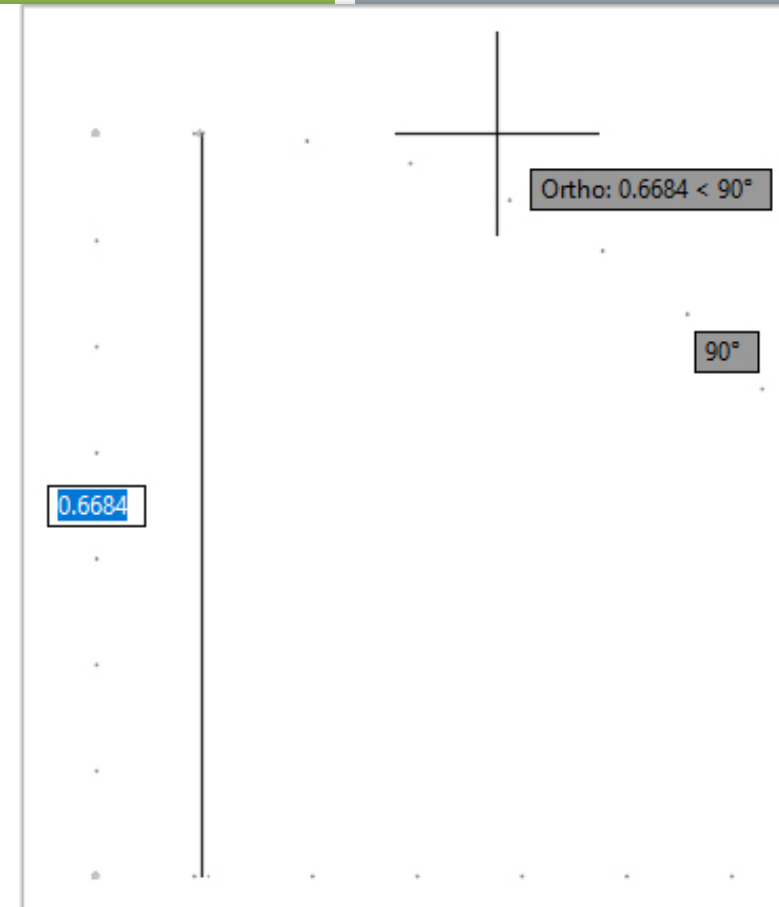
QUESTION NO. 46

In AutoCAD, it creates copies of objects arranged in a pattern. You can create copies of objects in a regularly spaced rectangular, polar, or path

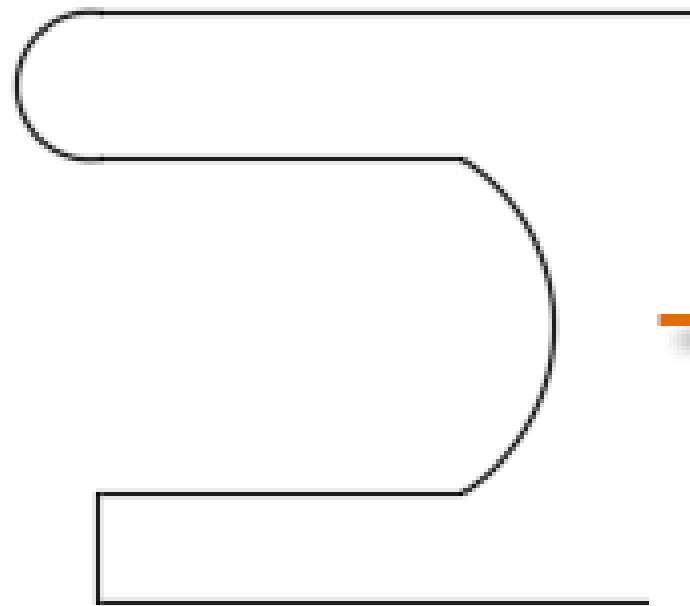
- a. Ortho
- b. Offset
- c. Array
- d. Fillet



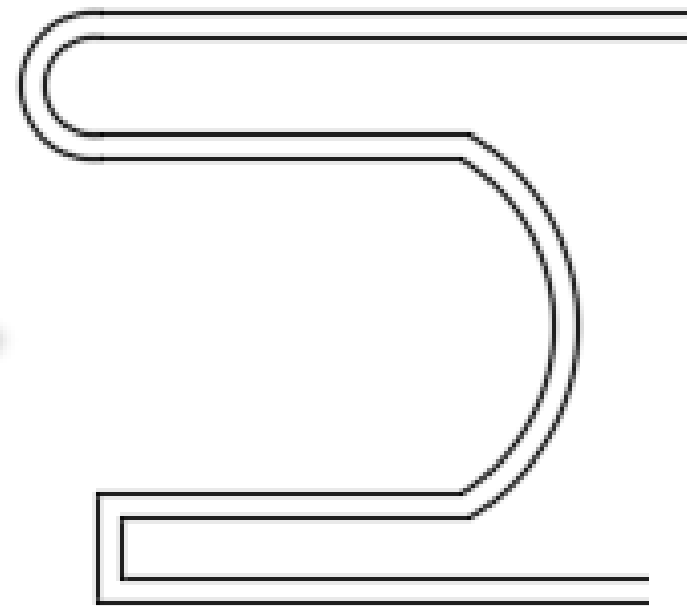
When not in Ortho mode, the line tool is directed by the cursor.



When in Ortho mode, the line tool follows the cursor on a vertical or horizontal axis. This can prevent you from accidentally changing the angle by moving the cursor because the line is constrained to vertical and horizontal angles.



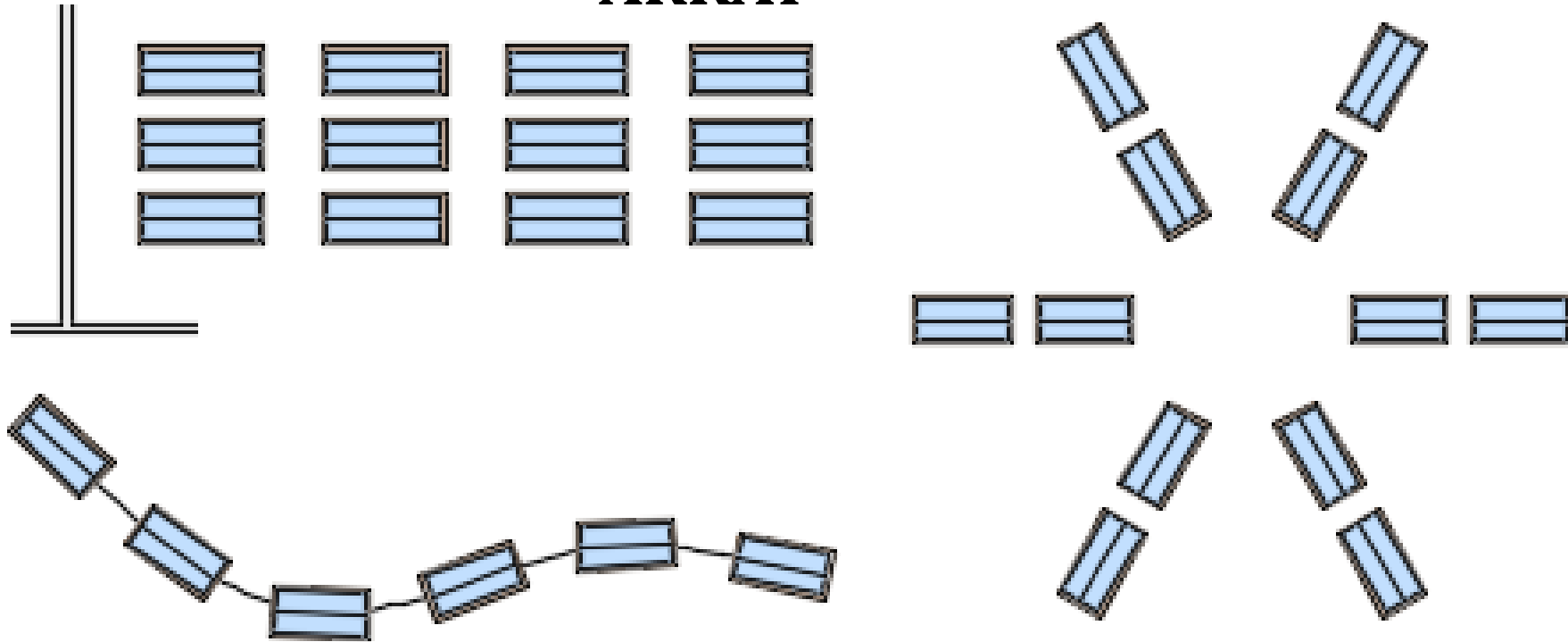
polyline



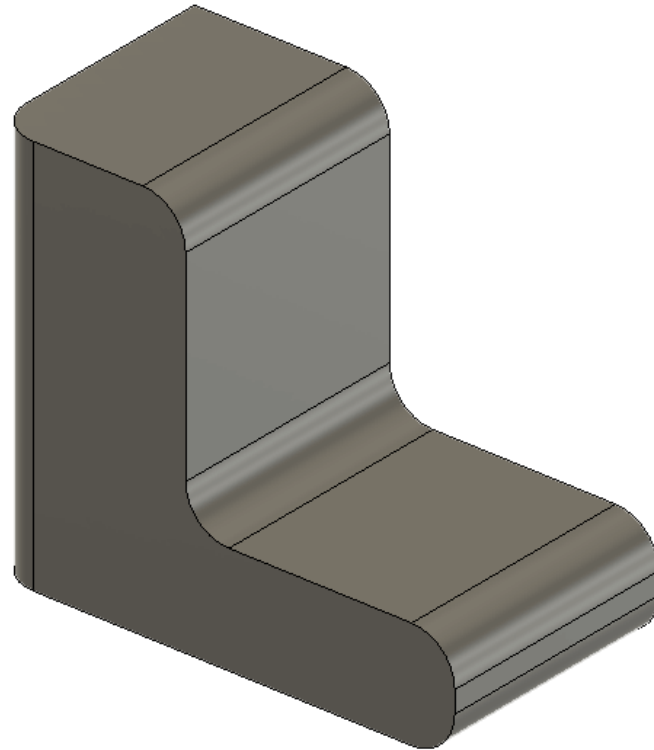
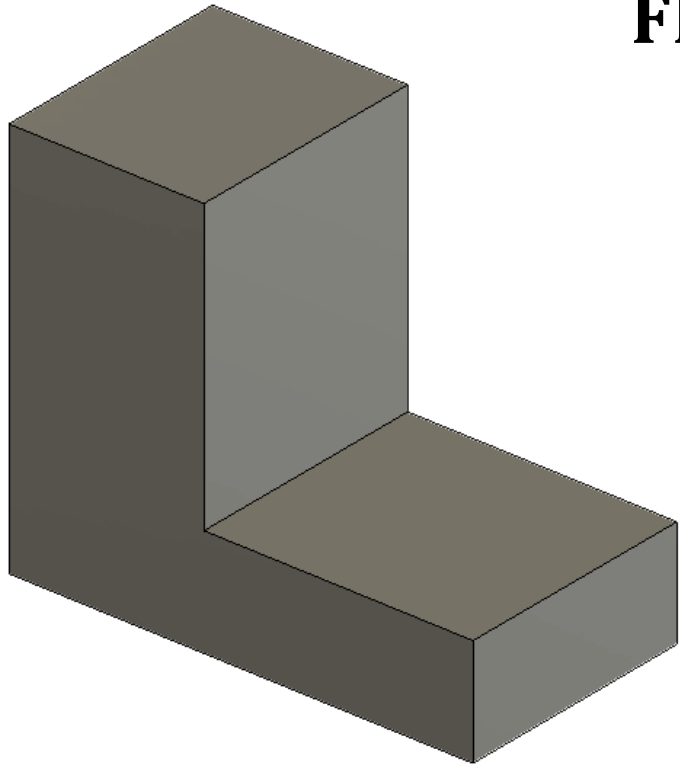
polyline with offset



ARRAY



FILLET



QUESTION NO. 46

In AutoCAD, it creates copies of objects arranged in a pattern. You can create copies of objects in a regularly spaced rectangular, polar, or path

- a. Ortho
- b. Offset
- c. Array
- d. Fillet

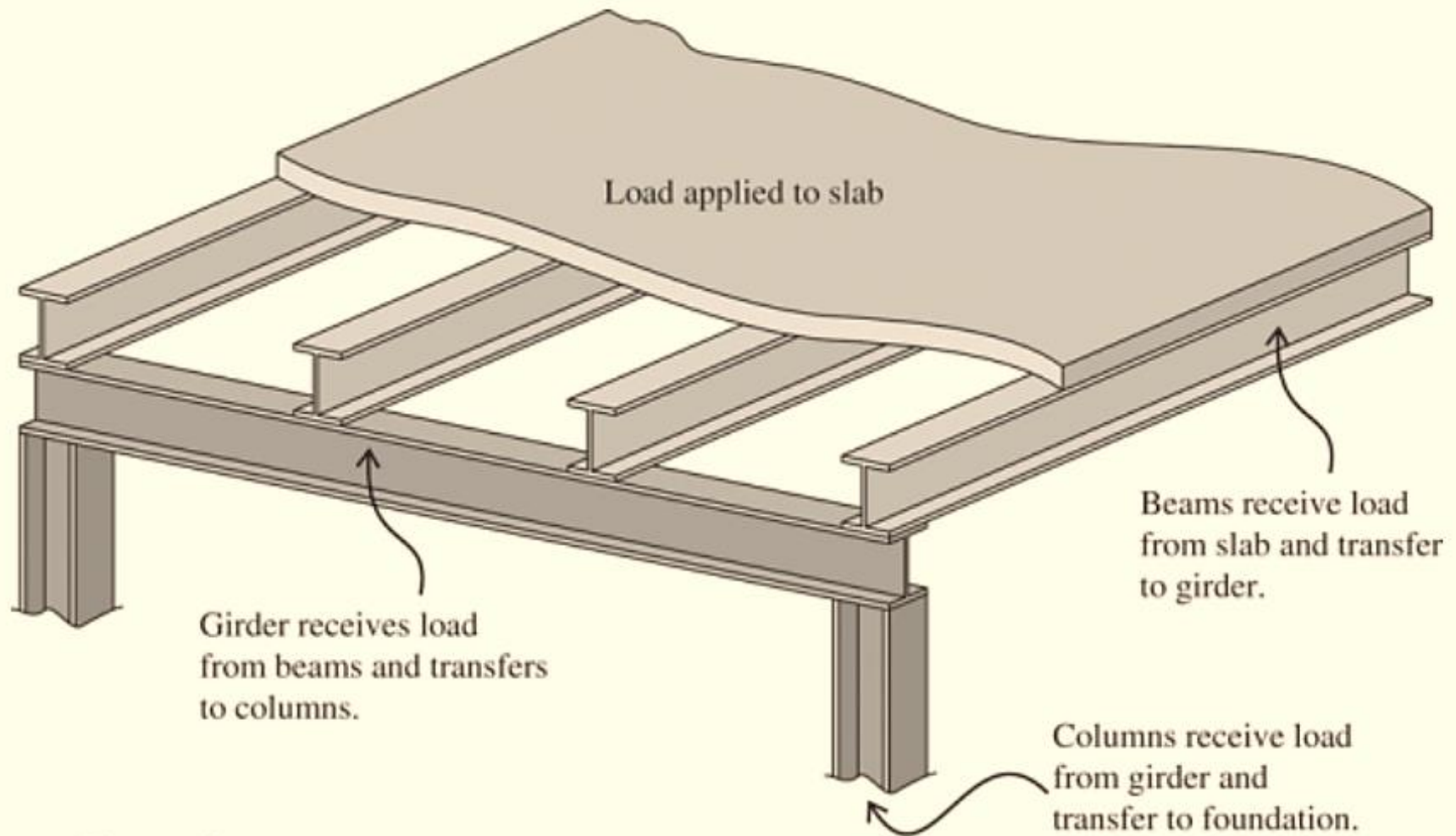
QUESTION NO. 47

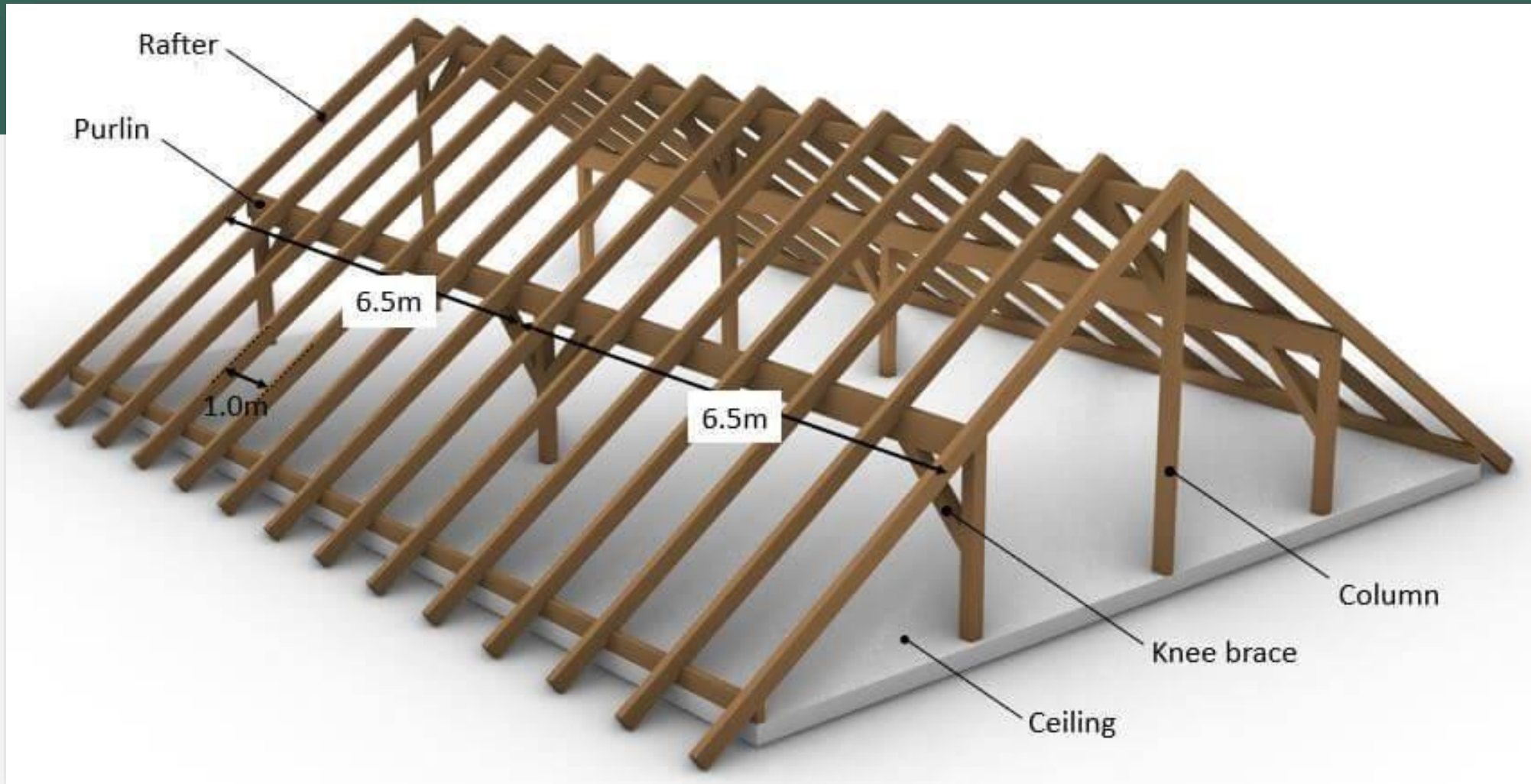
A structure the consists of compressive force action and supports only axial loadings is:

- a. Beam
- b. Column
- c. Purlins
- d. Girder



AIM TO TOP
REGION 100 TRAINING CENTER
ILIGAN





QUESTION NO. 47

A structure the consists of compressive force action and supports only axial loadings is:

- a. Beam
- b. Column
- c. Purlins
- d. Girder



QUESTION NO. 48

A beam needs 4 pcs of 20 mm RSB. If there is no available 20 mm RSB, how many 16 mm RSB can substitute for 20 mm?

- a. 4 pcs
- b. 5 pcs
- c. 6 pcs
- d. 7 pcs

QUESTION NO. 48

A beam needs 4 pcs of 20 mm RSB. If there is no available 20 mm RSB, how many 16 mm RSB can substitute for 20 mm?

- a. 4 pcs
- b. 5 pcs
- c. 6 pcs
- d. 7 pcs

QUESTION NO. 49

How many pieces of umbrella nails are there if there are 10 kilograms of umbrella nails?

- a. 10,000 pcs
- b. 11,000 pcs
- c. 12,000 pcs
- d. 12,500 pcs



QUANTITY OF ROOF ACCESSORIES IN KILOGRAMS

Materials	Number per kilogram
GI Roof Nails	120
GI Rivets	180
GI Washers	126
Lead Washers	75
Umbrella Nails	120

Reference: Simplified Construction Estimate by Max B. Fajardo Jr.

QUESTION NO. 49

How many pieces of umbrella nails are there if there are 10 kilograms of umbrella nails?

- a. 10,000 pcs (1,000 pcs)
- b. 11,000 pcs (1,100 pcs)
- c. 12,000 pcs (1,200 pcs)
- d. 12,500 pcs (1,250 pcs)

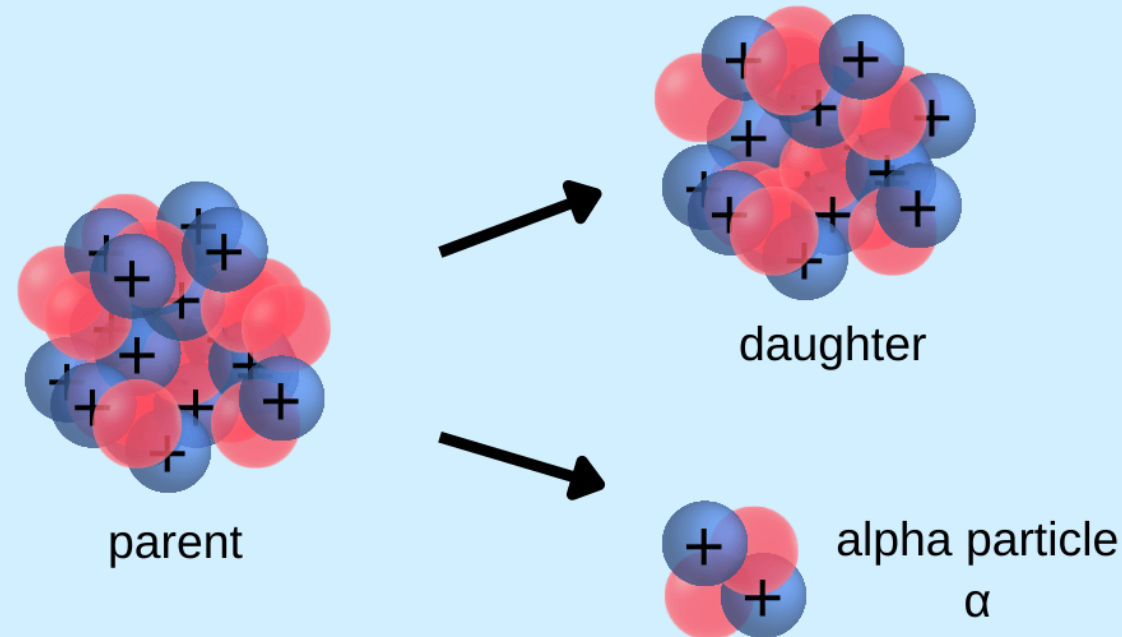
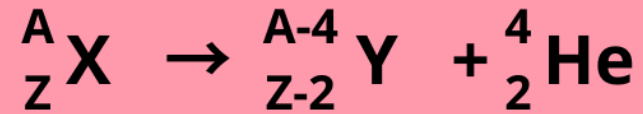
QUESTION NO. 50

It a positively charged particle made up of two protons and two neutrons and released by a radioactive nucleus

- a. Alpha radiation
- b. Beta radiation
- c. Ultra-violet radiation
- d. Gamma radiation

Alpha Particle

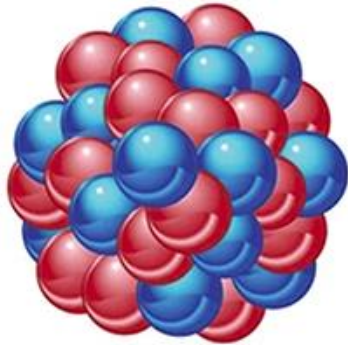
An alpha particle is a particle that is identical to a helium-4 nucleus. It contains two protons and two neutrons and has a +2 charge.



sciencenotes.org

What is **Beta Radiation**?

Beta Desintegration



Nucleous

 Neutron

 Proton

Beta Particle (Fast Electron)

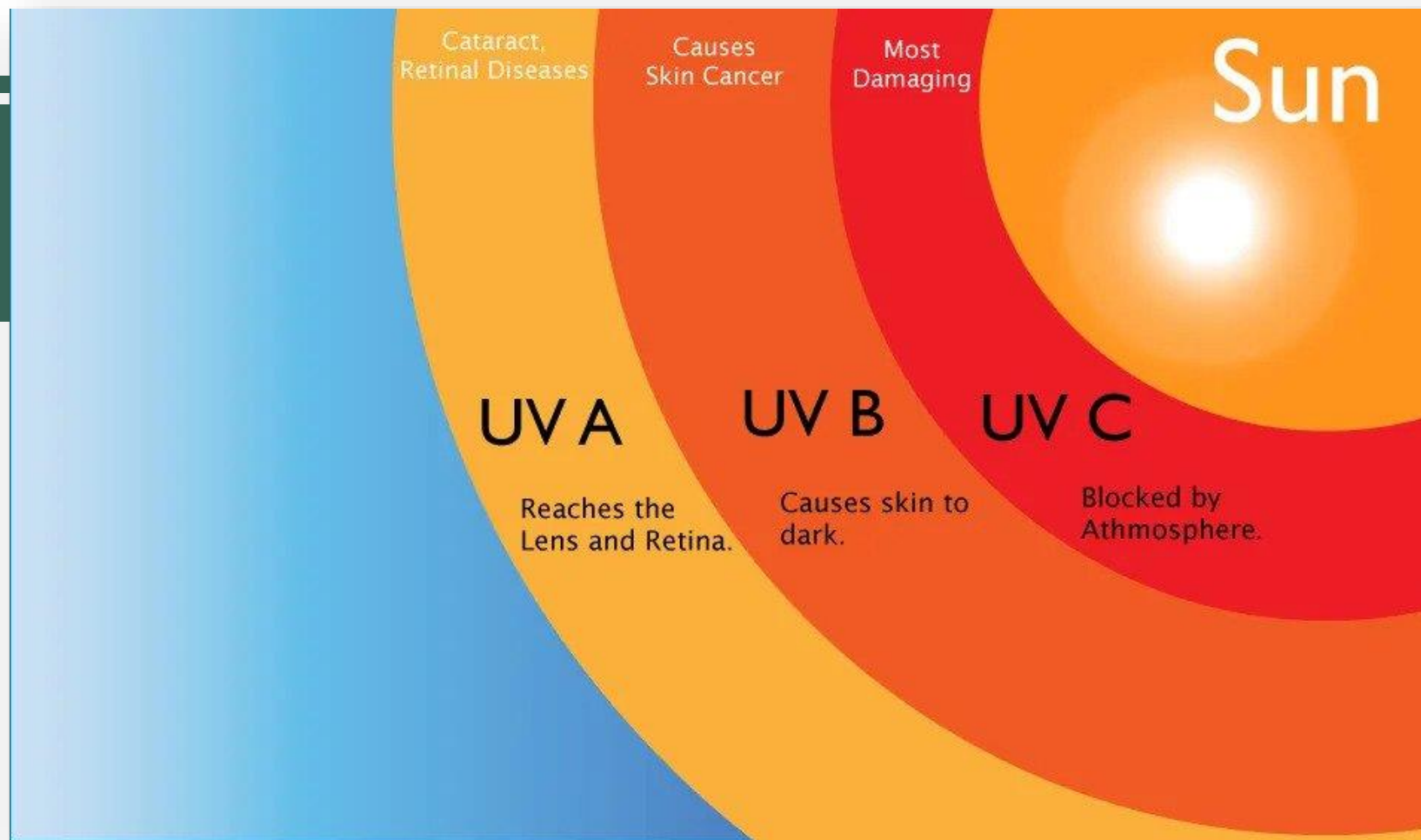


Gamma Ray

Beta radiation is a form of ionizing radiation composed of beta particles, which are electrons (β^-) or positrons (β^+).

They originate from nuclear processes, such as radioactive decay. Beta electrons are ejected from the atomic nucleus during the decay of a neutron into a proton, while positrons are emitted when a neutron converts into a proton.

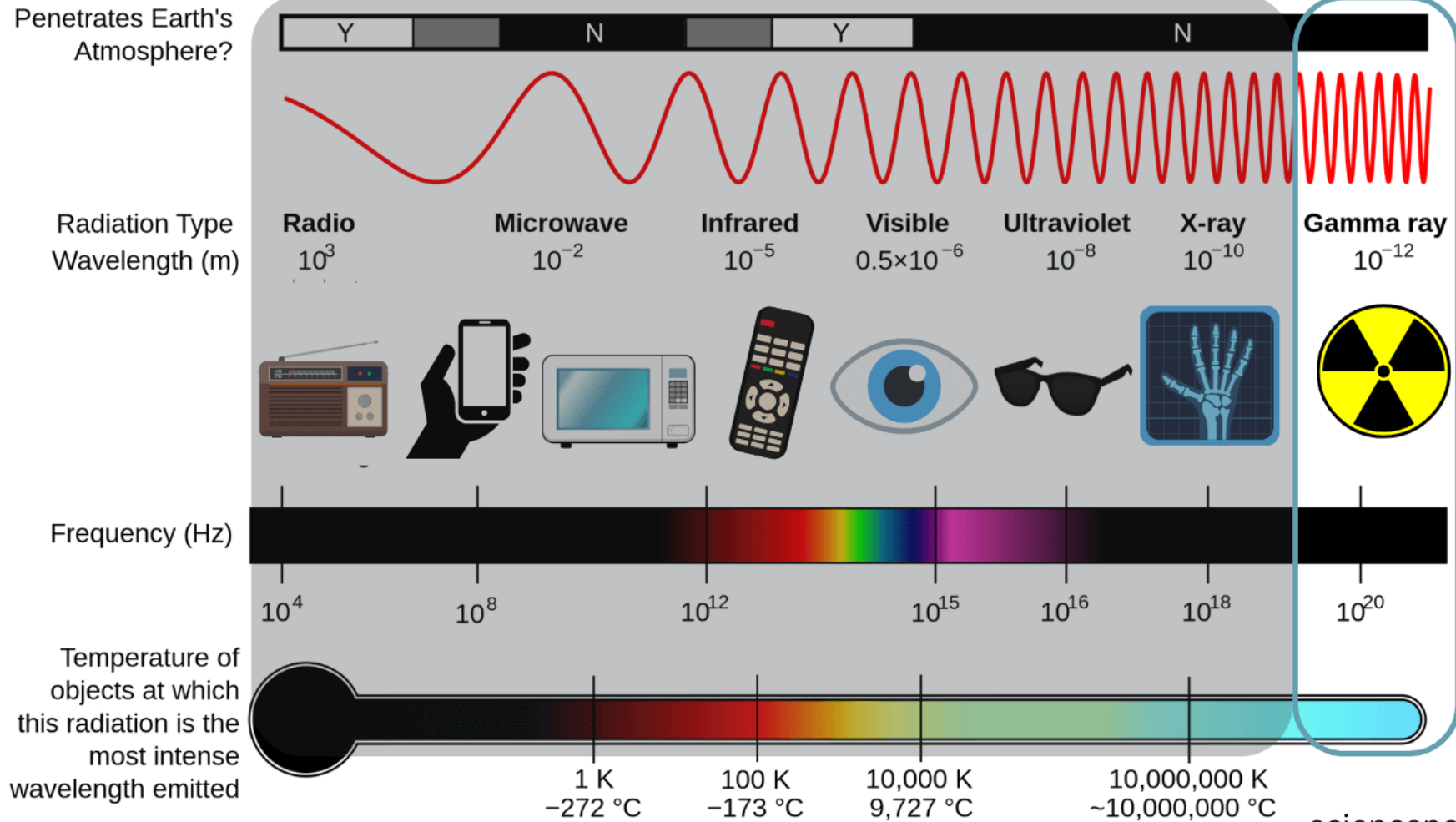
Beta radiation has moderate penetration ability and is used in various applications, such as nuclear medicine and scientific research.



Ultraviolet (UV) radiation is a type of electromagnetic radiation emitted by the sun and artificial sources. It falls between visible light and X-rays on the electromagnetic spectrum and is invisible to the human eye.

Gamma Rays or Gamma Radiation

A gamma ray is the type of electromagnetic radiation with the highest energy and shortest wavelength.



QUESTION NO. 50

It a positively charged particle made up of two protons and two neutrons and released by a radioactive nucleus

- a. Alpha radiation
- b. Beta radiation
- c. Ultra-violet radiation
- d. Gamma radiation

QUESTION NO. 51

Given the initial temperature of 45 F and initial pressure of 16 bar, the pressure changed to 35 bar, determine the change in temperature.

- a. 250
- b. 300
- c. 500
- d. 600

$$P_1 V_1 = P_2 V_2 \qquad \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

$$PV = nRT$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$



GAS LAW	FORMULA
Boyle's law	$P_1 V_1 = P_2 V_2$
Charles's law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay-lussac's law	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Combined law	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$
Ideal gas law	$PV = nRT$



QUESTION NO. 51

Given the initial temperature of 45 F and initial pressure of 16 bar, the pressure changed to 35 bar, determine the change in temperature.

- a. 250
- b. 300
- c. 500
- d. 600

QUESTION NO. 52

It is the production of food, fiber, or other plant or animal products using farming techniques that protect the environment, public health, human communities, and animal welfare.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

QUESTION NO. 52

It is the production of food, fiber, or other plant or animal products using farming techniques that protect the environment, public health, human communities, and animal welfare.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

QUESTION NO. 53

It is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

QUESTION NO. 53

It is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

QUESTION NO. 54

It is the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the subgrade. Treatment may include application of chemicalbased soil additives such as soil-stabilizers and/or any approved method.

- a. Aggregate Base Course
- b. Base Course
- c. Subbase Course
- d. Aggregate Subbase Course

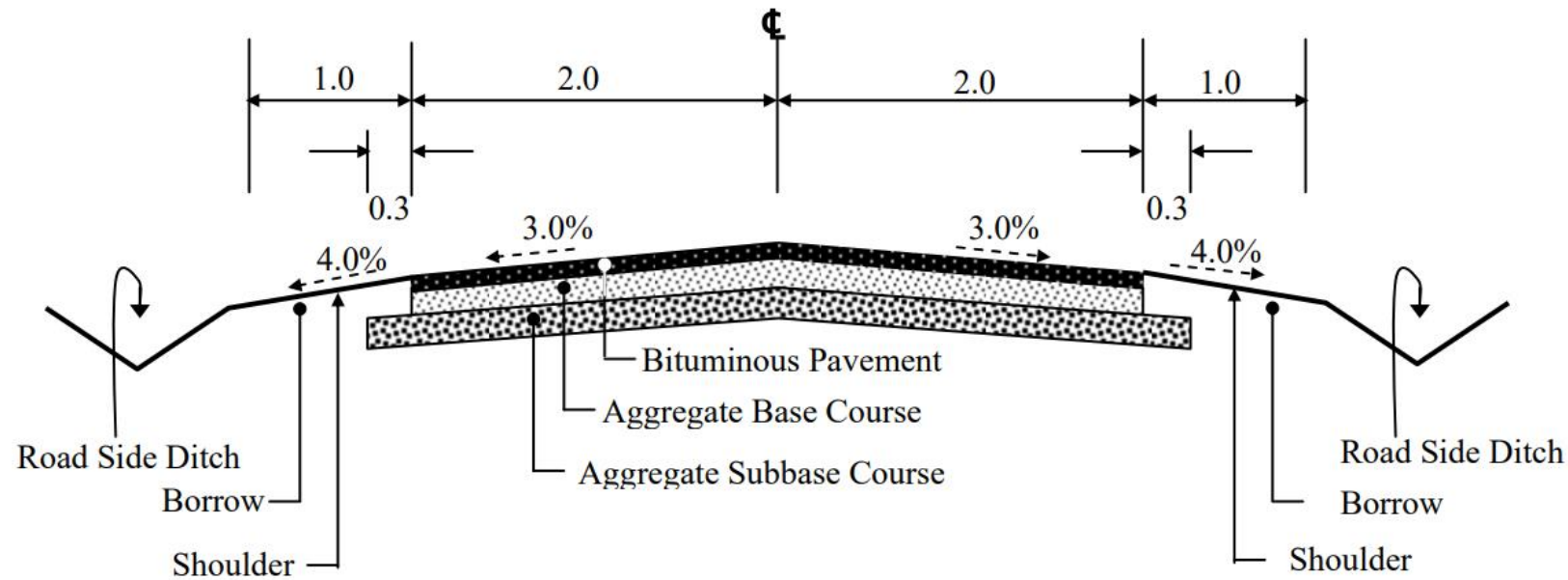


Figure 13 – Typical-Cross-Section of Two-Lane Bituminous Road

4.3 **base course**

the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the sub-grade. Treatment may include application of chemical-based soil additives such as soil-stabilizers and/or any approved method

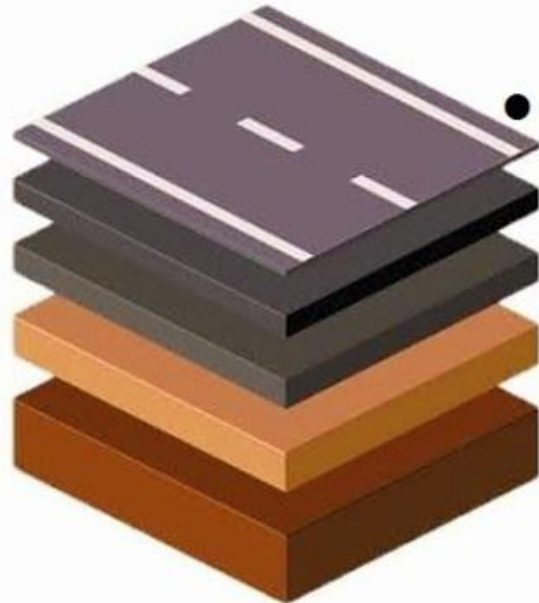
4.39 **Subbase course**

refers to the layer of the specified or selected materials of designated thickness in a pavement structure immediately above the sub-grade and below the base course

Structural Components of a Road



- Sub Grade
- Sub Base
- Base
- Surfacing



QUESTION NO. 54

It is the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the subgrade. Treatment may include application of chemical based soil additives such as soil-stabilizers and/or any approved method.

- a. Aggregate Base Course
- b. Base Course
- c. Subbase Course
- d. Aggregate Subbase Course

QUESTION NO. 55

It involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions

- a. Aerated static pile
- b. In-vessel system composting
- c. Windrow composting
- d. Bin composting



Aerated static pile composting is a method that uses forced air to maintain oxygen levels for efficient decomposition of organic waste without manual turning. It involves placing the compost materials on a bed of perforated pipes connected to a blower, which either pushes or pulls air through the pile.



In-vessel composting is accomplished inside an enclosed container or vessel. Type of vessel has been used include vertical tower, horizontal rectangular and circular tanks, and circular rotating tanks.

6.2.9.1 Windrow composting

6.2.9.1.1 Windrows should be 1 m – 2 m high and 2 m – 5 m wide at the base (Figure 1).

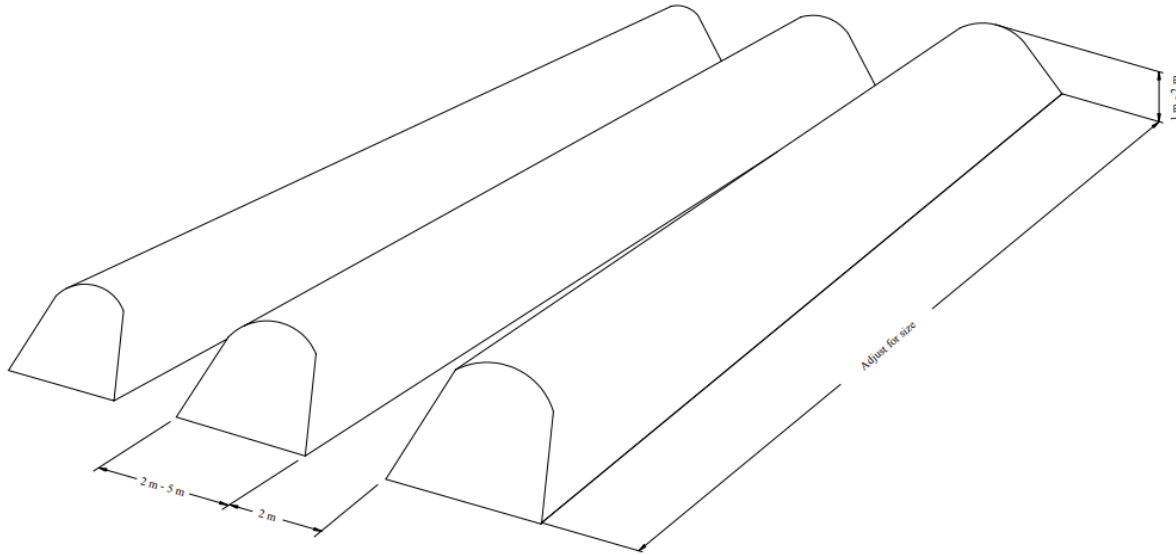


Figure 1 – Typical windrow composting



windrow composting involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions



6.2.2 Composting time

Composting time varies with C:N ratio, moisture content, climate, type of operation, management, and the types of wastes and amendments being composted. Typical composting time is shown in Table 1.

Table 1 – Typical composting time

Method	Composting time
Windrow	4 months
Aerated static pile	4 weeks
In-vessel	7 – 30 days



QUESTION NO. 55

It involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions

- a. Aerated static pile
- b. In-vessel system composting
- c. Windrow composting
- d. Bin composting

QUESTION NO. 56

The percent of moisture content of test materials for peanut sheller.

- a. 8%-10%, wet basis
- b. 12%-14%, wet basis
- c. 8%-10%, dry basis
- d. 12%-14%, dry basis

4.6 Test materials

4.6.1 Sample characteristics

Test materials to be used shall have the following characteristics:

4.6.1.1 Variety : commonly or locally grown

4.6.1.2 Moisture content : 12 % - 14 %, wet basis



QUESTION NO. 56

The percent of moisture content of test materials for peanut sheller.

- a. 8%-10%, wet basis
- b. 12%-14%, wet basis
- c. 8%-10%, dry basis
- d. 12%-14%, dry basis

QUESTION NO. 57

Measures velocity by the change of source light or sound frequency from the frequency of reflections from moving particles.

- a. Optical Strobe
- b. Doppler
- c. Bellow
- d. Platometer



A stroboscope, also called a strobe light or strobe tachometer, can be used to measure the rotation speed or revolution velocity and frequency of rotating, vibrating or moving parts, components and equipment. The flash frequency of a stroboscope light is adjustable to ensure accurate measurement of these parameters.



Doppler instruments are used to measure velocity and speed, leveraging the Doppler effect. This effect, the change in frequency of a wave due to relative motion between the source and observer.



Bellow is a deformable, sealed container that expands and contracts in response to pressure changes. It's primarily used as a pressure-sensing element in instruments like pressure gauges. Bellows can be made of metal or other flexible materials and are designed to convert pressure variations into measurable mechanical displacement.



A **planimeter**, also known as a **platometer**, is a measuring instrument used to determine the area of an arbitrary two-dimensional shape. It works by tracing the boundary of the shape with a tracing point, and the area is then calculated based on the movement of the tracing point.

QUESTION NO. 57

Measures velocity by the change of source light or sound frequency from the frequency of reflections from moving particles.

- a. Optical Strobe
- b. Doppler
- c. Bellow
- d. Platometer

QUESTION NO. 58

Ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent.

- a. Scattering loss
- b. Cylinder loss
- c. Separation loss
- d. Blower loss

3.1

blower loss

ratio of the weight of corn kernels blown by the sheller fan, to the weight of the total corn kernel input of the sheller, expressed in percent

3.20

scattering loss

ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent

3.21

separation loss

ratio of the weight of corn kernels that come out of the shelling chamber with the cobs at the cob outlet, to the weight of the total corn kernel input of the sheller, expressed in percent

3.29

unshelled loss

ratio of the weight of corn kernels that remained in the cobs of the corn fed into the shelling chamber, to the weight of the total corn kernel input of the sheller, expressed in percent



QUESTION NO. 58

Ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent.

- a. **Scattering loss**
- b. Cylinder loss
- c. Separation loss
- d. Blower loss

QUESTION NO. 59

A 2.4-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 4 minutes. It produced 600 kg of crude oil in 2 hours. What is its crude oil production rate?

- a. 300 kg/h
- b. 200 kg/min
- c. 600 kg/h
- d. 600 kg/min

QUESTION NO. 59

A 2.4-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 4 minutes. It produced 600 kg of crude oil in 2 hours. What is its crude oil production rate?

- a. 300 kg/h
- b. 200 kg/min
- c. 600 kg/h
- d. 600 kg/min

QUESTION NO. 60

A 10MW-25MW hydro power plant is classified as: ____.

- a. Micro hydro
- b. Small hydro
- c. Mini hydro
- d. Medium hydro



Small hydro power uses the flow of water to turn turbines that are connected to a generator for the production of electricity. Small hydro is divided into further categories depending on its size, such as **mini- (less than 1000kW)**, **micro-hydro (less than 100kW)** and **pico-hydro (less than 5kW)** (EHSA 2005); the definitions may vary according to manufacturers and countries, as there is no internationally accepted definition of small hydropower. In China, small hydropower refers to capacities of up to **25 MW**, in India of up to **15 MW** and in Sweden 'small' refers to up to **1.5 MW**. However, a capacity of up to **10 MW** is a generally accepted norm by the European Small Hydropower Association (ESHA), the European Commission, and UNIPED (International Union of Producers and Distributors of Electricity).

Reference: <https://www.ctc-n.org/technologies/small-hydropower>

Hydro plants are classified based on their capacities, as follows:

- (i) micro-hydro - 1 to 100 kW;
- (ii) mini-hydro - 101 kW to 10 MW; and
- (iii) large hydro - more than 10 MW.

Reference: <https://legacy.doe.gov.ph/hydropower>

QUESTION NO. 60

A 10MW-25MW hydro power plant is classified as: ____.

- a. Micro hydro
- b. **Small hydro**
- c. Mini hydro
- d. Medium hydro

QUESTION NO. 61

Convert the binary number 010101_2 to decimal number system

- a. 53_{10}
- b. 85_{10}
- c. 42_{10}
- d. 21_{10}

QUESTION NO. 61

Convert the binary number 010101_2 to decimal number system

- a. 53_{10}
- b. 85_{10}
- c. 42_{10}
- d. 21_{10}

QUESTION NO. 62

Which one is an integrated circuit used in instrumentation and control?

- a. TD 8062
- b. LM333
- c. ES8813
- d. CQ 556



QUESTION NO. 62

Which one is an integrated circuit used in instrumentation and control?

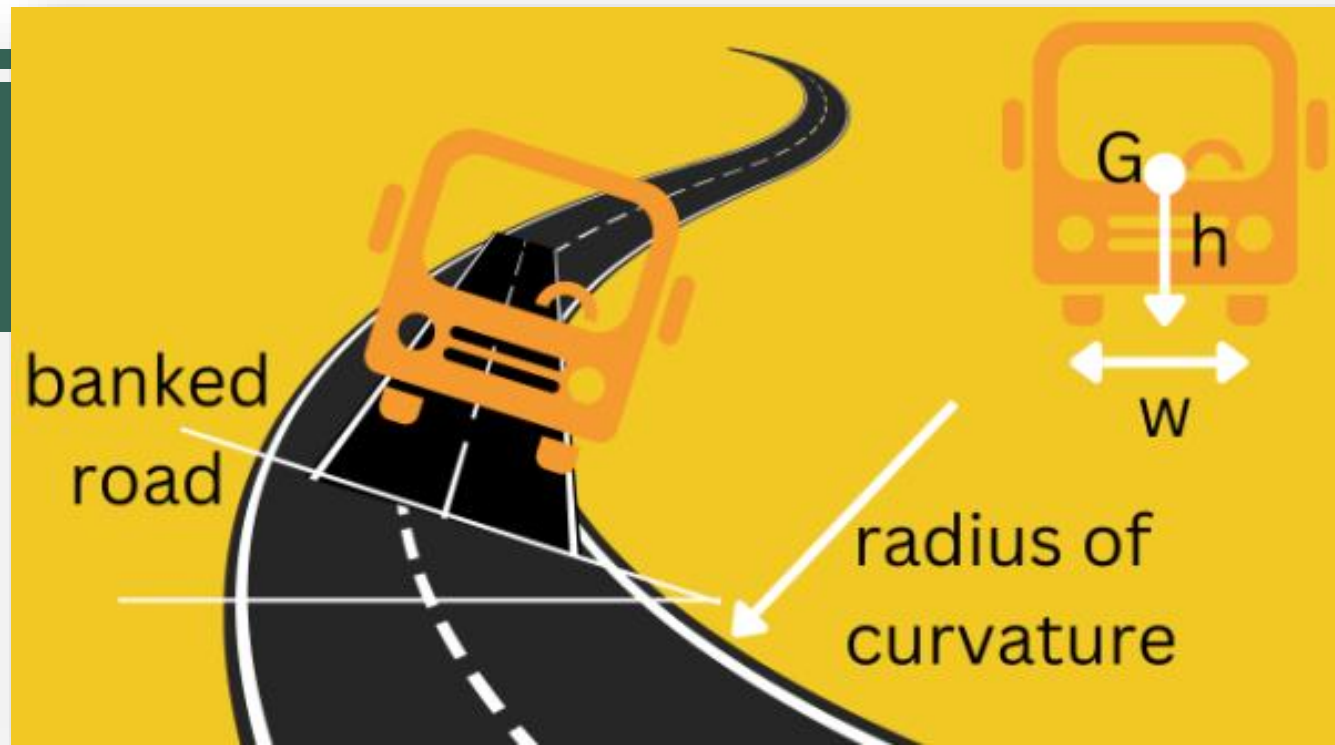
- a. TD 8062
- b. LM333
- c. ES8813
- d. CQ 556



QUESTION NO. 63

In PAES 421:2009, the minimum radius of curvature for flat, rolling and mountainous terrains respectively shall be.

- a. 75, 50 & 5 m
- b. 500, 150 & 50 m
- c. 150, 100 & 50 m
- d. 600, 200 & 100 m



Radius of Curvature

The minimum radius design for curvature shall be 500 meters for flat terrain, 150 meters for rolling terrain and 50 meters for mountainous terrain. Avoid minimum radius at the beginning and end of all straight sections with a length of more than 5 kilometers and or isolated curves. Curves with larger radius must be used where terrain permits.

QUESTION NO. 63

In PAES 421:2009, the minimum radius of curvature for flat, rolling and mountainous terrains respectively shall be.

- a. 75, 50 & 5 m
- b. 500, 150 & 50 m
- c. 150, 100 & 50 m
- d. 600, 200 & 100 m

QUESTION NO. 64

It refers to the extent or degree by which the bran layer of the brown rice is removed as a result of whitening

- a. Whitening degree
- b. Milling degree
- c. Milling recovery
- d. Milling index

3.11

milling capacity

quantity of paddy that the rice mill can process to a specified quality per total milling time, expressed in metric tons per hour

3.12

milling degree

extent or degree by which the bran layer of the brown rice is removed as a result of whitening

3.13

milling recovery

ratio of the weight of milled rice to the total weight of paddy, expressed in percent

3.14

milling recovery index

ratio of the milling recovery obtained in actual testing, to the milling recovery obtained from the laboratory test mill



QUESTION NO. 64

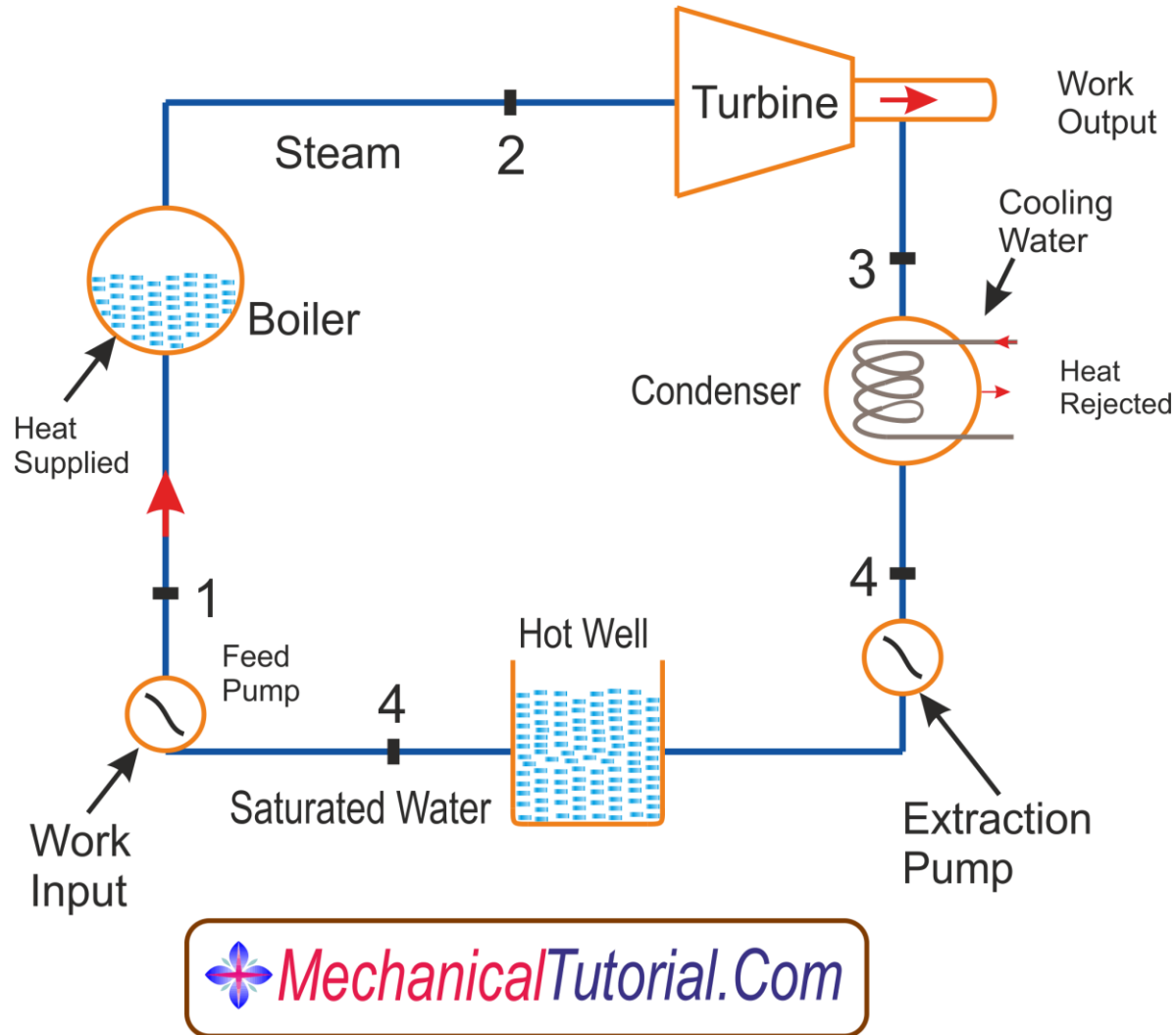
It refers to the extent or degree by which the bran layer of the brown rice is removed as a result of whitening

- a. Whitening degree
- b. **Milling degree**
- c. Milling recovery
- d. Milling index

QUESTION NO. 65

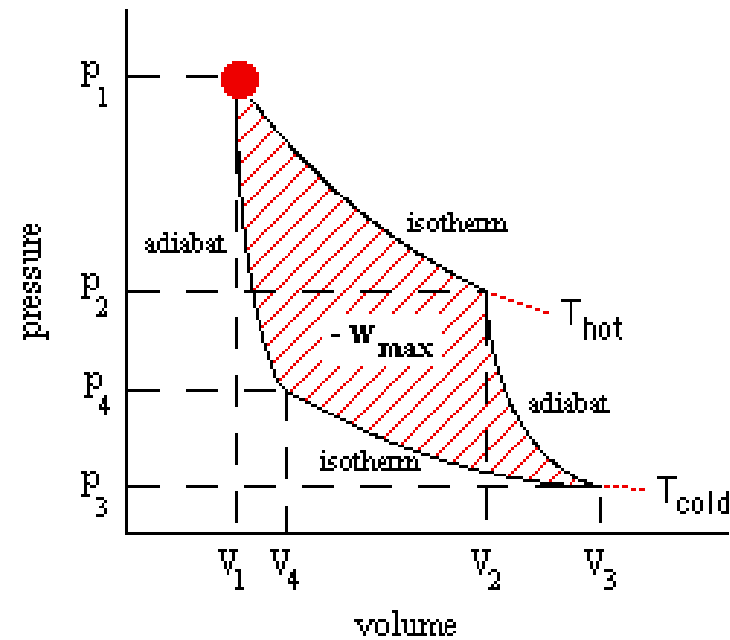
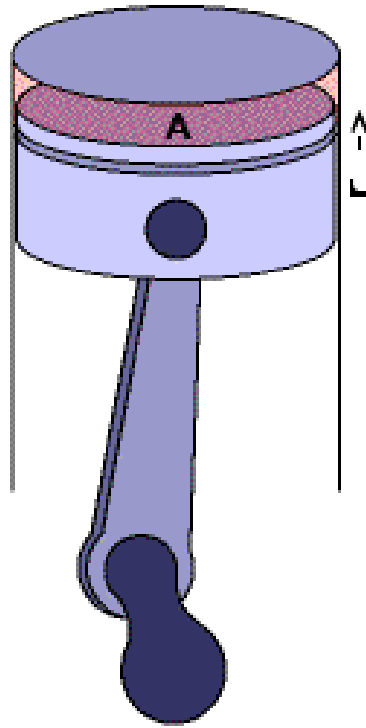
The most efficient and ideal gas cycle.

- a. Rankine Cycle
- b. Carnot Cycle
- c. Otto Cycle
- d. Brayton Cycle



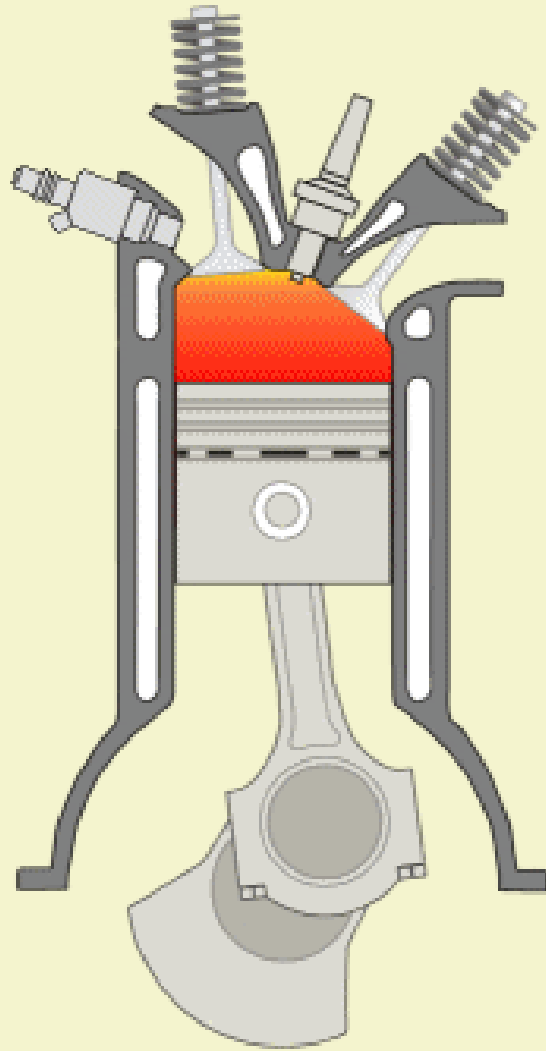
The Rankine cycle is the fundamental operating cycle of all power plants where an operating fluid is continuously evaporated and condensed.

Rankine Cycle is an ideal thermodynamic theoretical cycle, used in power plants, which **converts heat into mechanical energy working at a constant pressure.**



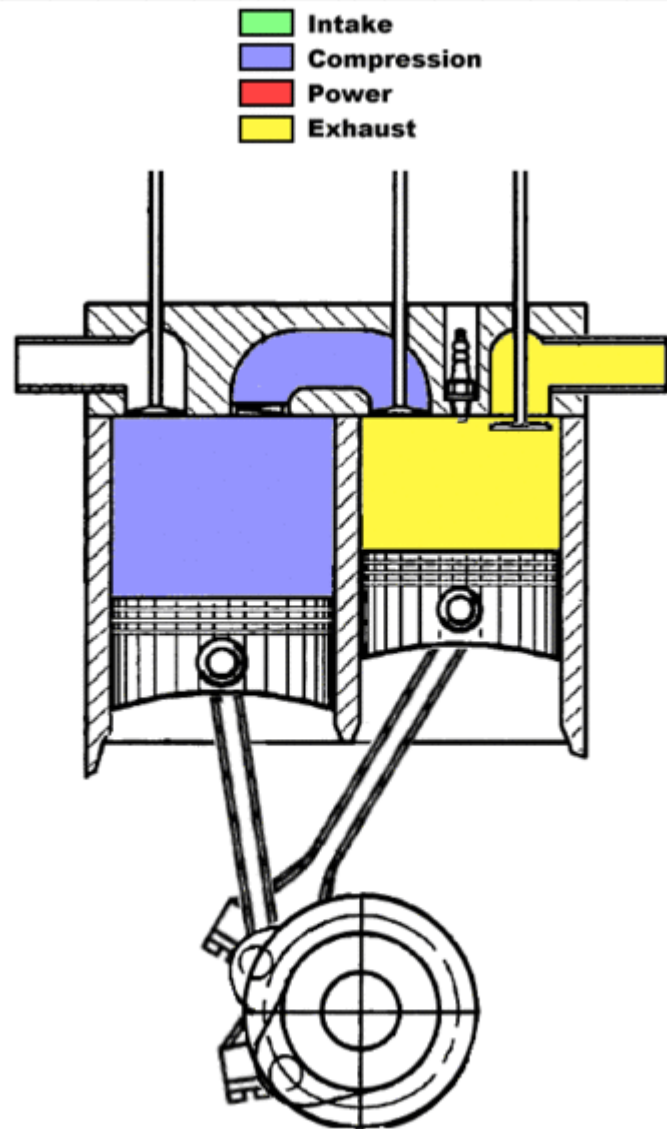
The Carnot cycle is the most efficient engine possible based on the assumption of the absence of incidental wasteful processes such as friction.

Otto Cycle with Sequential Fuel Injection (SFI)



Power

The **Otto cycle** is a thermodynamic cycle that explains how internal combustion engines convert chemical energy from gasoline into motion.



A Brayton cycle is a power conversion system that **uses a gas to produce thrust in jet engines or power in natural gas turbines** by compressing and heating air.

QUESTION NO. 65

The most efficient and ideal gas cycle.

- a. Rankine Cycle
- b. **Carnot Cycle**
- c. Otto Cycle
- d. Brayton Cycle



QUESTION NO. 66

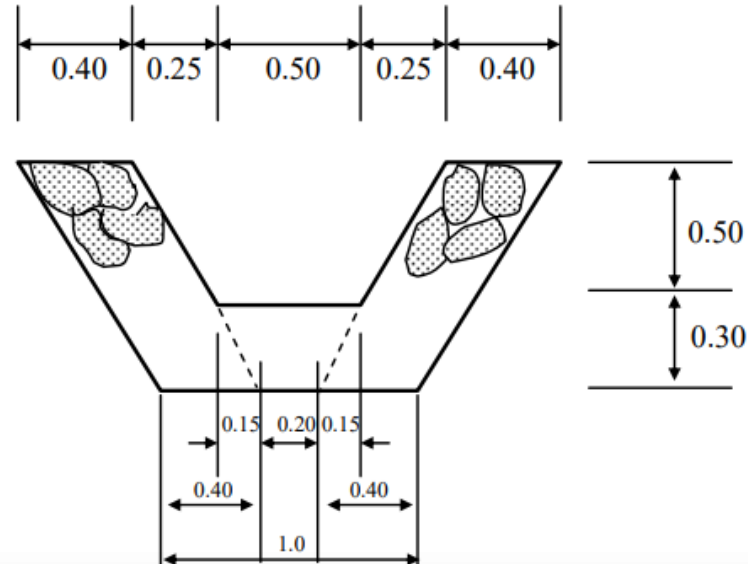
Standard thickness of concrete of trapezoidal ditch w/ riprap

- a. 0.25 m
- b. 0.15 m
- c. 0.30 m
- d. 0.20 m



a. Roadside Trapezoidal Ditch

This shall reduce the risk of erosion for a lower water velocity.
Below is a preferred designed ditch shaped with relatively flat front slope and back slope, and a wide flat bottom.



Construction Requirements

Trapezoidal ditch design shall be constructed with grouted riprap or with concrete lining.

It shall have a minimum thickness of 0.25 meters.



QUESTION NO. 66

Standard thickness of concrete of trapezoidal ditch w/ riprap

- a. **0.25 m**
- b. 0.15 m
- c. 0.30 m
- d. 0.20 m

QUESTION NO. 67

Lighting computation in a warehouse with 4x4 grid lighting layout with mounting height ratio of 1.0

- a. 4 m
- b. 16 m
- c. 2 m
- d. 8 m



AIM TO TOP

ILIGAN

QUESTION NO. 67

Lighting computation in a warehouse with 4x4 grid lighting layout with mounting height ratio of 1.0

- a. 4 m
- b. 16 m
- c. 2 m
- d. 8 m



AIM TO TOP

ILIGAN

QUESTION NO. 68

Contains less than or equal to 20% longitudinal streaks

- a. Well-milled rice
- b. Regular milled rice
- c. Overmilled rice
- d. Undermilled rice



3.9

regular milled rice

rice grain from which the hull, the germ, the outer bran layers and the greater part of the inner bran layers have been removed but part of the lengthwise streaks of the bran layers may still be present on 21% to 40% of the sample grains

3.10

undermilled rice

rice grain from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layer may still be present on more than 40 % of the sample grains

3.11

well-milled rice

rice grain from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layer may still be present on 1% to 20% of the sample grains



QUESTION NO. 68

Contains less than or equal to 20% longitudinal streaks

- a. **Well-milled rice**
- b. Regular milled rice
- c. Overmilled rice
- d. Undermilled rice



QUESTION NO. 69

Moisture gradient of corn and rice in mechanical dryer

- a. 0.5%
- b. 1.0%
- c. 2.5%
- d. 2.0%



[illegible]

QUESTION NO. 69

Moisture gradient of corn and rice in mechanical dryer

- a. 0.5%
- b. 1.0%
- c. 2.5%
- d. **2.0%**



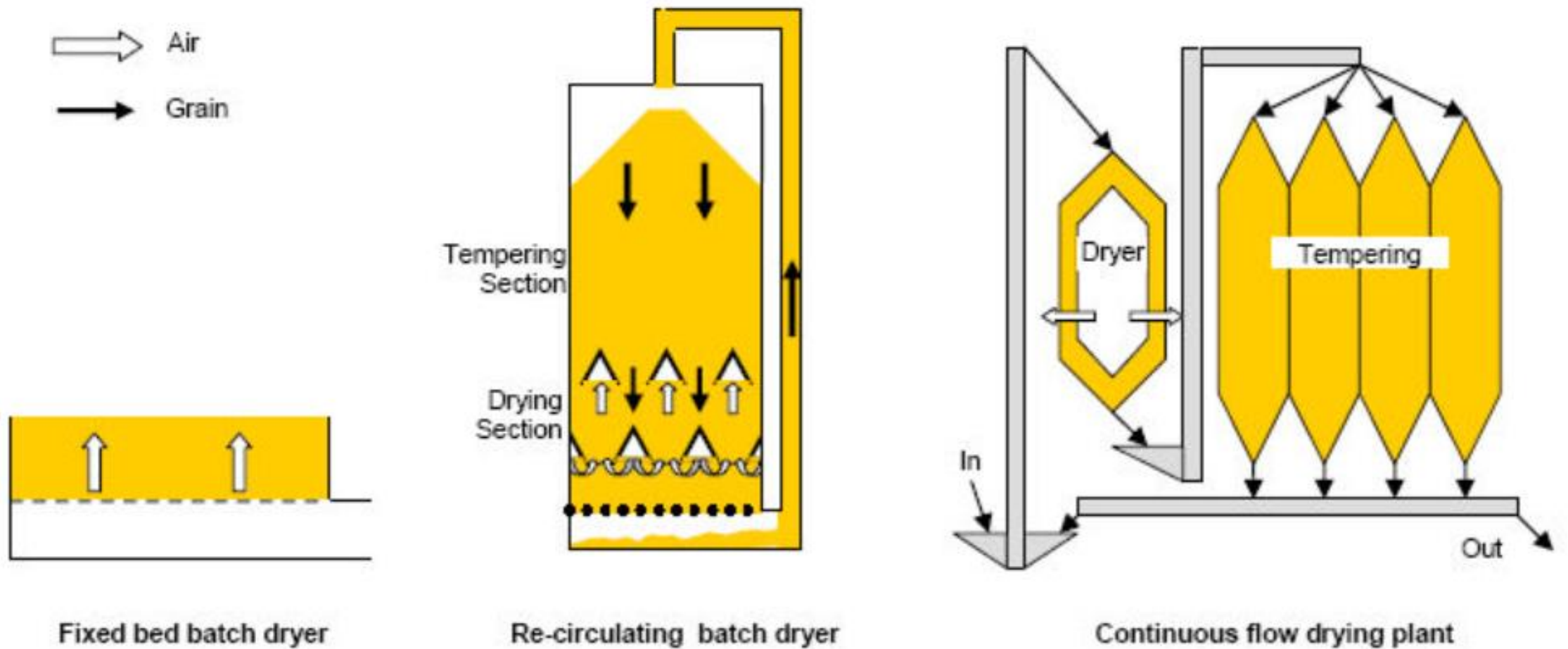
AIM TO TOP

ILIGAN

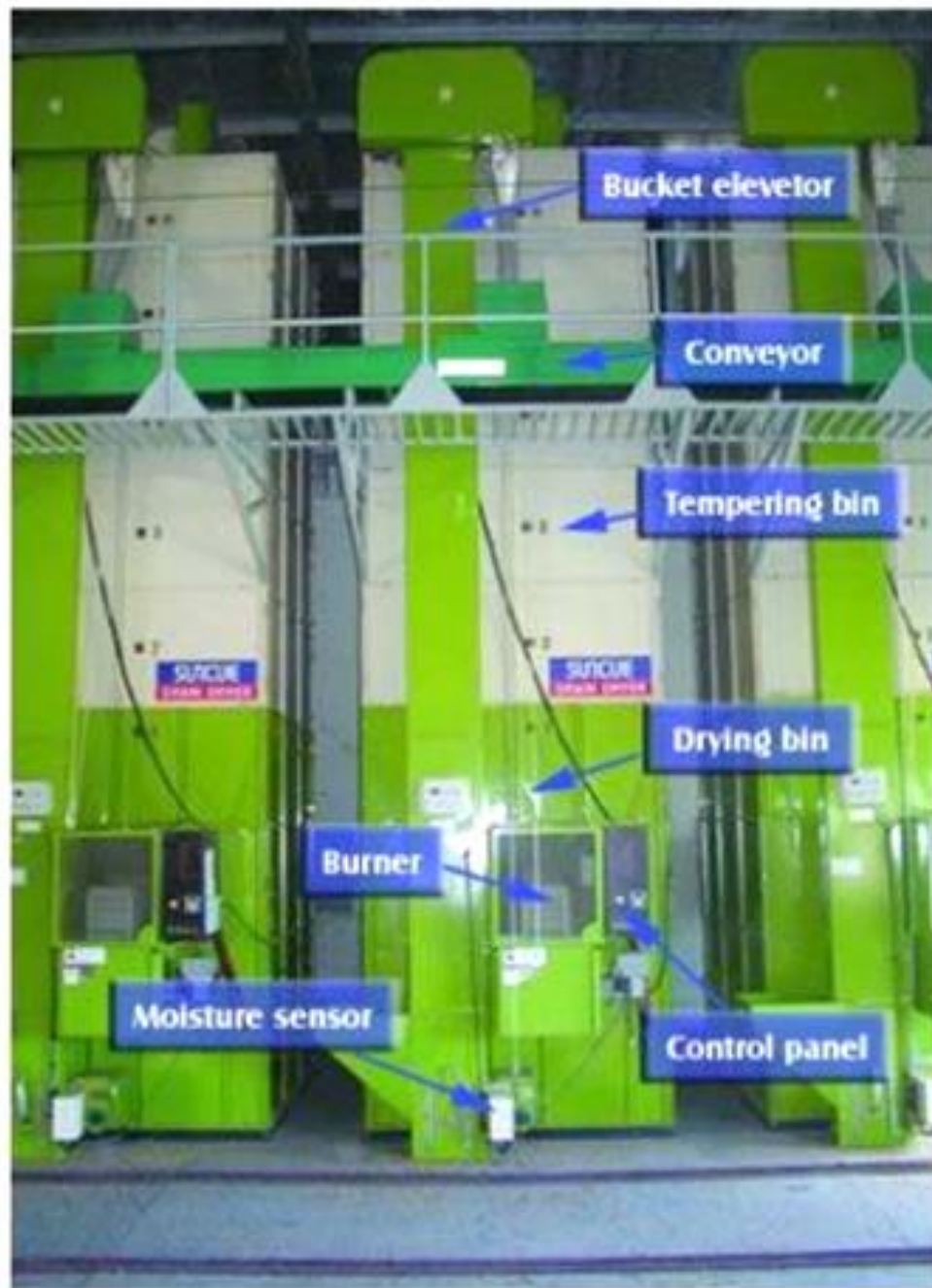
QUESTION NO. 70

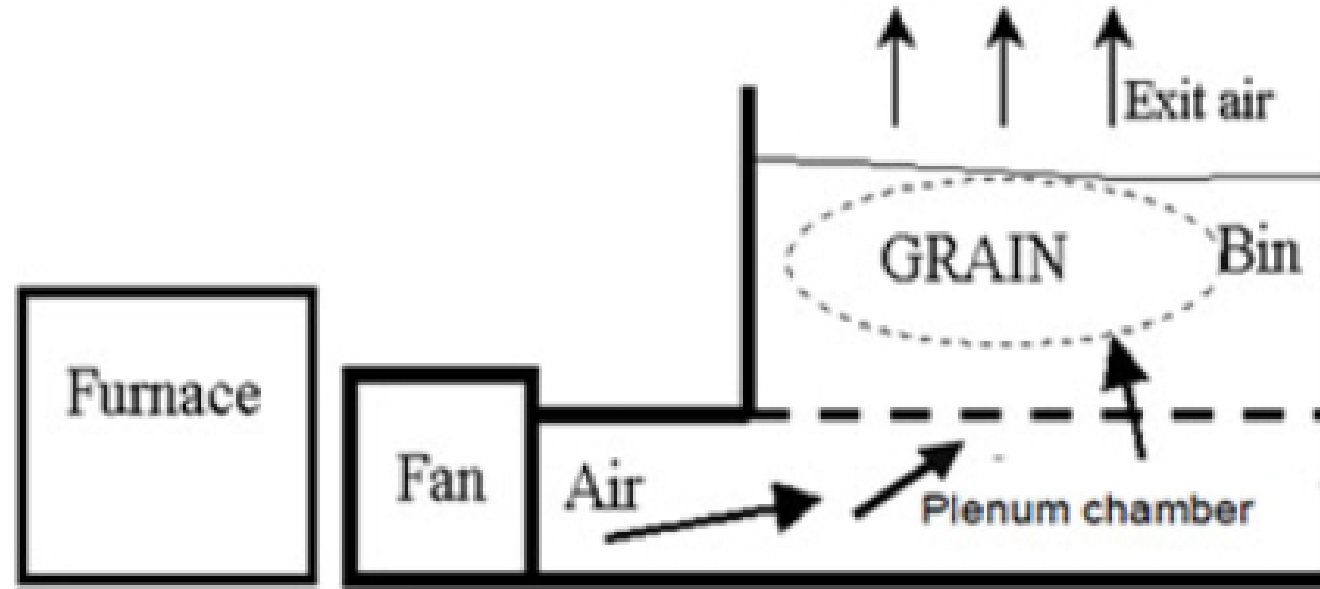
It is the chamber maintained under pressure for uniform distribution of the heated air through the grain mass

- a. dust collection system
- b. drying chamber
- c. plenum
- d. tempering bin



Dryer classification according to crop holding / crop flow





What is a flat bed dryer?

A dryer is a machine that removes the water from wet grains by forcing heated air through the grain bulk. In a flat bed dryer, the same quantity of grain is kept stationary in a holding bin until drying is completed. A flat bed dryer consists of a drying bin, an air distribution system with a plenum chamber, a fan, and a furnace or burner for heating the drying air.

QUESTION NO. 70

It is the chamber maintained under pressure for uniform distribution of the heated air through the grain mass

- a. dust collection system
- b. drying chamber
- c. **plenum**
- d. tempering bin



QUESTION NO. 71

The rate of radiation of a black body at 0°C is E watt. Then the rate of radiation of this black body at 273°C will be:

- a. $8E$
- b. $16E$
- c. $4E$
- d. E

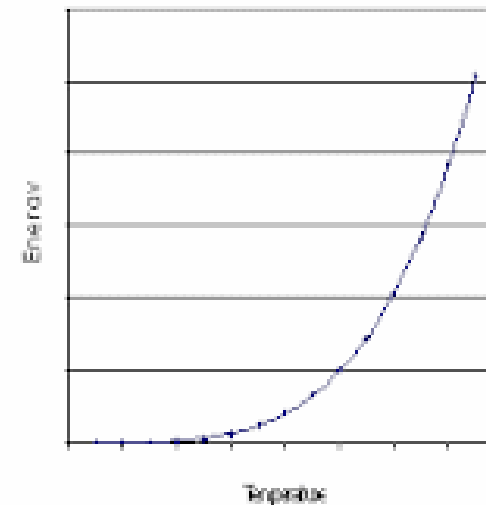
Stefan-Boltzmann Law

- Energy radiated by blackbody is proportional to the temperature to the 4th power

- $E = \sigma T^4$



Energy Temperature



QUESTION NO. 71

The rate of radiation of a black body at 0°C is E watt. Then the rate of radiation of this black body at 273°C will be:

- a. $8E$
- b. $16E$
- c. $4E$
- d. E

QUESTION NO. 72

In PAES 420:2002, the door height shall be provided with a clearance of at least ____ m above the tallest machinery

- a. 0.3 m
- b. 0.4 m
- c. 0.6 m
- d. 0.7 m



6.4 Door/Gate

6.4.1 Minimum dimension of a doorway shall be 3 m high and 3.7 m wide. For bigger equipment, a clearance of 0.6 m on both sides shall be added.

6.4.2 The door height shall be provided with a clearance of at least 0.3 m above the tallest machinery.

6.4.3 Overhead or rolling doors are recommended to be used than the hinged doors.

6.4.4 There shall be provision for a smaller service door so that the large door does not have to be opened for routine activities.



QUESTION NO. 72

In PAES 420:2002, the door height shall be provided with a clearance of at least ____ m above the tallest machinery

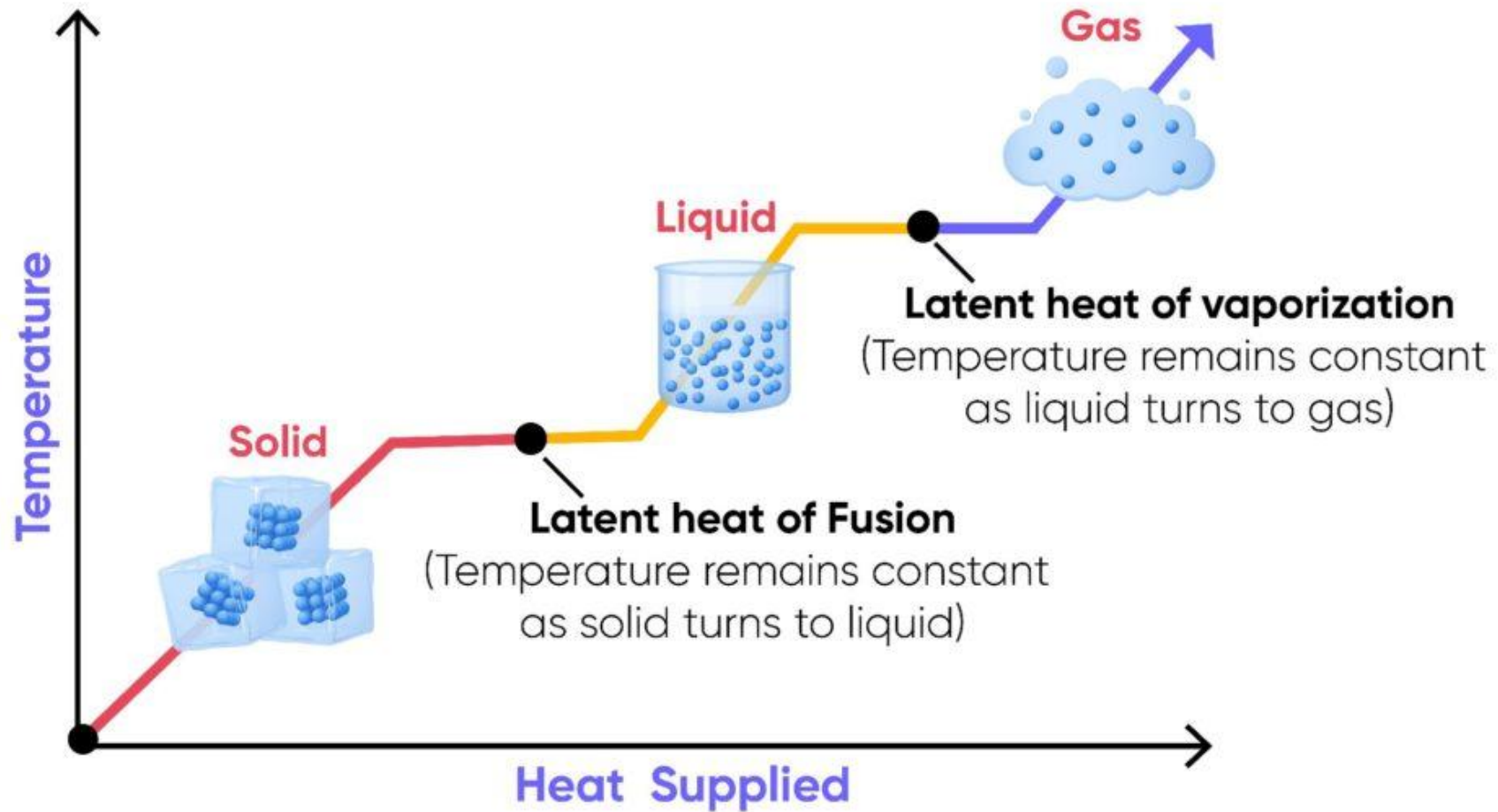
- a. 0.3 m
- b. 0.4 m
- c. 0.6 m
- d. 0.7 m

QUESTION NO. 73

Heat absorbed by a unit mass of a material at its boiling point to convert the material into a gas without temperature change

- a. Latent heat of sublimation
- b. Latent heat of deposition
- c. Latent heat of fusion
- d. Latent heat of vaporization

LATENT HEAT



QUESTION NO. 73

Heat absorbed by a unit mass of a material at its boiling point to convert the material into a gas without temperature change

- a. Latent heat of sublimation
- b. Latent heat of deposition
- c. Latent heat of fusion
- d. Latent heat of vaporization

QUESTION NO. 74

Which of the following is a good insulator

- a. Brass
- b. Mica
- c. Tin
- d. Platinum



QUESTION NO. 74

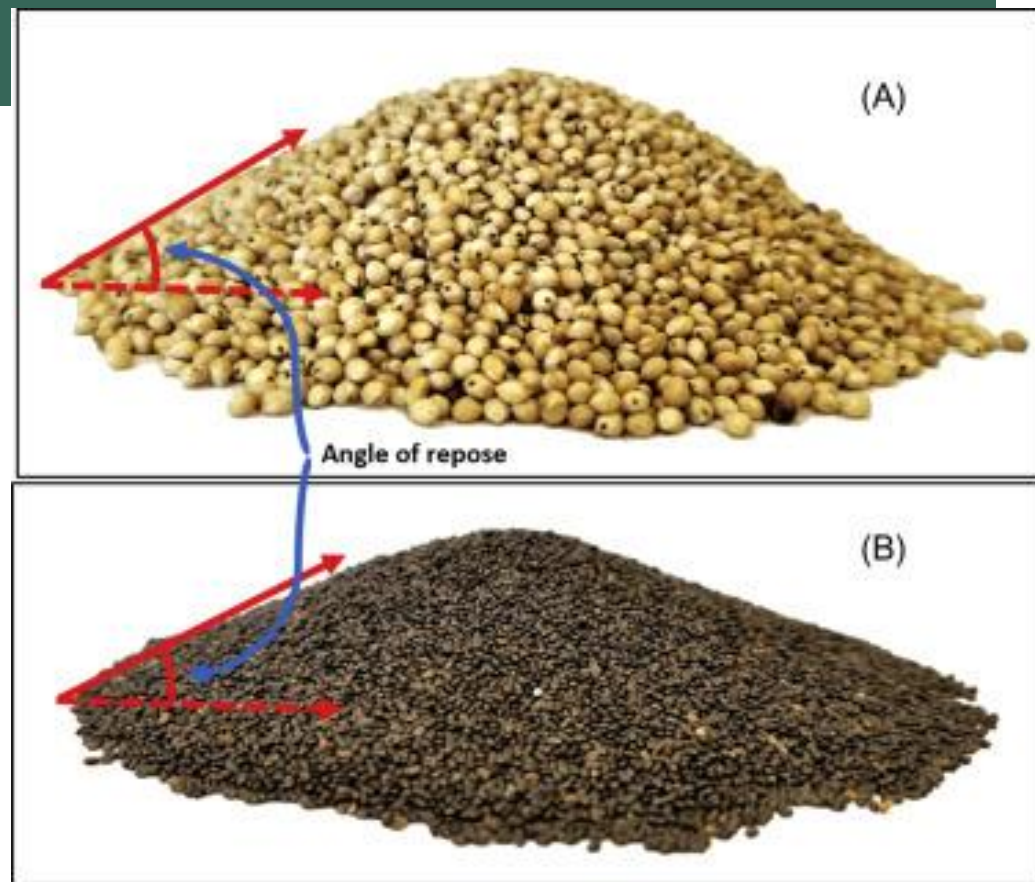
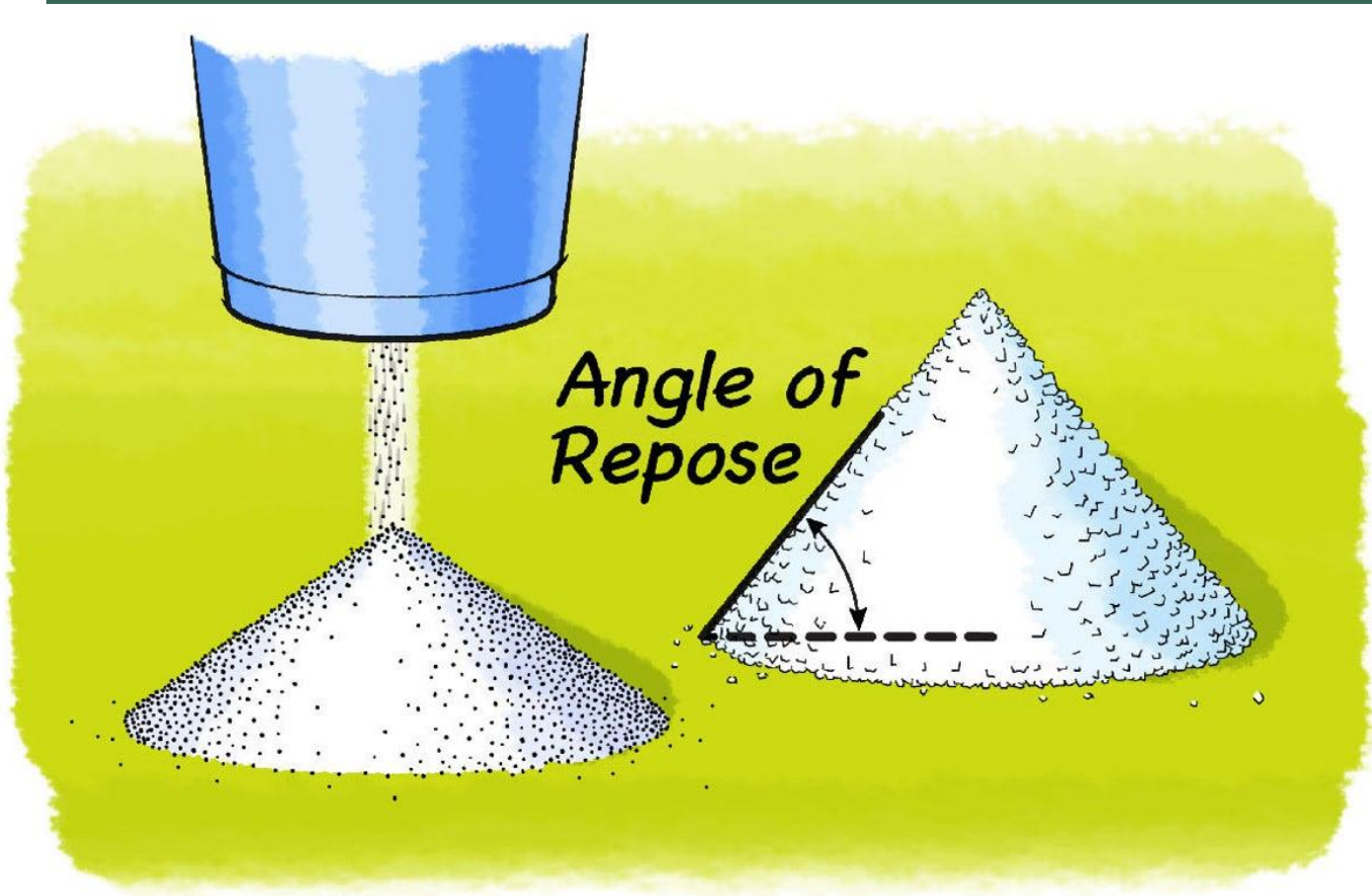
Which of the following is a good insulator

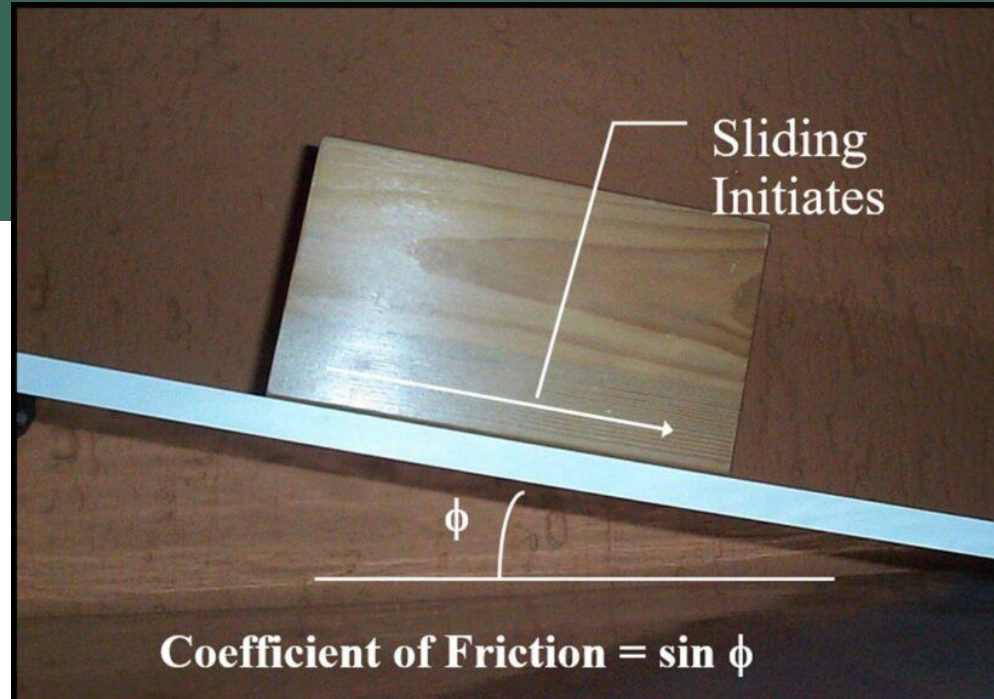
- a. Brass
- b. Mica
- c. Tin
- d. Platinum

QUESTION NO. 75

The angle measured from the horizontal at which product will start to move downwards over a smooth surface with gravity discharging the product is called a.

- a. angle of discharge
- b. angle of friction
- c. angle of repose
- d. hopper angle





3.3

angle of friction

angle of the material when the grains held starts to move



QUESTION NO. 75

The angle measured from the horizontal at which product will start to move downwards over a smooth surface with gravity discharging the product is called a.

- a. angle of discharge
- b. angle of friction
- c. angle of repose
- d. hopper angle

QUESTION NO. 76

In PAES 616:2016, precautionary measures in *Wastewater Re-Use for Irrigation*, which is false?

- a. Direct contact with wastewater shall be avoided
- b. Use of fine mist for sprinkler irrigation shall be used
- c. Potable and wastewater lines shall not cross connect
- d. Irrigation with wastewater shall be stopped immediately when algal bloom occurs

10 Precautionary Measures

10.1 Direct contact with wastewater shall be avoided.

10.2 Use of fine mist for sprinkler irrigation shall be avoided to minimize the risk of aerosol dispersion by wind drift.

10.3 Potable and wastewater lines shall not cross-connect. Wastewater pipeline shall be installed far enough from a parallel potable water pipeline.

10.4 Storage facilities shall be designed such that seepage is prevented and freeboard is adequate.

10.5 Irrigation with wastewater shall be stopped immediately when algal bloom occurs.



QUESTION NO. 76

In PAES 616:2016, precautionary measures in *Wastewater Re-Use for Irrigation*, which is false?

- a. Direct contact with wastewater shall be avoided
- b. Use of fine mist for sprinkler irrigation shall be used
- c. Potable and wastewater lines shall not cross connect
- d. Irrigation with wastewater shall be stopped immediately when algal bloom occurs

QUESTION NO. 77

Corn Grits with 0.86 mm to 1.09 mm length.

- a. Grit #16
- b. Grit #18
- c. Grit #10
- d. Grit #14



3.5.1

grit #10

milled corn kernels with particle size between 1.8 mm to 2.0 mm

3.5.2

grit # 12

milled corn kernels with particle size between 1.5 mm to 1.7 mm

3.5.3

grit # 14

milled corn kernels with particle size between 1.2 mm to 1.4 mm

3.5.4

grit # 16

milled corn kernels with particle size between 1.10 mm to 1.19 mm

3.5.5

grit # 18

milled corn kernels with particle size between 0.86 mm to 1.09 mm



AIM TO TOP
ILIGAN

QUESTION NO. 77

Corn Grits with 0.86 mm to 1.09 mm length.

- a. Grit #16
- b. Grit #18
- c. Grit #10
- d. Grit #14

QUESTION NO. 78

The amount of moisture in the grain expressed as a percentage of the total weight of the samples.

- a. Moisture content dry basis
- b. Moisture content wet basis
- c. Moisture content
- d. Equilibrium moisture content

QUESTION NO. 78

The amount of moisture in the grain expressed as a percentage of the total weight of the samples.

- a. Moisture content dry basis
- b. Moisture content wet basis
- c. Moisture content
- d. Equilibrium moisture content

QUESTION NO. 79

Recommended stacking density for corn in terms of number of bags per cubic meter.

- a. 10
- b. 9
- c. 12
- d. 15



AIM TO TOP

ILIGAN

Table 2 – Recommended stack heights

Type of grain	Maximum stack height in layers	Stack height, m
Wheat, barley and corn	18	4.57
Paddy	16	6
Rice	16	5.5

Table 3 – Recommended stacking density

Stacking Density	Number of bags per cubic meter
Paddy	10
Rice	15
Corn	12



QUESTION NO. 79

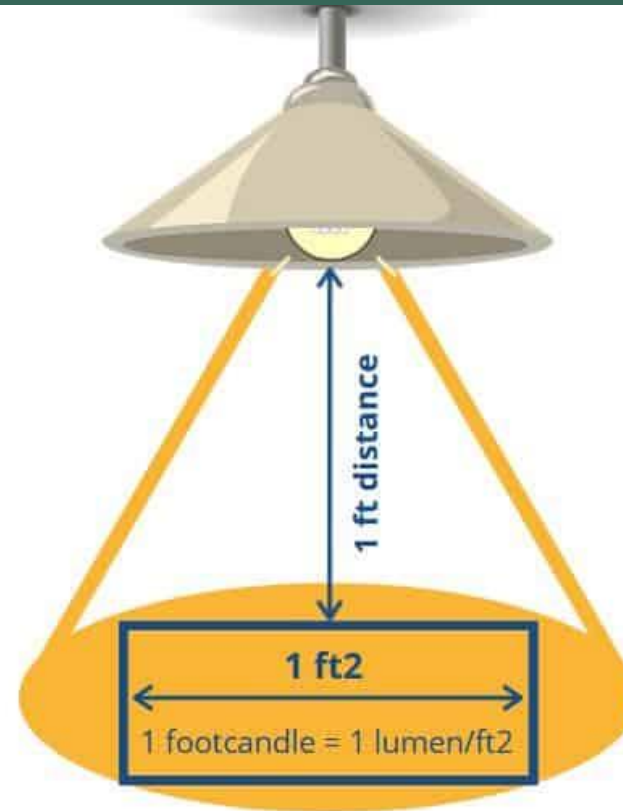
Recommended stacking density for corn in terms of number of bags per cubic meter.

- a. 10
- b. 9
- c. **12**
- d. 15

QUESTION NO. 80

A 1m x 100m poultry handling area is to be lighted at an average maintained footcandle of 50 fc. How many 2-lamp fixtures of 120cm long F40 T12 WW rapid start fluorescent lamps are required assuming the CU is 0.35 and the LLF is 0.75?

- a. 3 fixtures
- b. 6 fixtures
- c. 2 fixtures
- d. 4 fixtures



$$\text{No. of fixtures} = \frac{\text{Illumination} \times \text{Area}}{\text{Lamp per fixture} \times \text{lumens} \times \text{cu} \times \text{mf}}$$

QUESTION NO. 80

A 1m x 100m poultry handling area is to be lighted at an average maintained footcandle of 50 fc. How many 2-lamp fixtures of 120cm long F40 T12 WW rapid start fluorescent lamps are required assuming the CU is 0.35 and the LLF is 0.75?

- a. 3 fixtures
- b. 6 fixtures
- c. 2 fixtures
- d. 4 fixtures

QUESTION NO. 81

The heat exchange when a material changes its state of molecular aggregation without change in temperature.

- a. Latent heat
- b. Specific heat
- c. Sensible heat
- d. All of the above

LATENT HEAT VERSUS SENSIBLE HEAT

Latent heat is the amount of energy that is either absorbed or released during a phase transition at a constant temperature

Defined for a system with phase transition

Describes the changes in the internal energy of matter

Related to a system at a constant temperature

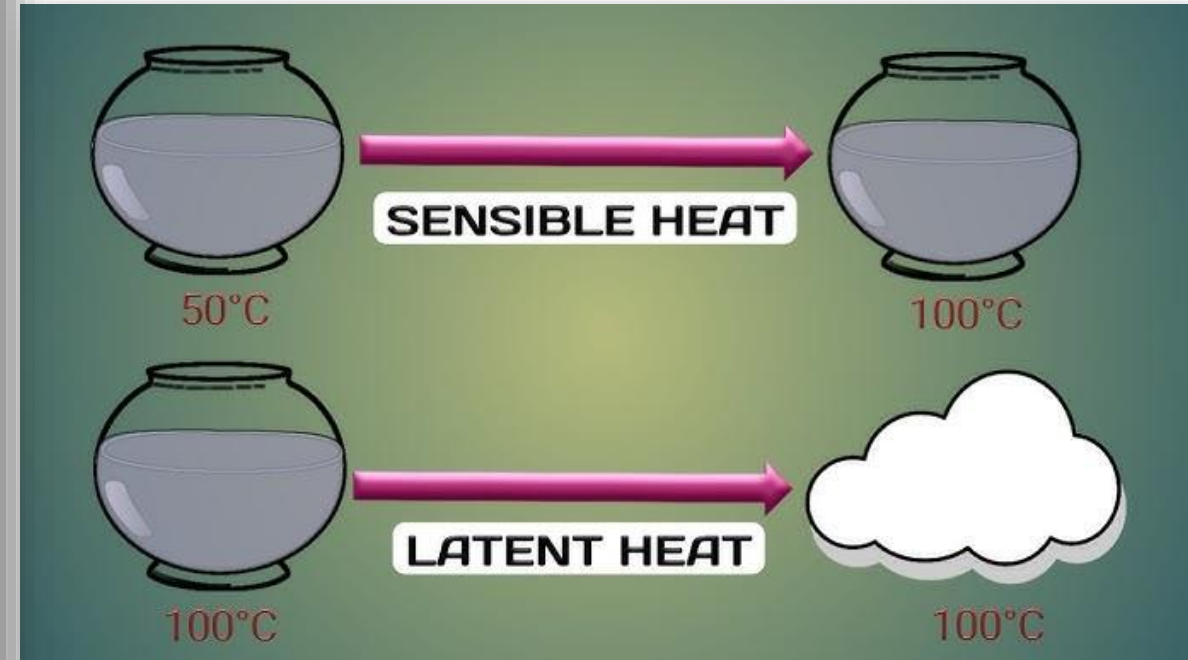
Sensible heat is the energy required to change the temperature of a substance with no phase change

Defined for a system without phase transition

Describes the energy exchange between matter and surroundings

Related to a system with changes in temperature

Visit www.pediaa.com



QUESTION NO. 81

The heat exchange when a material changes its state of molecular aggregation without change in temperature.

- a. **Latent heat**
- b. Specific heat
- c. Sensible heat
- d. All of the above

QUESTION NO. 82

A systematic approach to identifying, evaluating, and controlling food safety hazards.

- a. CPES
- b. HACCP
- c. GMP
- d. SSOP

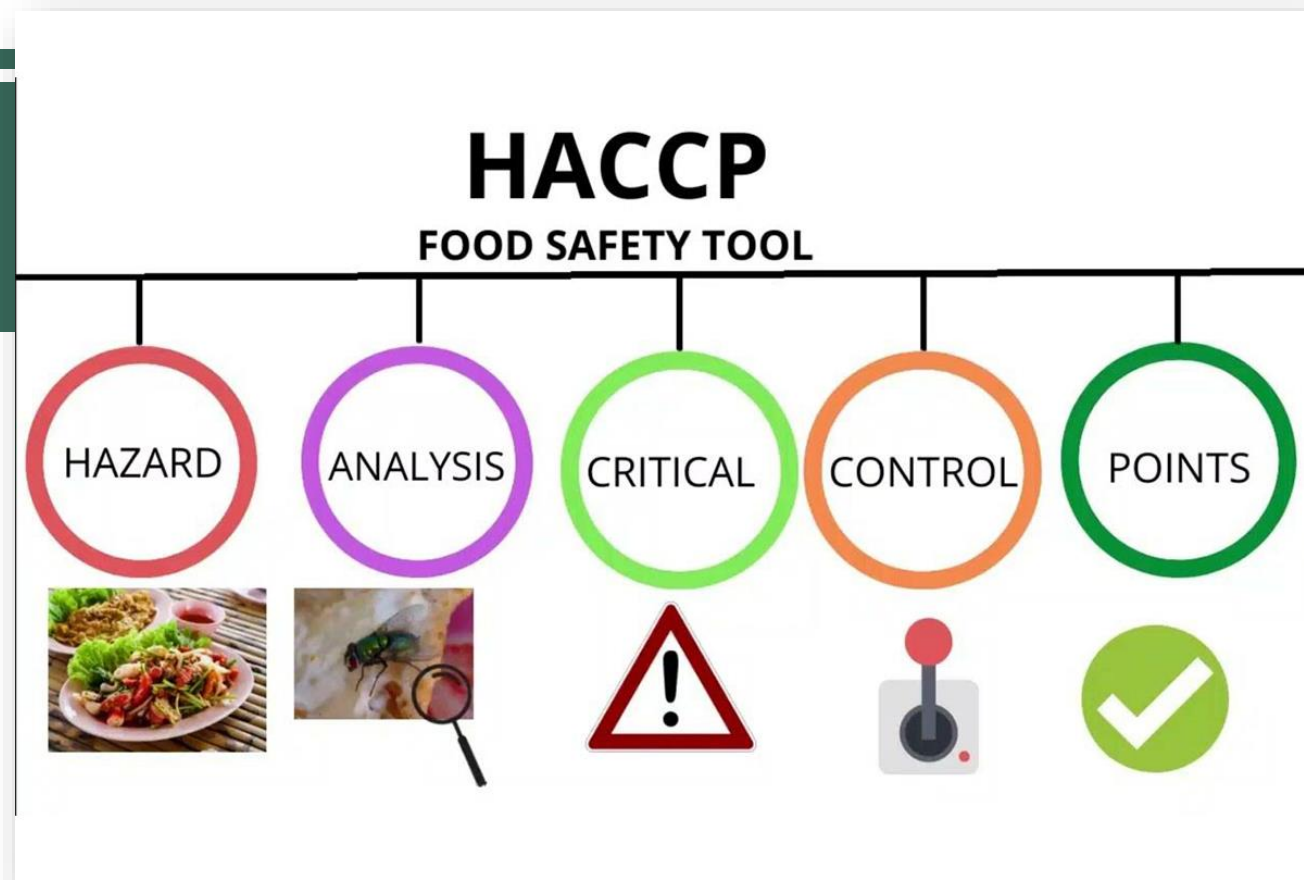


PCAF is the Implementing Unit of the Constructors' Performance Evaluation System (CPES) under the Department of Agriculture.

The CPES aims to ensure that all infrastructure projects carried out by the Department of Agriculture meet necessary standards and requirements.



Constructors' Performance Evaluation System (CPES) is a system of grading the performance of a constructor for a specific kind of projects using a set of criteria.



HACCP is a management system in which food safety is addressed through the analysis and control of biological, chemical, and physical hazards.

The 5 P's of Good Manufacturing Practices (GMP)



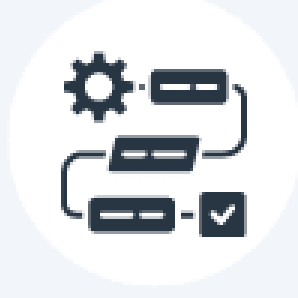
People

Comprehend roles and responsibility



Products

Clear specifications at every phase of production



Processes

Properly documented, simple, and consistent



Procedures

Guidelines for undertaking critical processes



Premises

Cleanliness and equipment calibration at all times

GMP stands for Good Manufacturing Practices, a system that ensures that manufactured products—such as food, cosmetics, and pharmaceutical goods—are consistently produced and controlled according to set quality standards.

A **Sanitation Standard Operating Procedure (SSOP)**

is a written document of procedures or programs used to maintain equipment and the environment in a sanitary condition for food processing. It is a step-by-step description of cleaning and sanitizing procedures and specifies

- ✓ what is to be cleaned
- ✓ how it is to be cleaned,
- ✓ how often it is to be cleaned, and
- ✓ what records are used to monitor the procedures.



QUESTION NO. 82

A systematic approach to identifying, evaluating, and controlling food safety hazards.

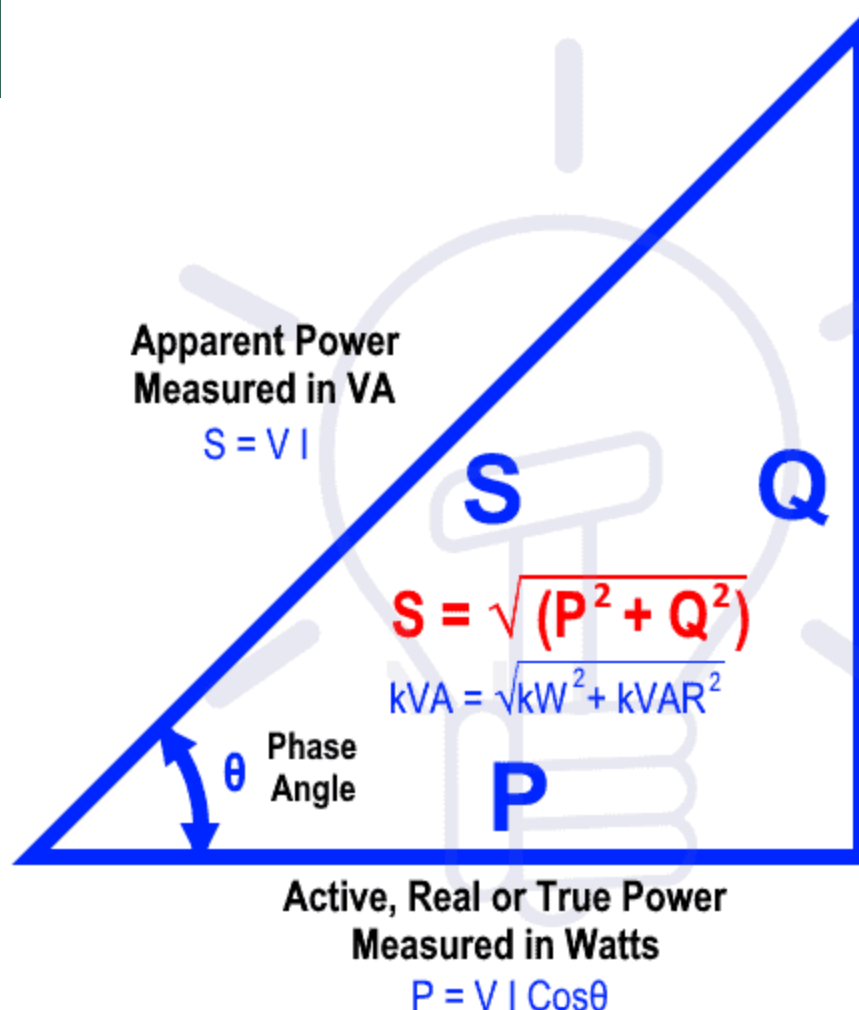
- a. CPES
- b. **HACCP**
- c. GMP
- d. SSOP

QUESTION NO. 83

For the purely inductive circuit, the power factor is:

- a. 0
- b. <1
- c. >1
- d. 1

Power Triangle & Power Factor



$$P = VI \cos\theta$$

$$\cos\theta = \frac{P}{VI}$$

$$\cos\theta = \frac{kW}{kVA}$$

$$\cos\theta = \frac{\text{True Power}}{\text{Apparent Power}}$$

$$\cos\theta = \frac{R}{Z}$$

$$\cos\theta = \frac{kW}{kVA} = \text{Power Factor}$$

$$\text{Power Factor} = \frac{\text{Active Power}}{\text{Reactive Power}}$$

www.electricaltechnology.org



AIM TO TOP
ILIGAN

QUESTION NO. 83

For the purely inductive circuit, the power factor is:

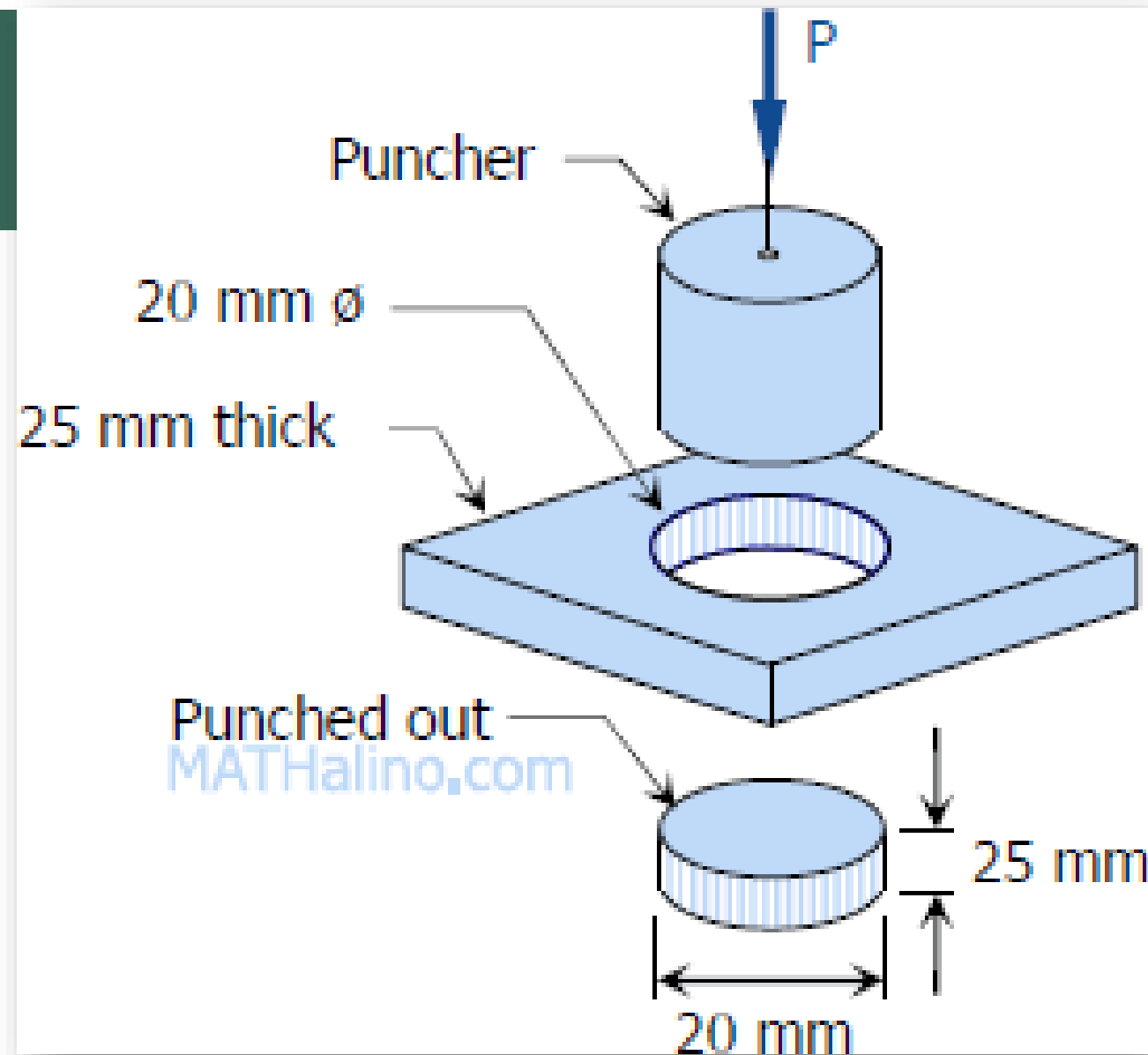
- a. 0
- b. <1
- c. >1
- d. 1

QUESTION NO. 84

What force is required to punch a 20 mm diameter hole in a plate that is 25 mm thick? The shear strength is 350 MPa

- a. 550 kN
- b. 400 kN
- c. 500 kN
- d. 450 kN





QUESTION NO. 84

What force is required to punch a 20 mm diameter hole in a plate that is 25 mm thick? The shear strength is 350 MPa

- a. 550 kN
- b. 400 kN
- c. 500 kN
- d. 450 kN

QUESTION NO. 85

In PAES 418:2002, area in sqm per ton of commodity in primary processing plant is:

- a. 10 sqm per ton
- b. 15 sqm per ton
- c. 20 sqm per ton
- d. 25 sqm per ton



5 Space requirements

5.1 Capacity of the processing plant shall be calculated based on the volume of commodity being processed. An area of 20 square meter per ton of commodity shall be provided.

5.2 The storage area of a processing plant shall have a capacity for temporary storage of raw materials for 2-5 processing days.

5.3 Additional area shall be for equipment, working tables and space for circulation.



QUESTION NO. 85

In PAES 418:2002, area in sqm per ton of commodity in primary processing plant is:

- a. 10 sqm per ton
- b. 15 sqm per ton
- c. 20 sqm per ton
- d. 25 sqm per ton



QUESTION NO. 86

When the surface of an air-cooling evaporator falls below 0°C , frost will form. Frost is:

- a. Air and ice
- b. Air and water
- c. Frozen water
- d. Ice foam



AIM TO TOP

ILIGAN

QUESTION NO. 86

When the surface of an air-cooling evaporator falls below 0°C , frost will form. Frost is:

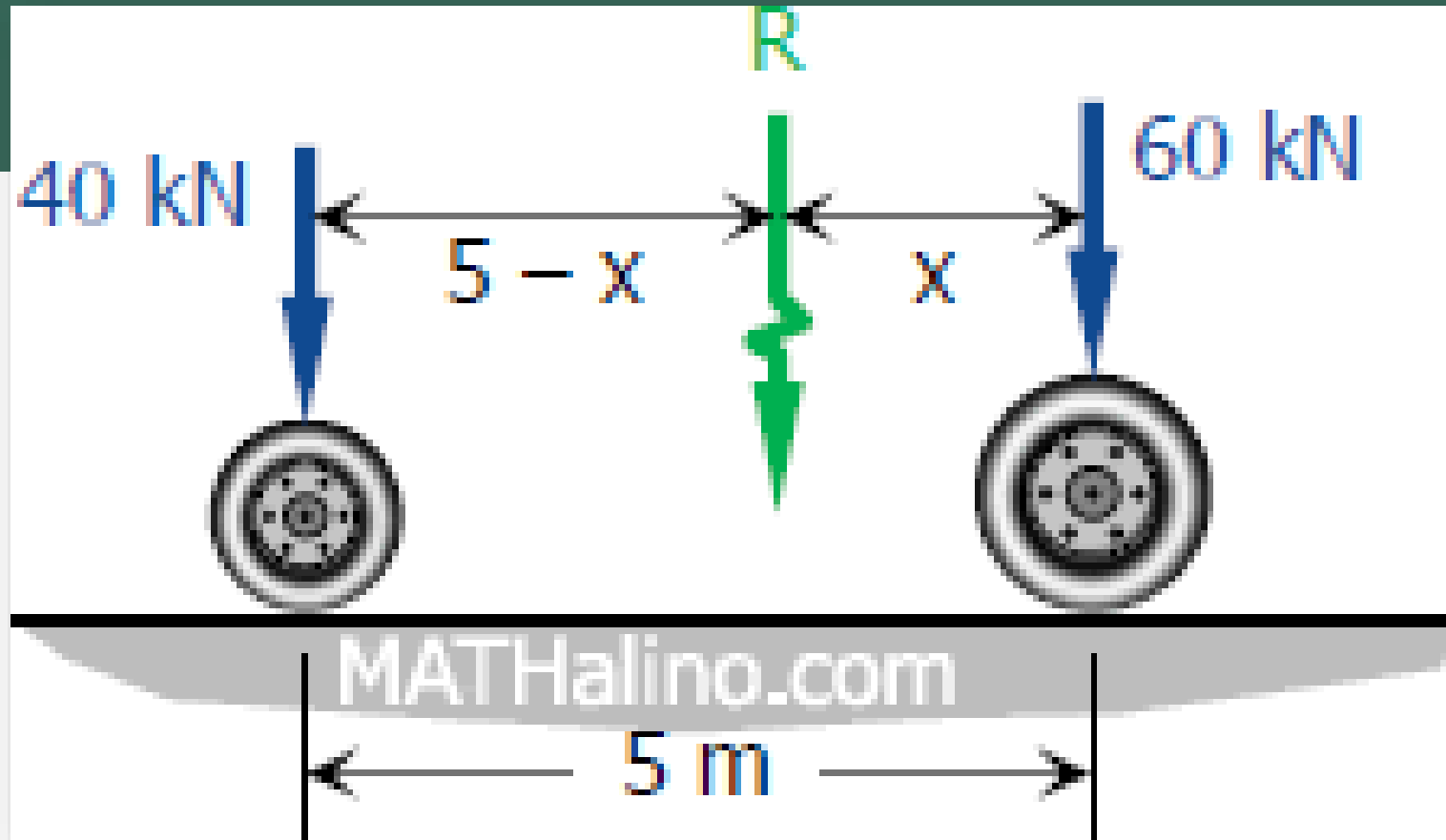
- a. Air and ice
- b. Air and water
- c. Frozen water
- d. Ice foam

QUESTION NO. 87

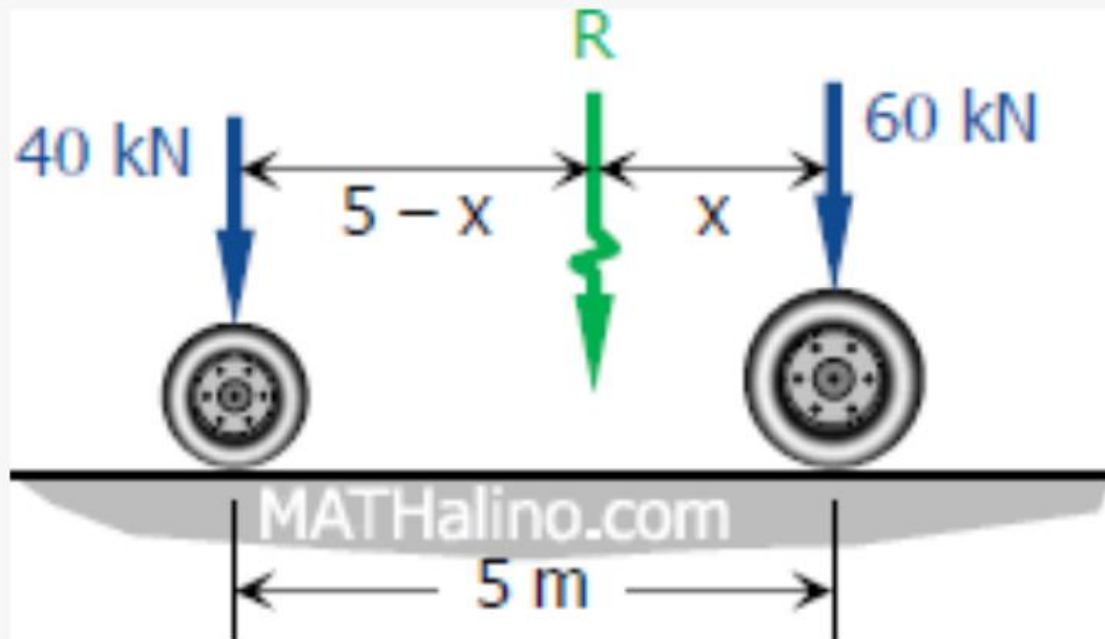
A truck with axle loads of 40 kN and 60 kN on a wheel base of 5 m rolls across a 10-m span. Compute the maximum bending moment.

- a. 160 kN·m
- b. 122.5 kN·m
- c. 245 kN·m
- d. 320 kN·m





Solution to Problem 453 | Moving Loads | Strength of Materials Review at MATHalino



$$R = 40 + 60 = 100 \text{ kN}$$

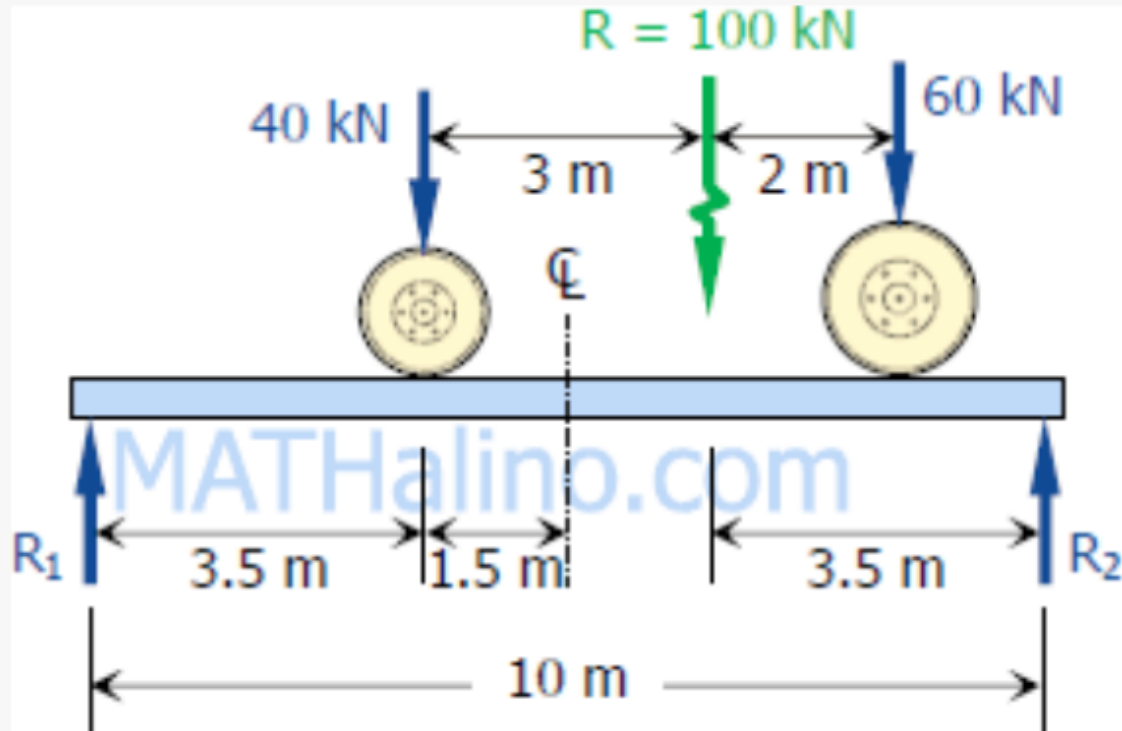
$$xR = 40(5)$$

$$x = 200/R$$

$$x = 200/100$$

$$x = 2 \text{ m}$$

For maximum moment under 40 kN wheel:



$$\Sigma M_{R2} = 0$$

$$10R_1 = 3.5(100)$$

$$R_1 = 35 \text{ kN}$$

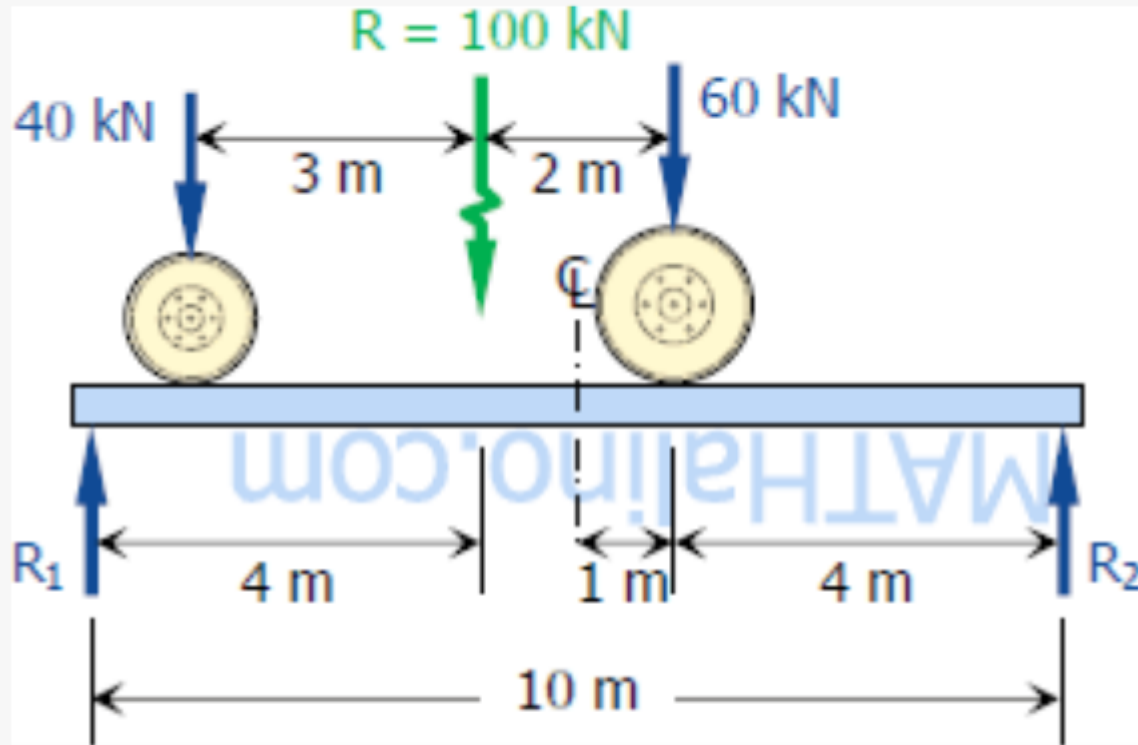
$$M_{\text{To the left of 40 kN}} = 3.5R_1$$

$$M_{\text{To the left of 40 kN}} = 3.5(35)$$

$$M_{\text{To the left of 40 kN}} = 122.5 \text{ kN} \cdot \text{m}$$



For maximum moment under 60 kN wheel:



$$\Sigma M_{R1} = 0$$

$$10R_2 = 4(100)$$

$$R_2 = 40 \text{ kN}$$

$$M_{To \text{ the right of } 60 \text{ kN}} = 4R_2$$

$$M_{To \text{ the right of } 60 \text{ kN}} = 4(40)$$

$$M_{To \text{ the right of } 60 \text{ kN}} = 160 \text{ kN} \cdot \text{m}$$

Thus, $M_{max} = 160 \text{ kN} \cdot \text{m}$ *answer*

QUESTION NO. 87

A truck with axle loads of 40 kN and 60 kN on a wheel base of 5 m rolls across a 10-m span. Compute the maximum bending moment.

- a. 160 kN·m
- b. 122.5 kN·m
- c. 245 kN·m
- d. 320 kN·m

QUESTION NO. 88

Biochemical Oxygen Demand (BOD₅) in mg/L for wastewater re-use:

- a. <500
- b. <200
- c. <150
- d. <50

Table 1. Limits on wastewater quality for irrigation

Parameter	Unit	Landscape Irrigation	Method of Analysis	Crop Irrigation		
				Food eaten raw and not commercially processed	Food crops commercially processed	Non-food crops
For crop productivity and protection of environment						
Bicarbonates ¹	mg/L	<500		<500	<500	<500
Biochemical Oxygen Demand (BOD ₅)	mg/L	<150	Azide Modification (Dilution Technique)	<150	<150	<150
Electrical Conductivity	µS/cm	<2000		<1000	<1000	<2000
Free Residual Chlorine	mg/L	<1		<1	<1	<1
pH		6.5 – 8.0	Glass Electrode Method	6.5 – 8.0	6.5 – 8.0	6.5 – 8.0
Sodium Adsorption Ratio (SAR)		<18		<18	<18	<18
Sodium (Na)	meq/L	<3		<3	<3	<3
Total Nitrogen (TN) ¹	mg/L	<30		<30	<30	<30
Total Phosphorus (TP) ¹	mg/L	<30		<30	<30	<30
Total Suspended Solids	mg/L	<140	Gravimetric Method	<140	<140	<140
For protection of animal and human health						
Ascaris	MPN/100 mL	0		0	0	0
Fecal Coliform	MPN/100 mL	<200	Multiple-Tube Fermentation Technique or Membrane Filter	Not detectable ²	<200	<200
Nematodes	MPN/100 mL	0		0	0	0
¹ None to moderate degree of restriction						
² Not detectable – means the total number of total fecal coliform organisms shall not exceed 14 MPN/100 ML in any sample						

ADOPTED AND MODIFIED from Department of Agriculture. 2007. Guidelines on the Procedures and Technical Requirements for the Issuance of a Certification Allowing the Safe Re-Use Of Wastewater for Purposes of Irrigation and Other Agricultural Uses and DENR Administrative Order 35 Series Of 1990 – Revised Effluent Regulations.

QUESTION NO. 88

Biochemical Oxygen Demand (BOD₅) in mg/L for wastewater re-use:

- a. <500
- b. <200
- c. <150
- d. <50

QUESTION NO. 89

In an inductive circuit, the inductance is 1.5H. What is the inductive resistance at 50Hz source?

- a. 471 ohms
- b. 461 ohms
- c. 416 ohms
- d. 417 ohms

QUESTION NO. 89

In an inductive circuit, the inductance is 1.5H. What is the inductive resistance at 50Hz source?

- a. 471 ohms
- b. 461 ohms
- c. 416 ohms
- d. 417 ohms

QUESTION NO. 90

Corn initially at 24% is to be dried to 14%. If the initial weight of the shelled corn is 1000 kilos, what is its final weight?

- a. 893.72 kg
- b. 783.72 kg
- c. 883.72 kg
- d. 383.72 kg



Calculating moisture content from wet weight and dry weight

Moisture Content of grain is usually determined on wet basis (wb), scientists sometimes use dry basis (db):

$$MC_{wb} = \frac{W_i - W_f}{W_i} \times 100$$

$$MC_{db} = \frac{W_i - W_f}{W_f} \times 100$$

MC_{wb} = Moisture content wet basis [%]

MC_{db} = Moisture content dry basis [%]

W_i = Initial weight

W_f = Final weight

Reference: <http://www.knowledgebank.irri.org/step-by-step-production/postharvest/harvesting/measuring-moisture-content/moisture-content-calculations>

Conversion from MC_{wb} to MC_{db} and back

Sometimes it is needed go convert from moisture content dry basis to moisture content wet basis.

$$MC_{wb} = \frac{100 \times MC_{db}}{100 + MC_{db}}$$

$$MC_{db} = \frac{100 \times MC_{wb}}{100 - MC_{wb}}$$

Reference: <http://www.knowledgebank.irri.org/step-by-step-production/postharvest/harvesting/measuring-moisture-content/moisture-content-calculations>

Weight loss during drying

During drying, paddy grain will lose weight due to loss of moisture:

$$W_f = W_i \times \frac{100 - MC_i}{100 - MC_f}$$

W_i = Initial weight [g]

W_f = Final weight [g]

Reference: <http://www.knowledgebank.irri.org/step-by-step-production/postharvest/harvesting/measuring-moisture-content/moisture-content-calculations>

QUESTION NO. 90

Corn initially at 24% is to be dried to 14%. If the initial weight of the shelled corn is 1000 kilos, what is its final weight?

- a. 893.72 kg
- b. 783.72 kg
- c. 883.72 kg
- d. 383.72 kg

QUESTION NO. 91

In PAES 411:2000, the gut room/tripery shall be provided with lighting intensity of ____ lux.

- a. 200
- b. 220
- c. 540
- d. 110



AIM TO TOP
REGIONAL TRAINING CENTER
ILIGAN

8.6 Gut Room/Tripery

8.6.1 For green offal, a separate area shall be provided for the emptying and cleansing of stomach and intestines and for the preparation of casings tripe or edible fat.

8.6.2 Illumination of 220 lux shall be provided.



QUESTION NO. 91

In PAES 411:2000, the gut room/tripery shall be provided with lighting intensity of ____ lux.

- a. 200
- b. 220
- c. 540
- d. 110



AIM TO TOP

ILIGAN

QUESTION NO. 92

Final weight for 9 tons of rice to be dried from 24% to 14%.

- a. 7.9 ton
- b. 8.9 ton
- c. 9.9 tons
- d. 10.9 tons

QUESTION NO. 92

Final weight for 9 tons of rice to be dried from 24% to 14%.

- a. **7.9 ton**
- b. 8.9 ton
- c. 9.9 tons
- d. 10.9 tons

QUESTION NO. 93

In PAES 409:2002, how many milking stalls are required for a milking herd having 21-30 animals with a milking machine?

- a. 1
- b. 2
- c. 3
- d. 4

Table 1 – Recommended number of milking stalls, machines and operators

Animals	No. of milking stalls	No. of operator		No. of milking machines* (if used)
		Without milking machine	With milking machine	
1-10	1	1	1	1
11-20	2	2	2	1
21-30	3	3	2	1

* Two buckets



AIM TO TOP
ILIGAN

QUESTION NO. 93

In PAES 409:2002, how many milking stalls are required for a milking herd having 21-30 animals with a milking machine?

- a. 1
- b. 2
- c. **3**
- d. 4

QUESTION NO. 94

Determine the volume of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5.

- a. 445 m^3
- b. 469 m^3
- c. 424 m^3
- d. 438 m^3



B.6 Gas production potential

Table B.4 –Gas production potential of various types of manure in m³/kg

Manure	Retention Period days			
	25	30	35	50
Pig	0.058	0.063	0.068	0.077
Cow	0.030	0.034	0.037	0.043
Buffalo	0.030	0.034	0.037	0.043
Horse	0.045	0.051	0.056	0.065
Chicken	0.060	0.065	0.069	0.078



QUESTION NO. 94

Determine the volume of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5.

- a. 445 m^3
- b. 469 m^3
- c. 424 m^3
- d. 438 m^3

Estimate the biogas production in cubic meters per day from a large balloon type digester using 1:1 water-manure ratio. The source farm has 1,100 heads of porkers of mixed ages. Use a retention period of 30 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 30-day manure retention period is $0.063 \text{ m}^3/\text{kg}$.

A. 97

B. 153

C. 125

D. 193

Answer: B

Solution:

$$P = NMG$$

where P = biogas production potential, N = No. of heads, M = manure production, G = Specific gas production of the manure for a specific retention period

$$\begin{aligned} P &= (1,100 \text{ heads})(2.2 \text{ kg/head})(0.063 \text{ m}^3/\text{kg}) \\ &= 152.46 \text{ m}^3 \end{aligned}$$



How many heads of breeding cattle are needed as source of manure to generate 80 m^3 of biogas a day if the retention period is 25 days and water-manure ratio is 1:1. The mean daily manure production of breeding cattle is 13 kg/head while its specific gas production for 25 day manure retention period is half of that of chicken dung.

A. 192

B. 205

C. 206

D. 139

Answer: C

Solution:

$$N = P/(MG)$$

*where N = No. of heads, C = biogas production, M = manure production,
G = Half of specific gas production of chicken dung for a specific retention period
= $1/2 \times 0.06 \text{ m}^3/\text{kg}$ or $0.03 \text{ m}^3/\text{kg}$*

$$C = (80 \text{ m}^3)/[(13 \text{ kg/head})(0.03 \text{ m}^3/\text{kg})]$$

= 205.13 heads, to avoid gas shortage round up to 206 heads

QUESTION NO. 95

What is the equivalent size in mm^2 of a 250 MCM cable?

- a. 126.67 mm^2
- b. 136.67 mm^2
- c. 146.67 mm^2
- d. 156.67 mm^2

QUESTION NO. 95

What is the equivalent size in mm^2 of a 250 MCM cable?

- a. 126.67 mm^2
- b. 136.67 mm^2
- c. 146.67 mm^2
- d. 156.67 mm^2

QUESTION NO. 96

Burning efficiency of the biomass furnace shall be at least:

- a. 98%
- b. 95%
- c. 90%
- d. 80%



AIM TO TOP

ILIGAN

- 6.4** Heating system efficiency of the furnace shall be at least 65% for direct-fired and 50% for indirect-fired. (PAES 201:2000)
- 6.5** Burning efficiency of the biomass furnace shall be at least 95%.
- 6.6** Furnace efficiency shall be at least 65%.



QUESTION NO. 96

Burning efficiency of the biomass furnace shall be at least:

- a. 98%
- b. 95%
- c. 90%
- d. 80%



AIM TO TOP
ASSESSMENT AND MONITORING TOOL FOR TRAINING QUALITY
ILIGAN

QUESTION NO. 97

Stoichiometric air requirement for pyrolysis

- a. 10-20%
- b. 20-30%
- c. 30-40%
- d. 40-50%



QUESTION NO. 97

Stoichiometric air requirement for pyrolysis

- a. 10-20%
- b. 20-30%
- c. 30-40%
- d. 40-50%

QUESTION NO. 98

The C/N Ratio of agricultural waste

- a. 10:20
- b. 20:30
- c. 30:40
- d. 40:50



AIM TO TOP

ILIGAN

Table 2 - Typical carbon to nitrogen ratios of common composting materials

Material	C:N ratios
Cattle manure (with straw)	25–30
Cattle feedlot	13
Corn & sorghum stover	60–100
Cucumber	20
Dairy manure	10–18
Garden wastes	20–60
Grain rice	36
Grass clippings	12–25
Green leaves	30–60
Horse	19
Lamb	10
Pig manure	5–8
Potato tops	25
Poultry manure (fresh)	6–10



Table 2 – continuation...

Material	C:N ratios
Poultry manure (henhouse litter)	12–18
Residue of mushroom culture	40
Rice straw	48–115
Saw dust	300–723
Seaweed	19
Soybean residues	20–40
Straw	40–80
Sugar cane (trash)	50
Tomato leaves	13
Tomatoes	25–30
Veal	2
Watermelon	20
Water hyacinth	20-30
Weeds	19
Wood chips	100–441



QUESTION NO. 98

The C/N Ratio of agricultural waste

- a. 10:20
- b. 20:30
- c. 30:40
- d. 40:50



AIM TO TOP

ILIGAN

QUESTION NO. 99

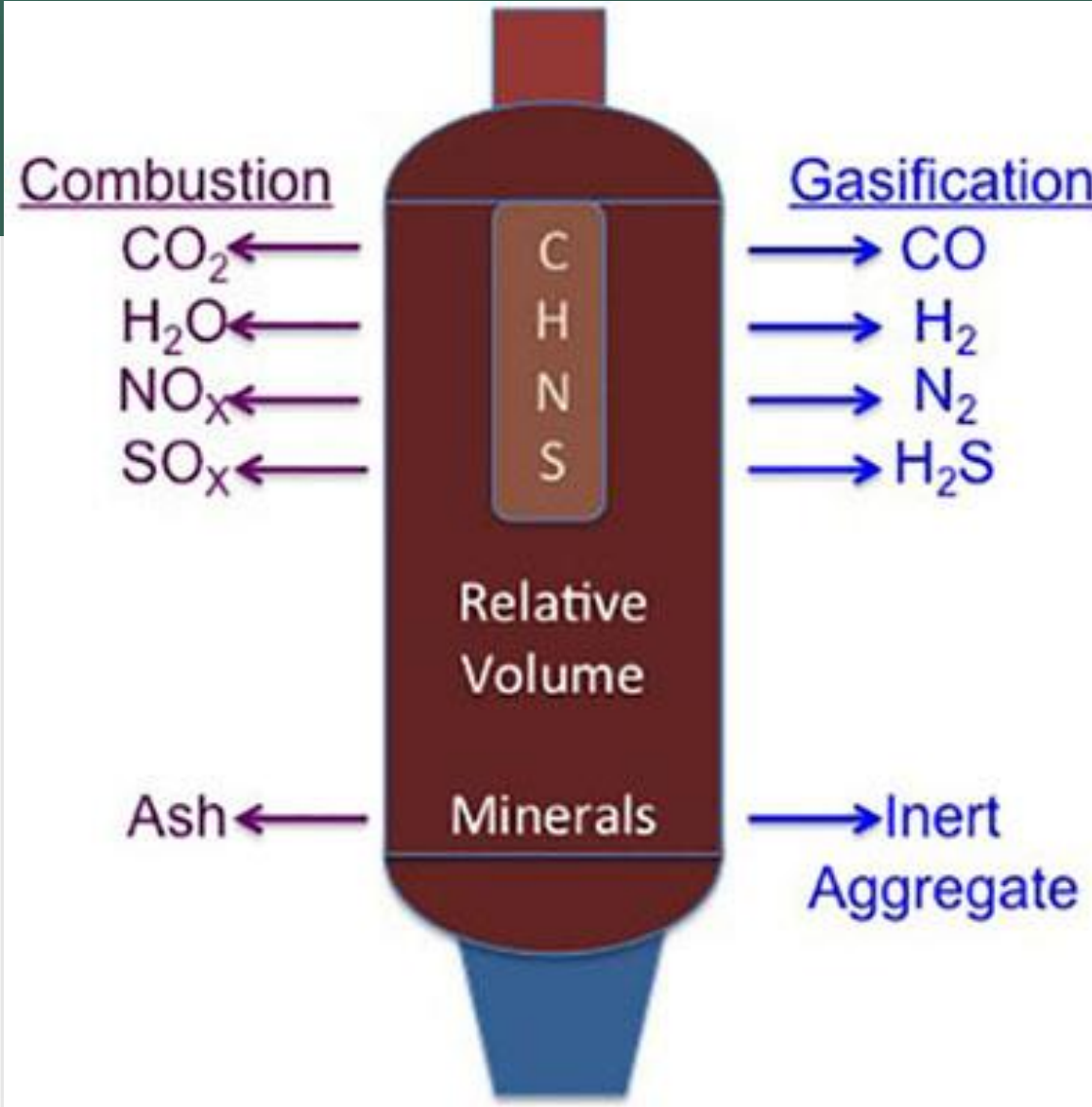
Main product of gasification

- a. CH_4
- b. CO
- c. O_2
- d. CO_2



AIM TO TOP

ILIGAN



QUESTION NO. 99

Main product of gasification

- a. CH_4
- b. CO
- c. O_2
- d. CO_2

QUESTION NO. 100

The heating system efficiency for rice hull direct-fired mechanical grain dryer

- a. 95%
- b. 80%
- c. 90%
- d. 65%



Table 1 – Performance Criteria for Mechanical Grain Dryer

CRITERIA	Performance Data	
	Rice	Corn
1. Moisture gradient, %, max.		
- Batch type	4.0	4.0
- Recirculating type	2.0	2.0
2. Drying rate, %/h	0.8 (Batch Type) 1.0 (Continuous flow)	2.0 (Shelled Corn) 0.4 (Ear Corn)
3. Product quality*		
- Cracked grain, % increase, max.	3.0 (Batch Type) 2.0 (Continuous flow)	10
- Head rice, % decrease, max.	5.0	N/A
- Hulled/damaged grain, % increase, max.	2.0	N/A
- Broken/split kernels, % increase, max.	N/A	7
4. Drying Loss/Spillage, %, max.	1.0	1.0
5. Drying efficiency, % min.	75	75
6. Heating system efficiency, % min		
- Petroleum based fuel direct-fired	90	90
indirect-fired	75	75
- Biomass fuel direct-fired	65	65
indirect-fired	50	50
7. Blower flow rate, cmm/kg (cfm/ton)	0.05 (1750)	0.05 (1750)
8. Static pressure, mm H ₂ O (in.H ₂ O)	25 (1)	25 (1)

* allowable difference between the laboratory analysis and machine's performance



QUESTION NO. 100

The heating system efficiency for rice hull direct-fired mechanical grain dryer

- a. 95%
- b. 80%
- c. 90%
- d. 65%